Modem verification · coverage map

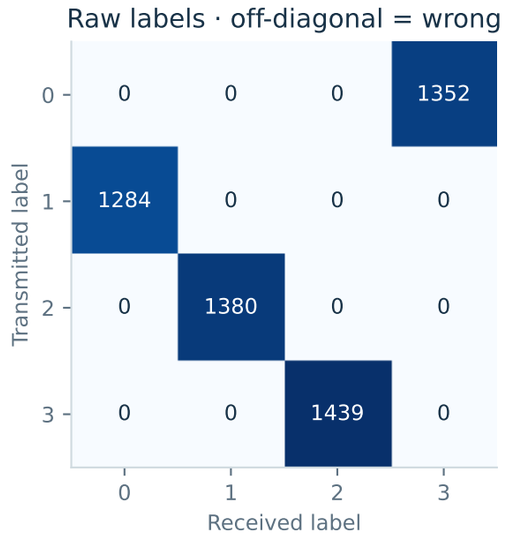
20 original models / 60 portable channel cases / 17 mapper blocks / 1 transmitter reference

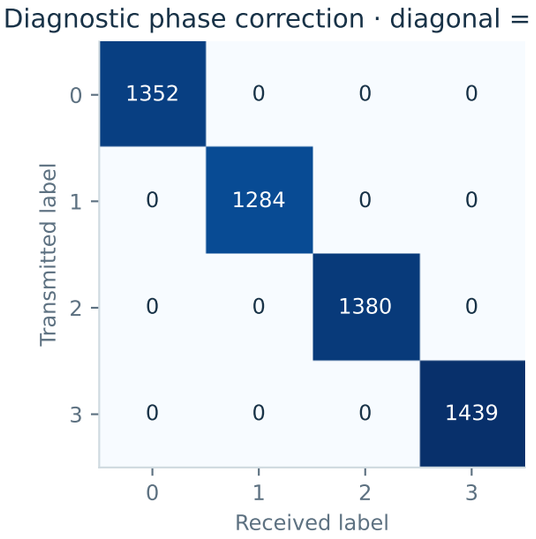
	Original compile	Adapted MATLAB run	Exact repeat	Portable 3 cases	Portable UVM	Mapper reference	MSK TX reference	Full RX equivalence
64-QAM · 2008	UPDATE	PASS	PASS	PASS	PASS	PASS	—	—
BPSK · Costas	UPDATE	PASS	PASS	PASS	PASS	PASS	—	—
8-PSK · legacy DQPSK archive	UPDATE	PASS	PASS	PASS	PASS	PASS	—	—
DQPSK · Farrow / fast AGC	UPDATE	PASS	PASS	PASS	PASS	PASS	—	—
GMSK · coherent original	PASS	PASS	PASS	PASS	PASS	—	—	—
MSK · modulator	PASS	PASS	PASS	PASS	PASS	—	TX ONLY	—
Pi/4 QPSK · 8 SPS / 2021	PASS	PASS	PASS	PASS	PASS	—	—	—
Pi/4 QPSK · fourth-power sync	PASS	PASS	PASS	PASS	PASS	—	—	—
8-PSK · fast feedforward	UPDATE	PASS	PASS	PASS	PASS	PASS	—	—
8-PSK · feedback / AGC	UPDATE	PASS	PASS	PASS	PASS	PASS	—	—
16-QAM · Gardner / Farrow	UPDATE	PASS	PASS	PASS	PASS	PASS	—	—
16-QAM · decision feedback EQ	UPDATE	PASS	PASS	PASS	PASS	PASS	—	—
16-QAM · variable delay	UPDATE	PASS	PASS	PASS	PASS	PASS	—	—
16-QAM · linear EQ	UPDATE	PASS	PASS	PASS	PASS	PASS	—	—
QPSK · phase noise / Costas	PASS	PASS	PASS	PASS	PASS	PASS	—	—
QPSK · fast acquisition	UPDATE	PASS	PASS	PASS	PASS	PASS	—	—
QPSK · feedback / AGC	UPDATE	PASS	PASS	PASS	PASS	PASS	—	—
QPSK · feedforward / Farrow	UPDATE	PASS	PASS	PASS	PASS	PASS	—	—
QPSK · phase tracking	UPDATE	PASS	PASS	PASS	PASS	PASS	—	—
SIMO · maximal ratio combining	UPDATE	PASS	PASS	PASS	PASS	PASS	—	—

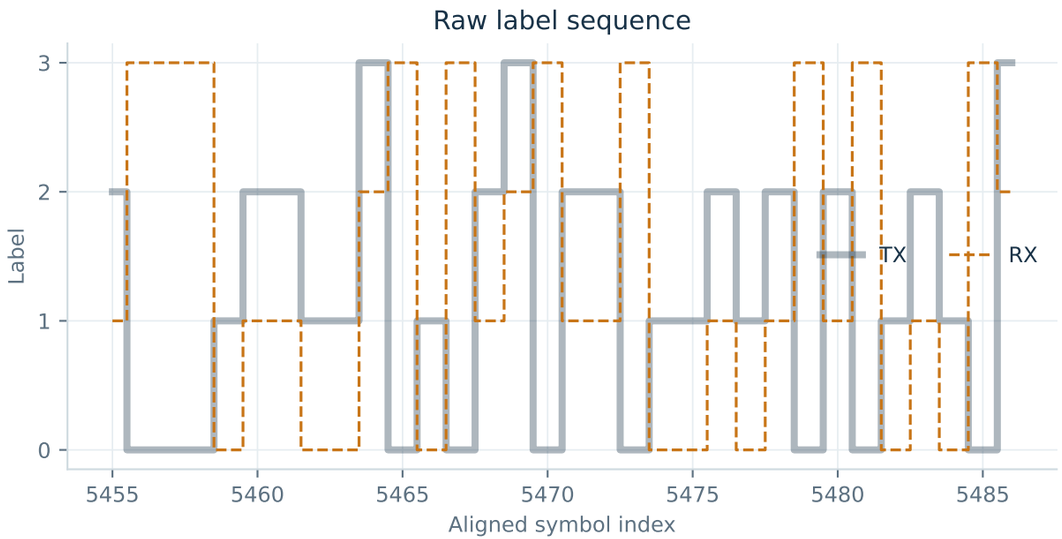
PASS applies only to the named check. UPDATE means the original requires compatibility changes in R2020b.
Gray cells are not established or not tested; the MSK reference covers TX after its scrambler at 32 samples/symbol.
Portable UVM checks agreement with Python. Original receiver correctness and complete historical equivalence remain unverified.

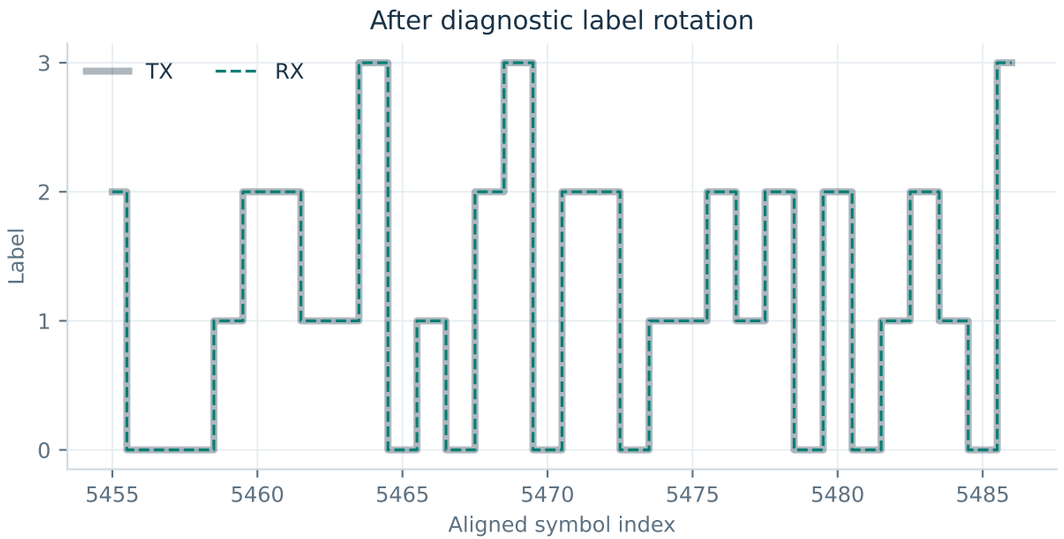
QPSK · expose the phase ambiguity

HELD-OUT CAPTURE / 5455 symbols / raw SER 100% / corrected diagnostic SER 0%





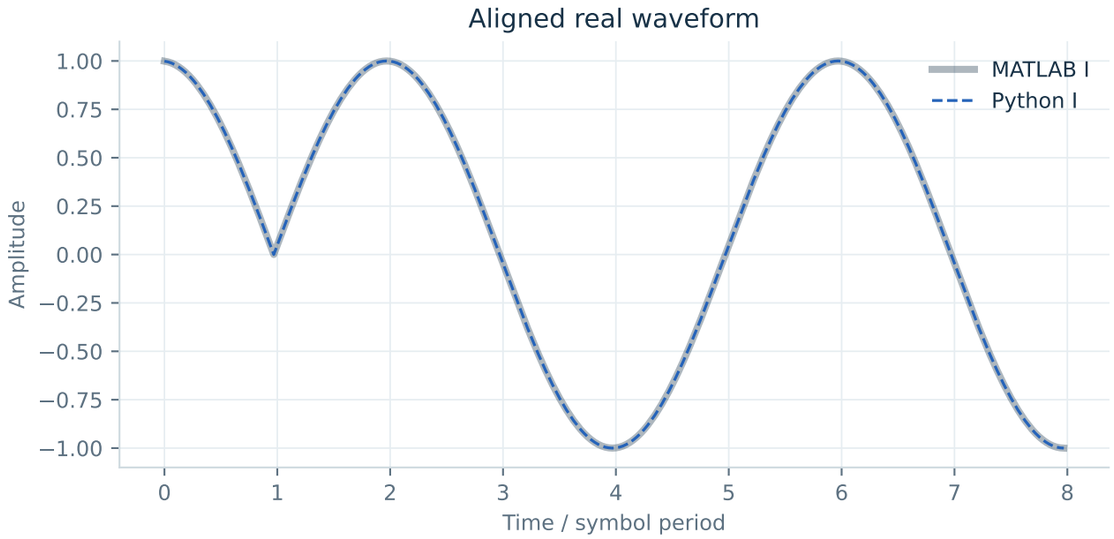


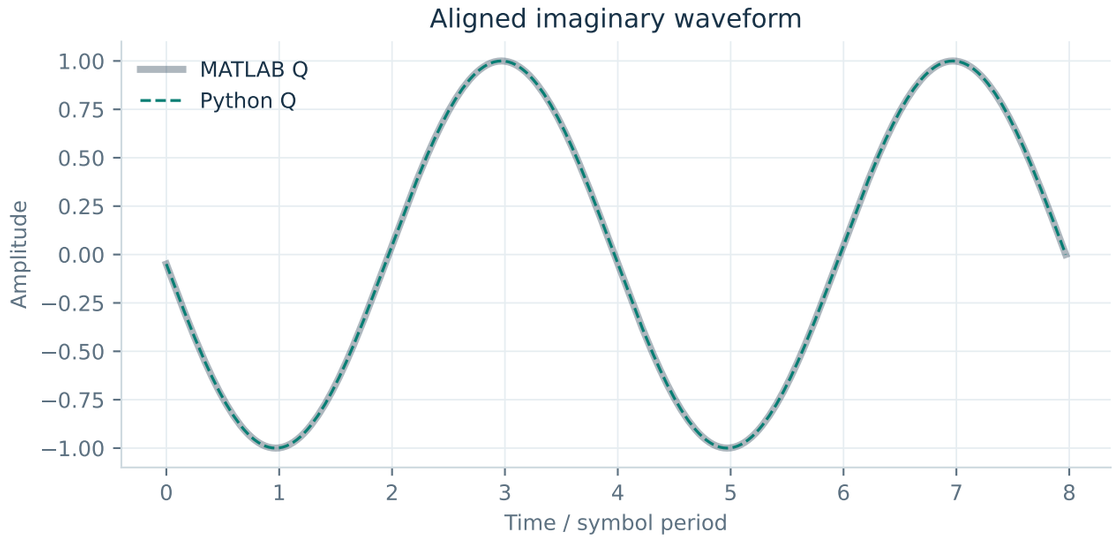


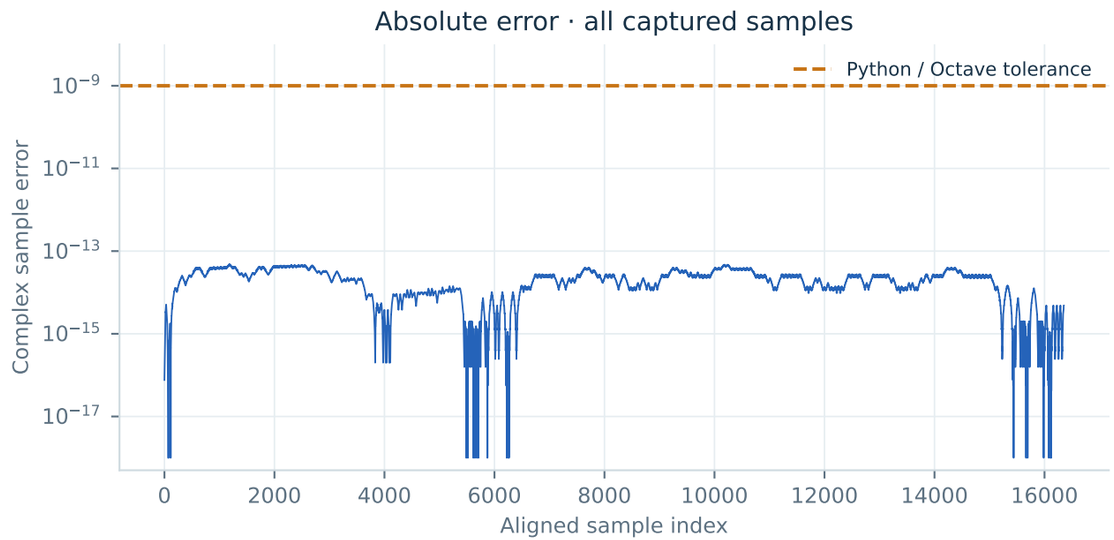
Source: original MATLAB QPSK phase-noise capture. Delay 13 comes from the archived error counter. Offset 3 modulo four was inferred before the held-out interval. The correction below is diagnostic only; it was not inserted into the receiver. The original counter reports 10910/10910 symbol errors.

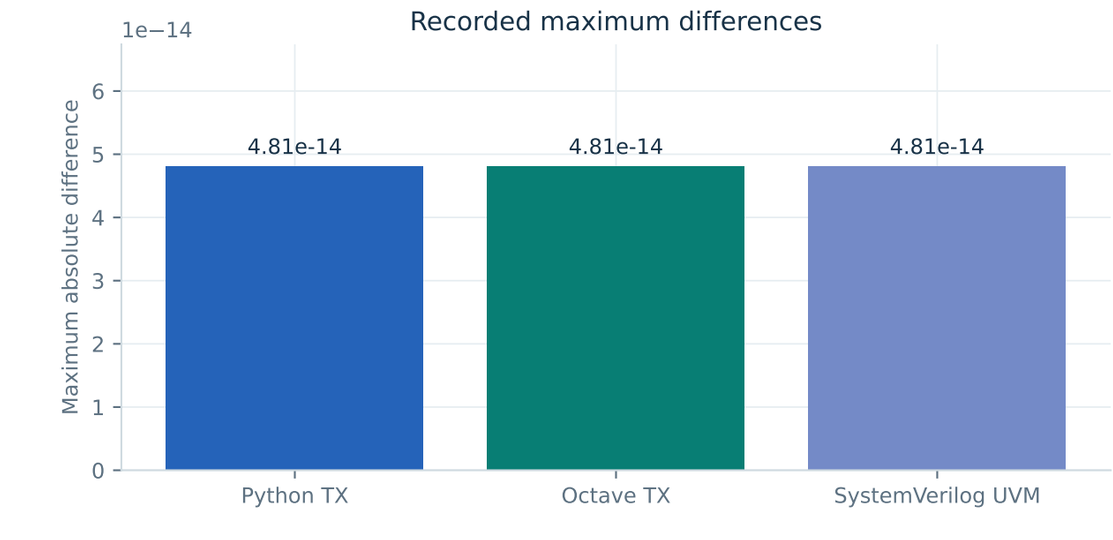
MSK · original waveform agreement

MATLAB REFERENCE / 16,352 samples / maximum TX error 4.81e-14





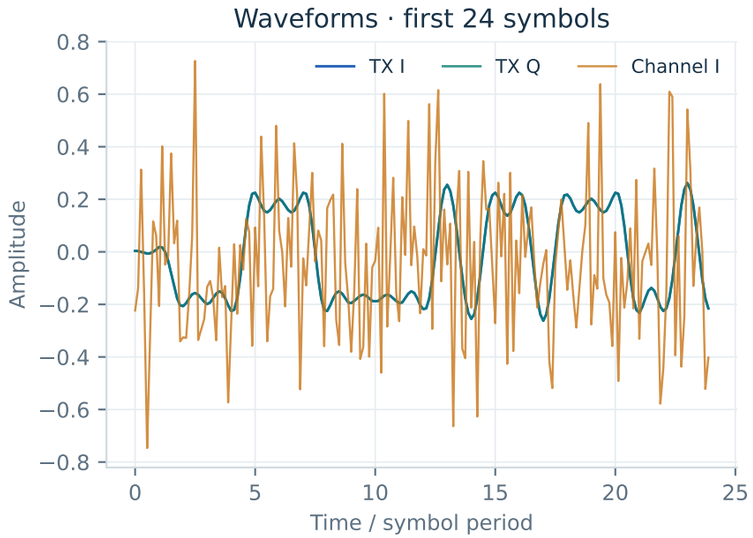


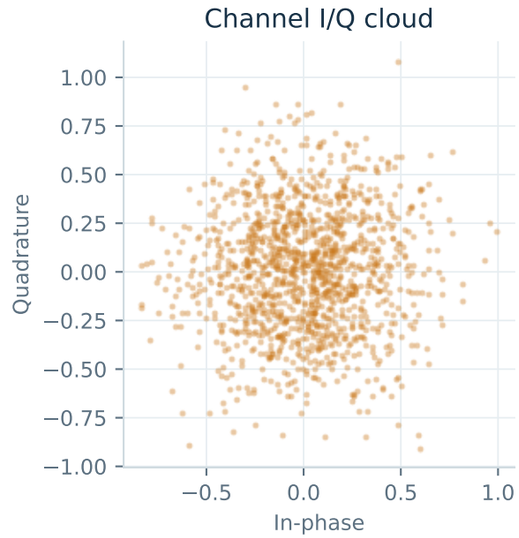


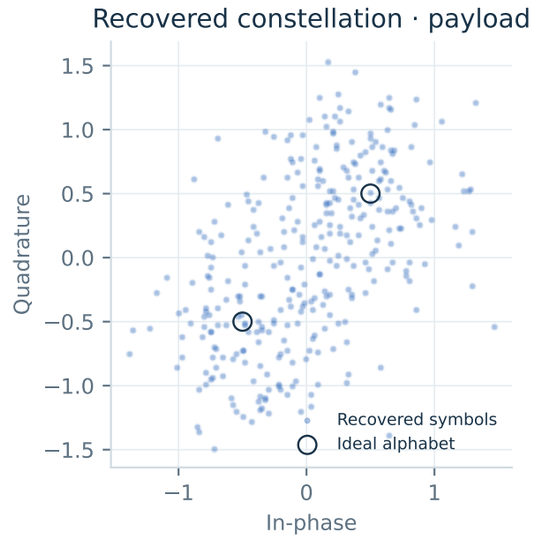
Source: actual MATLAB MSK samples after the original scrambler; 511 captured bits, 32 samples/symbol, fixed 33-sample startup alignment. No fitted gain or phase. Zero errors are plotted at 1e-18 on the log axis. The SystemVerilog maximum is from the recorded UVM run and includes Python-based RX comparisons.

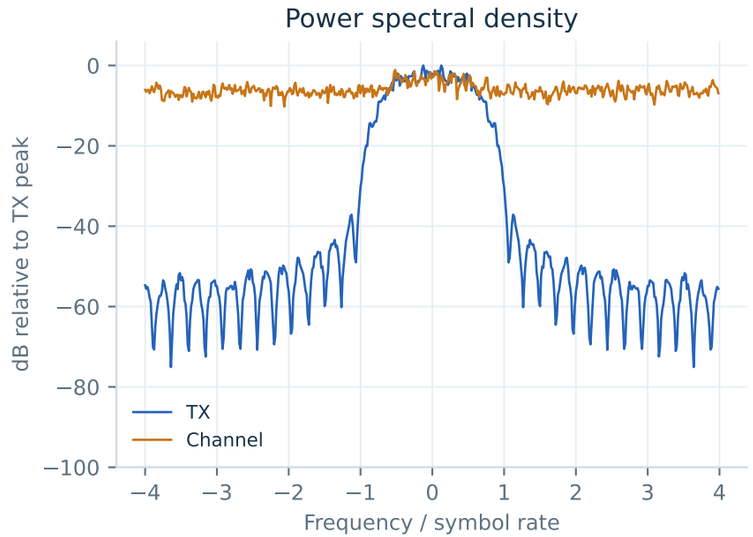
BPSK · Costas

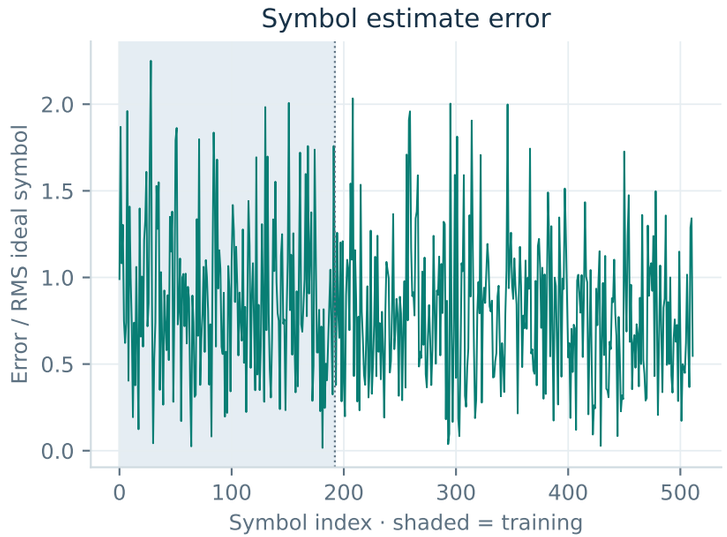
PORTABLE MODEL / Negative control · -8 dB sample SNR / 50 errors in 320 payload symbols

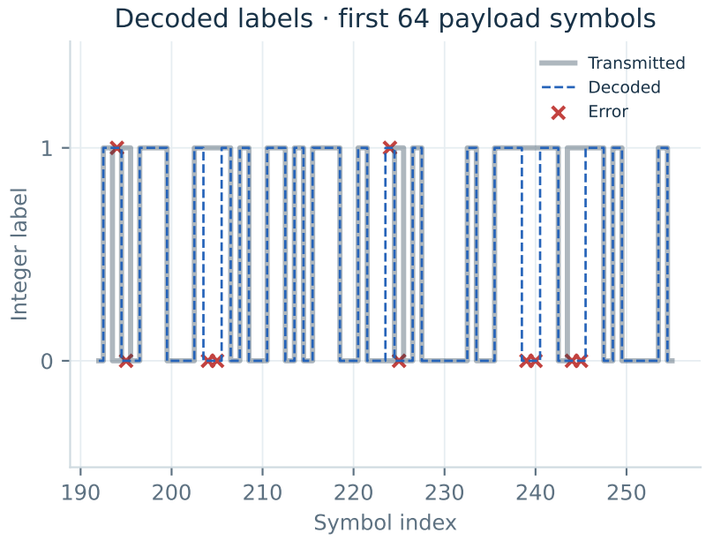








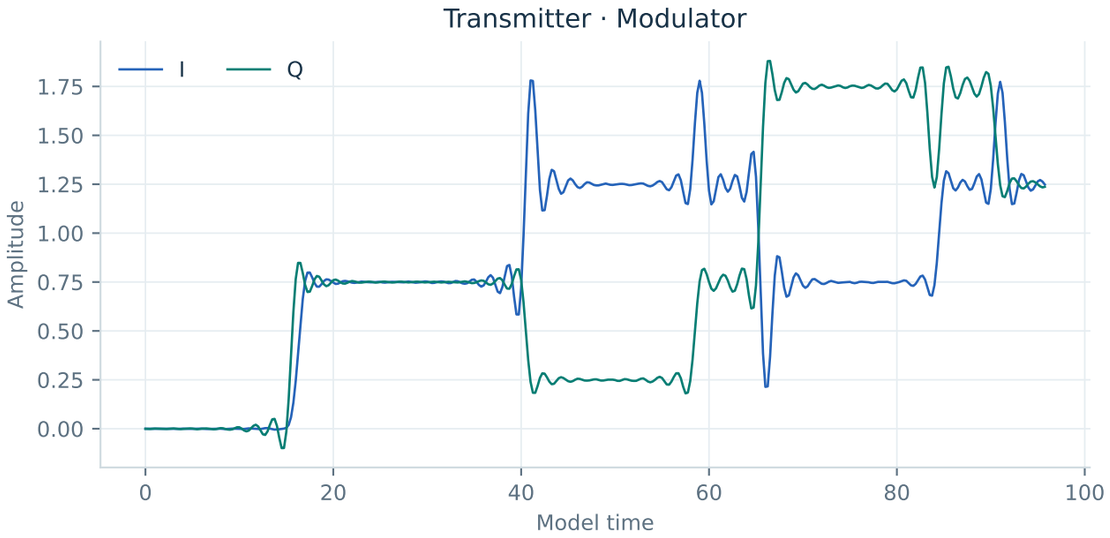


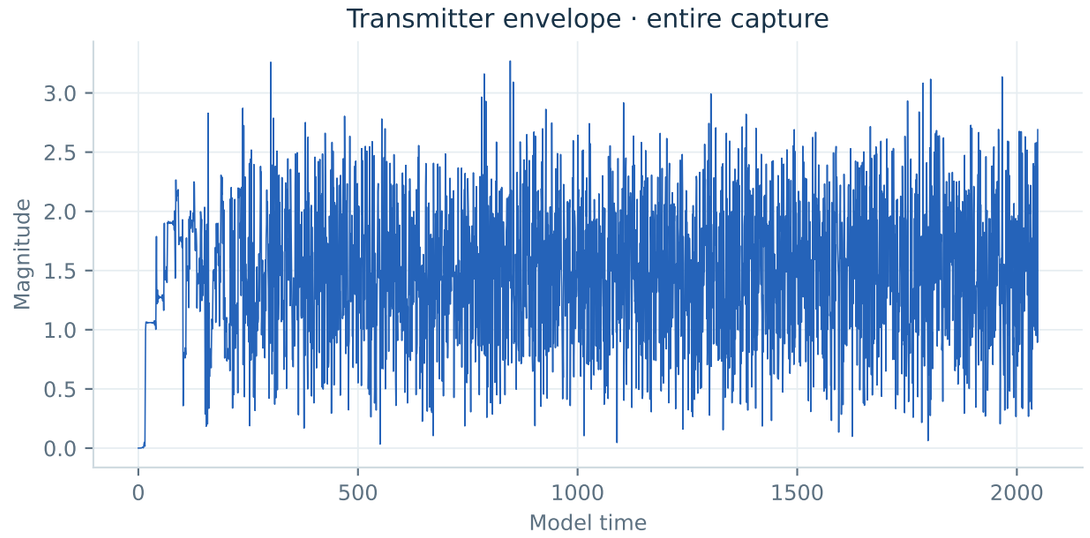


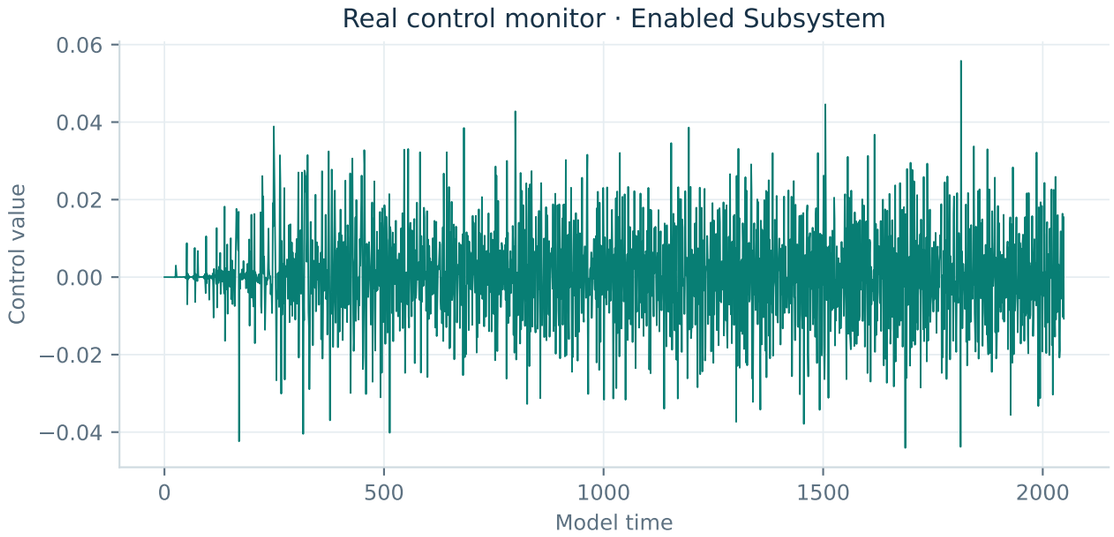
Source: fresh Python negative control using a saved TX payload; numerical signals are in negative_control_data.npz. The first 192 symbols are training and are excluded from BER/SER. Channel points are oversampled, not symbol decisions. Spectra share the TX peak reference. These figures do not establish original MATLAB receiver equivalence.

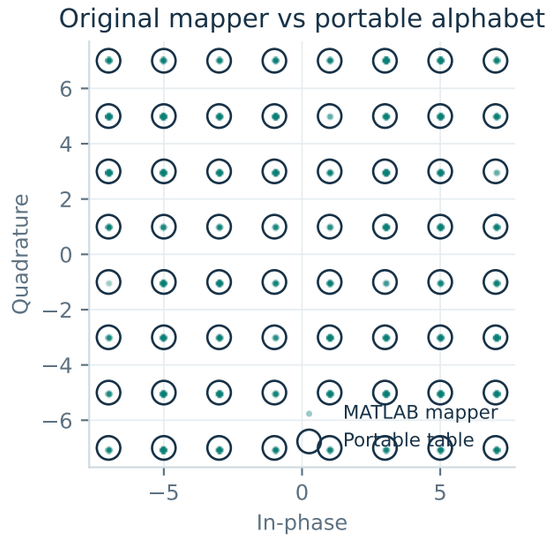
64-QAM · 2008

ORIGINAL MATLAB CAPTURE / runtime PASS / exact repeat PASS / solver: ode45





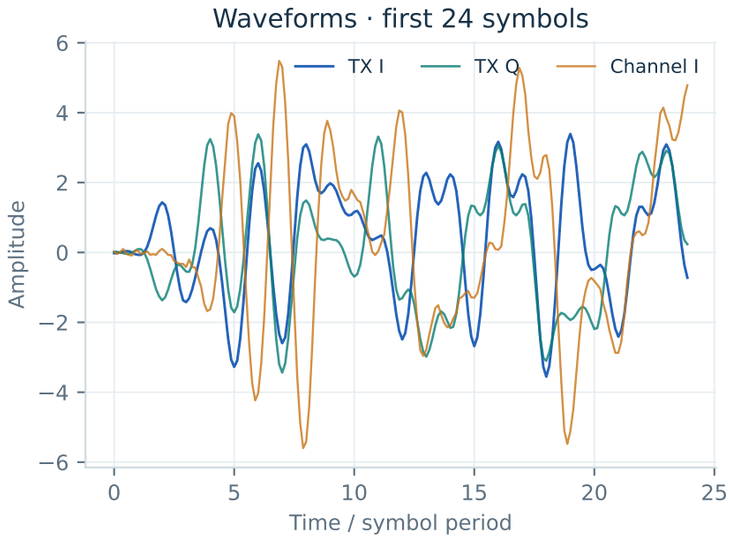


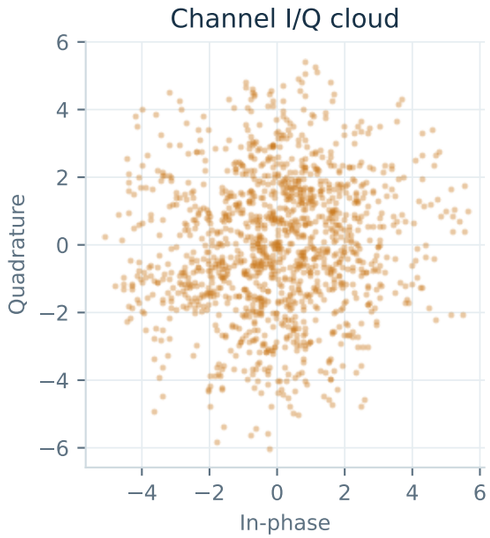


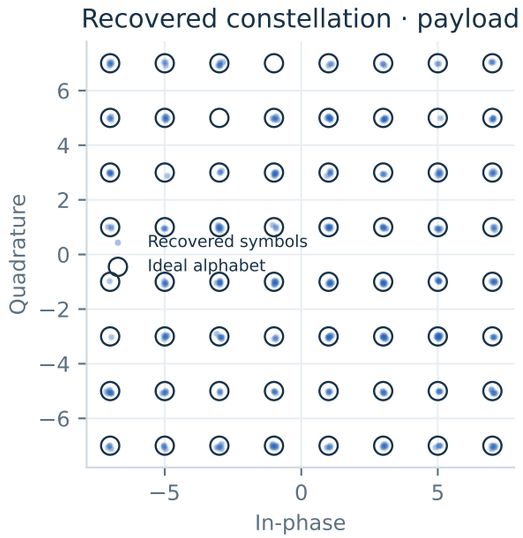
Actual R2020b signals with documented runtime settings. Native receiver monitors can contain held or oversampled values; their point clouds are not a BER measurement. Time is in the model's units. Simulation and repeatability do not establish receiver equivalence. Model: QAM64_2008_11_21

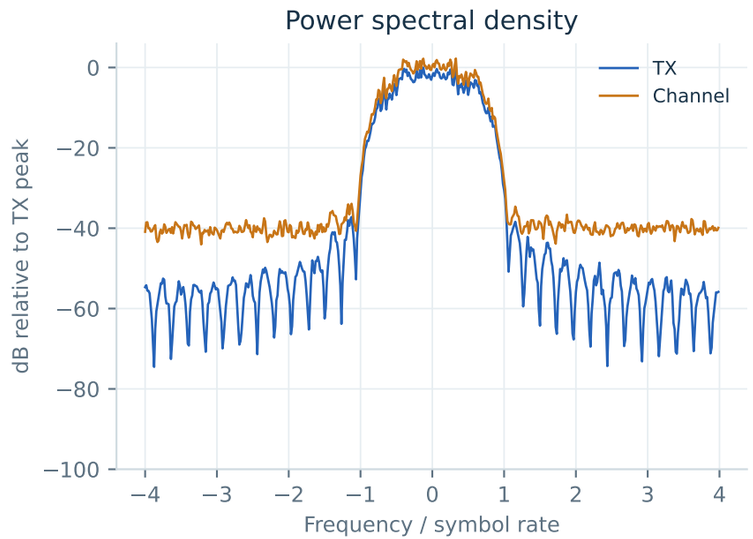
64-QAM · 2008

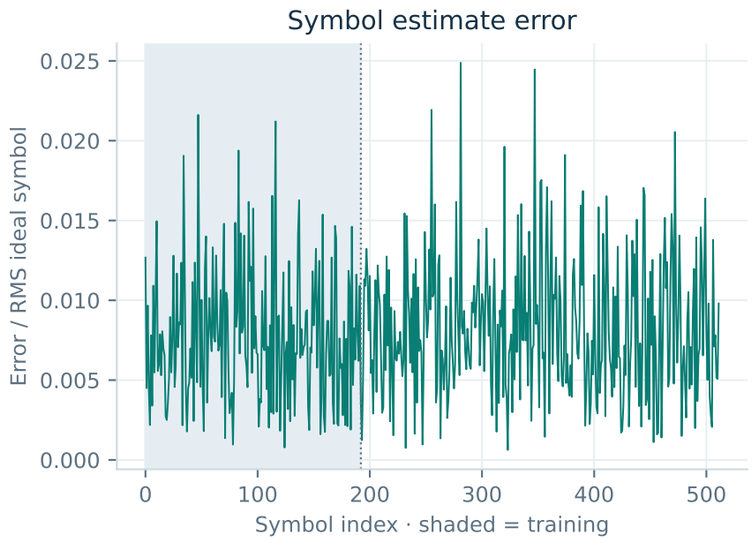
PORTABLE MODEL / Variant stress channel / 0 errors in 320 payload symbols

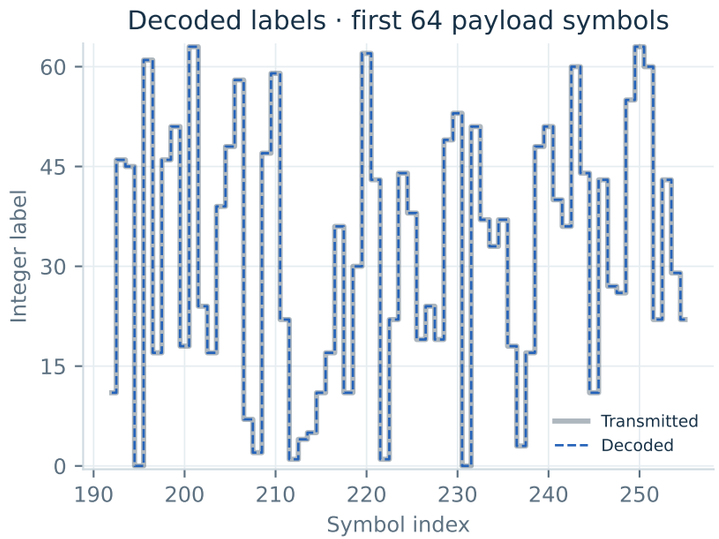








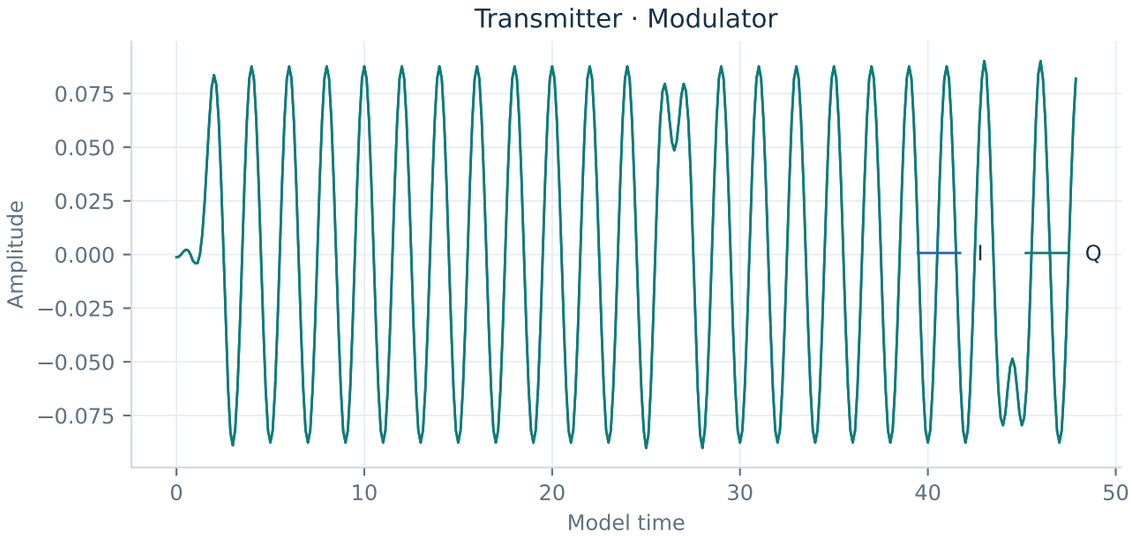


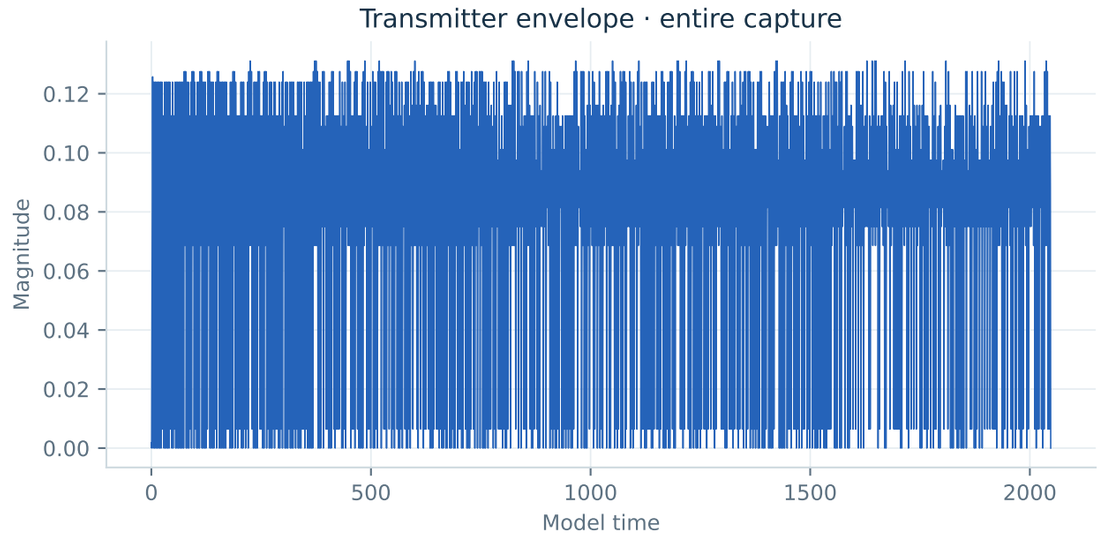


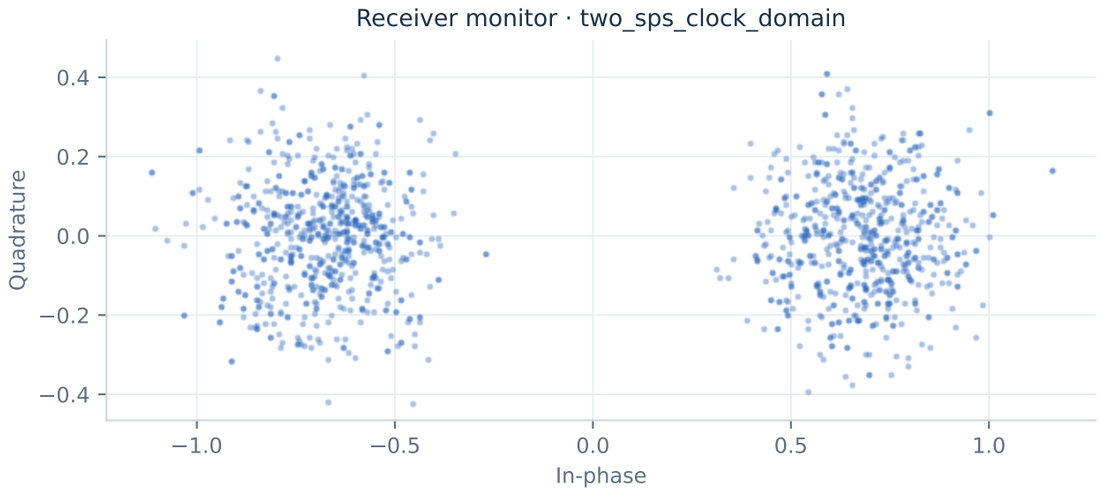
Source: saved portable regression vectors, replayed with the current Python core. The first 192 symbols are training and are excluded from BER/SER. Channel points are oversampled, not symbol decisions. Spectra share the TX peak reference. These figures do not establish original MATLAB receiver equivalence.

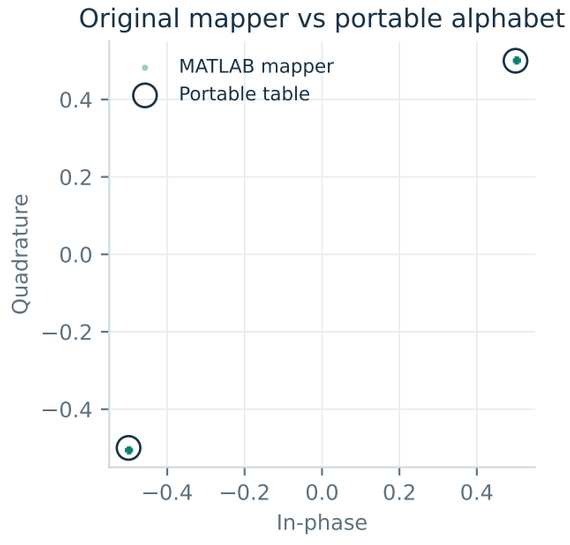
BPSK · Costas

ORIGINAL MATLAB CAPTURE / runtime PASS / exact repeat PASS / solver: FixedStepDiscrete





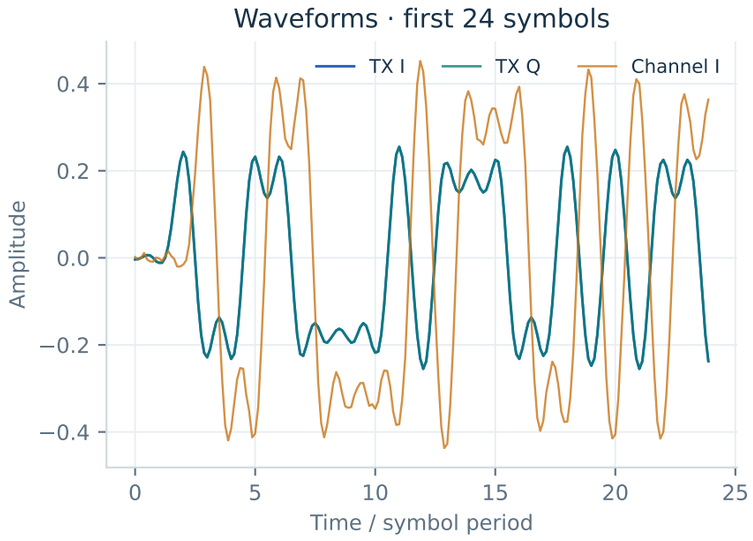


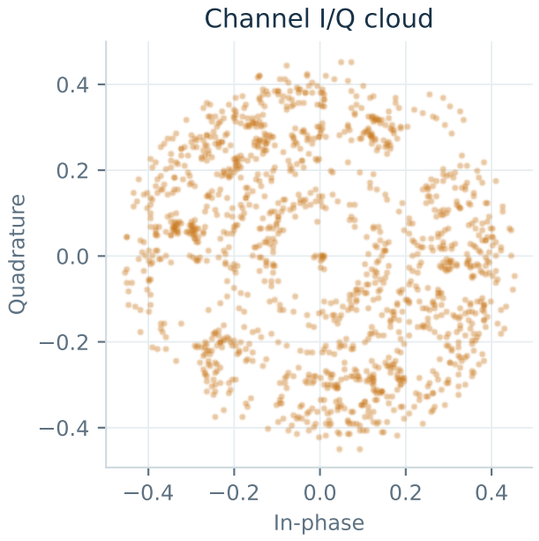


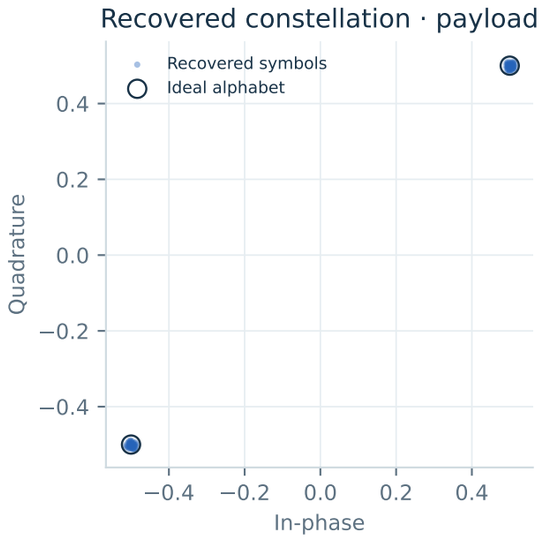
Actual R2020b signals with documented runtime settings. Native receiver monitors can contain held or oversampled values; their point clouds are not a BER measurement. Time is in the model's units. Simulation and repeatability do not establish receiver equivalence. Model: bpsk_costas_2008_08_25

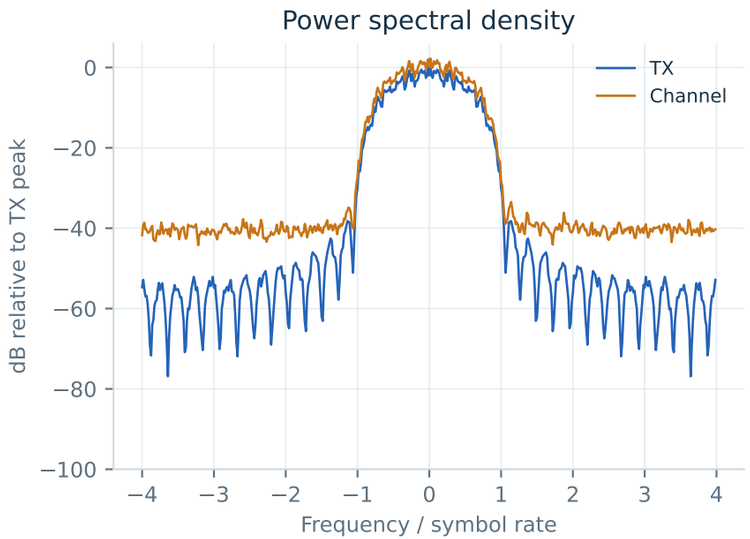
BPSK · Costas

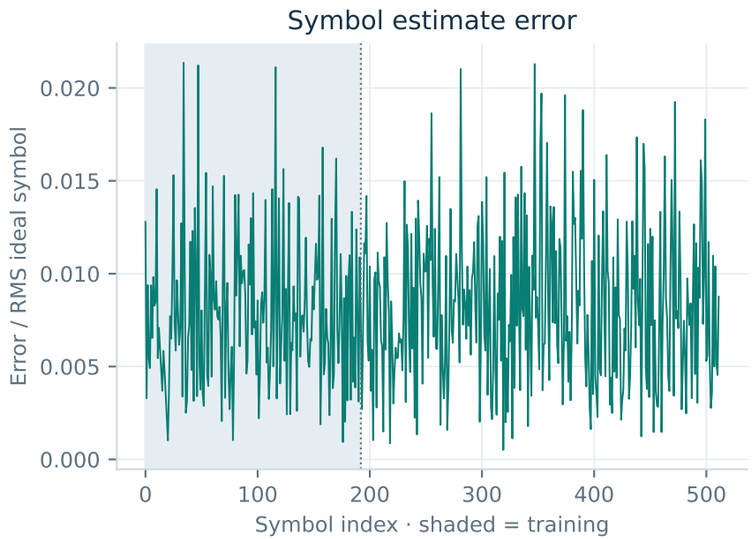
PORTABLE MODEL / Variant stress channel / 0 errors in 320 payload symbols

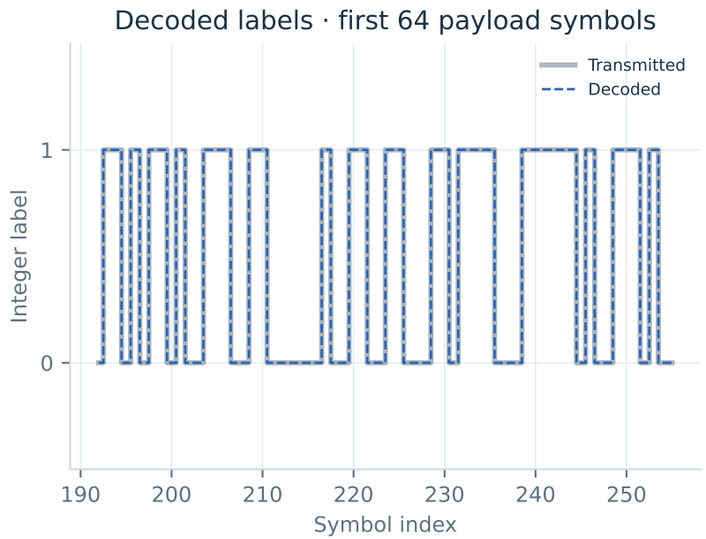








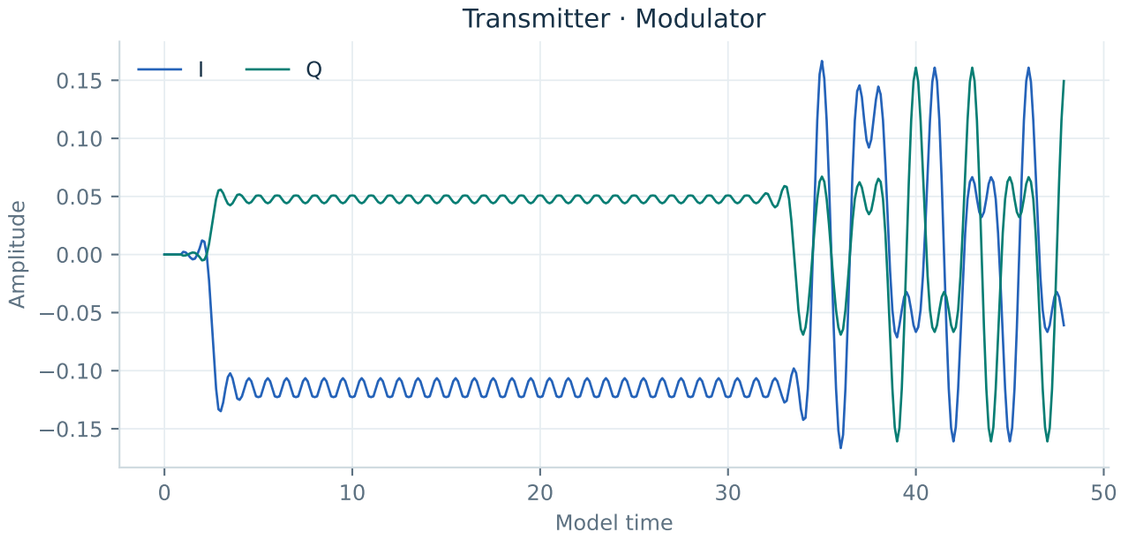


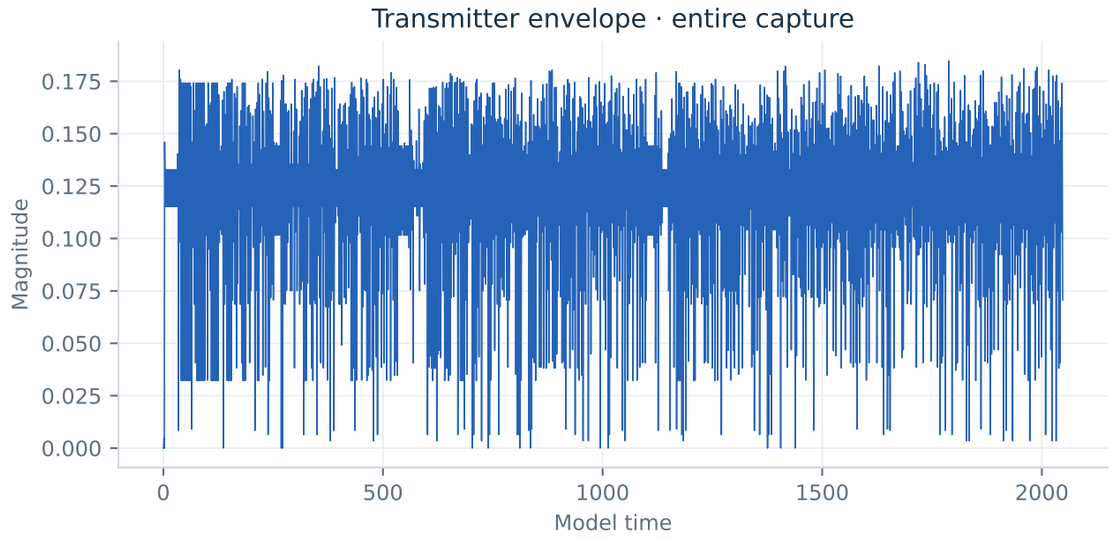


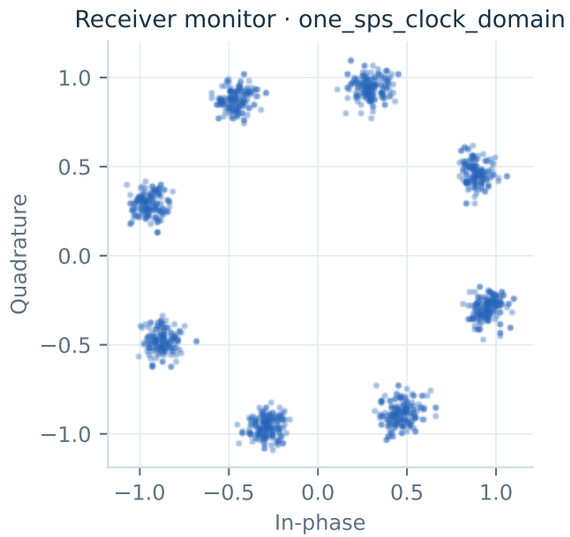
Source: saved portable regression vectors, replayed with the current Python core. The first 192 symbols are training and are excluded from BER/SER. Channel points are oversampled, not symbol decisions. Spectra share the TX peak reference. These figures do not establish original MATLAB receiver equivalence.

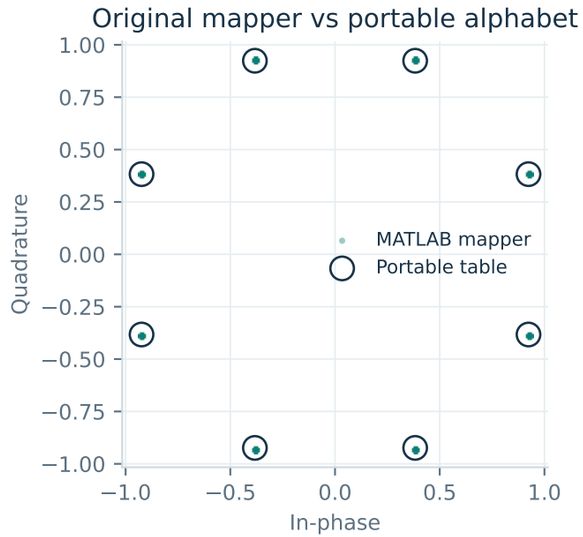
8-PSK · legacy DQPSK archive

ORIGINAL MATLAB CAPTURE / runtime PASS / exact repeat PASS / solver: FixedStepDiscrete





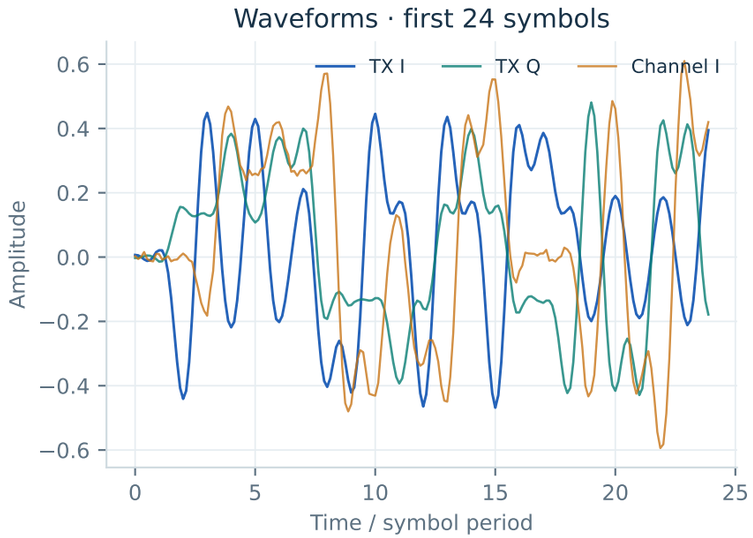


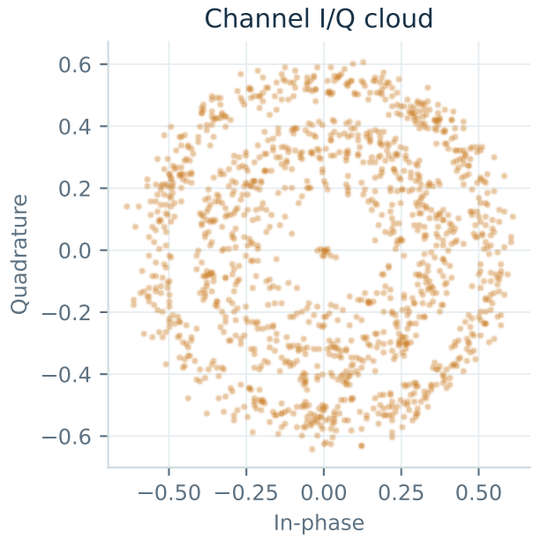


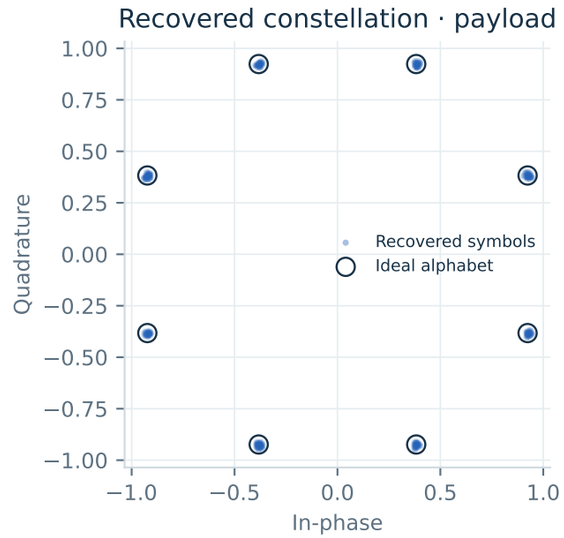
Actual R2020b signals with documented runtime settings. Native receiver monitors can contain held or oversampled values; their point clouds are not a BER measurement. Time is in the model's units. Simulation and repeatability do not establish receiver equivalence. Model: dqpsk_19_09_2006

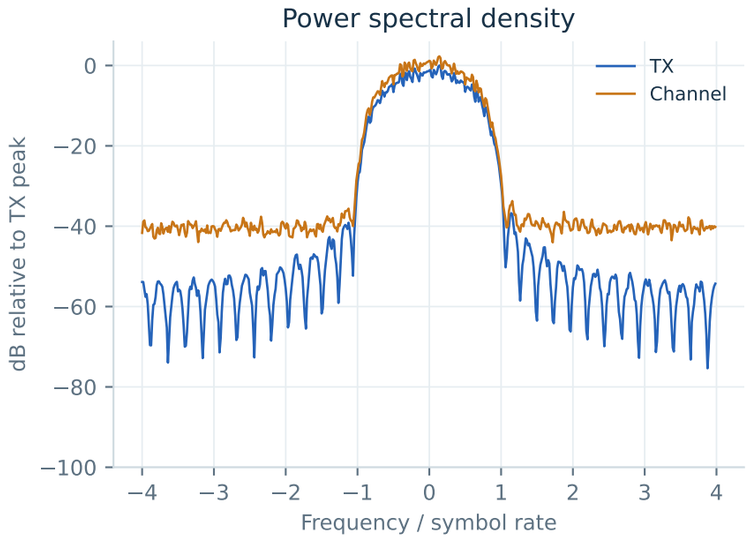
8-PSK · legacy DQPSK archive

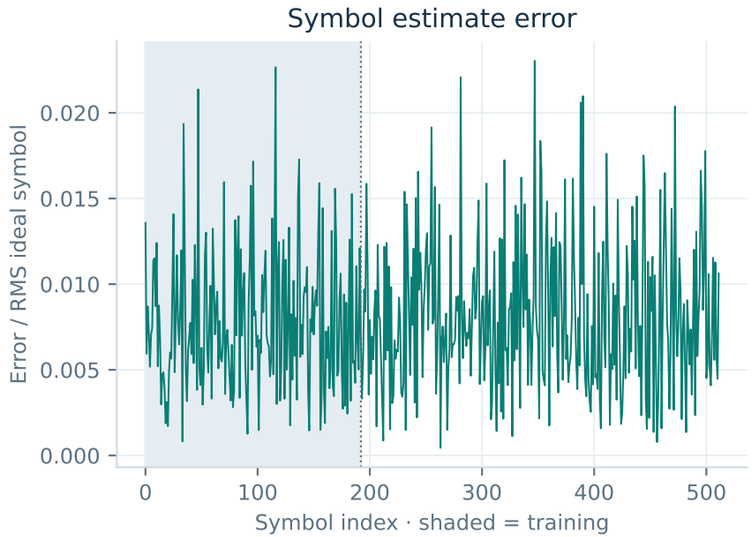
PORTABLE MODEL / Variant stress channel / 0 errors in 320 payload symbols

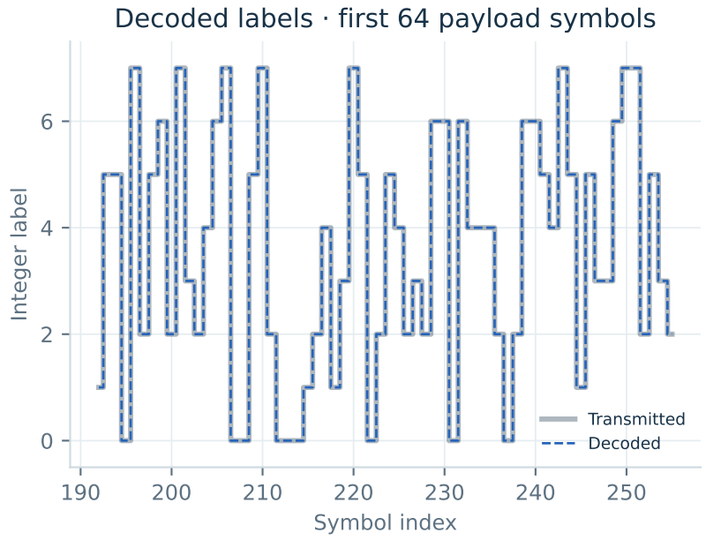








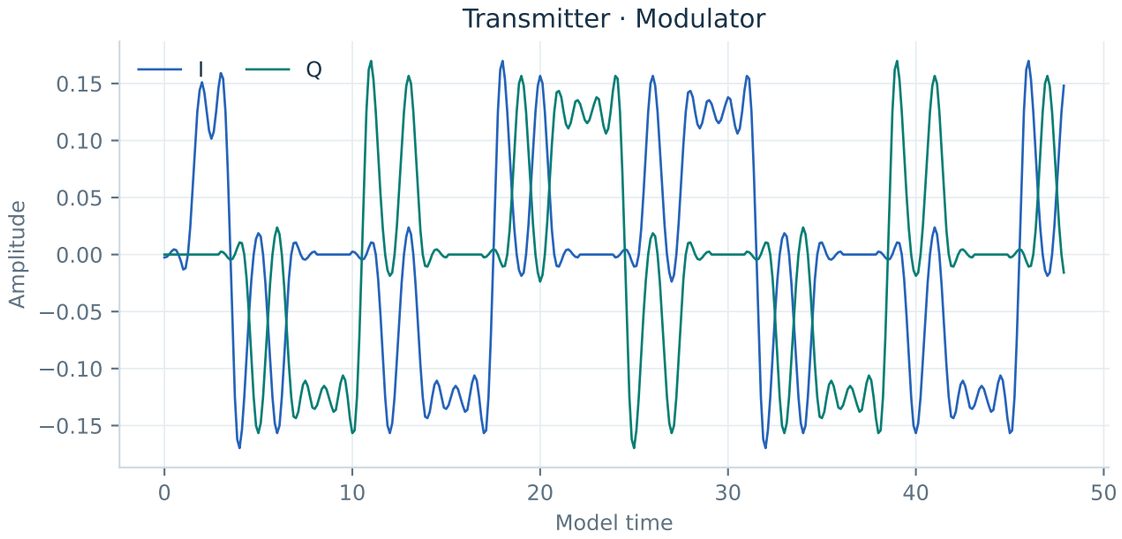


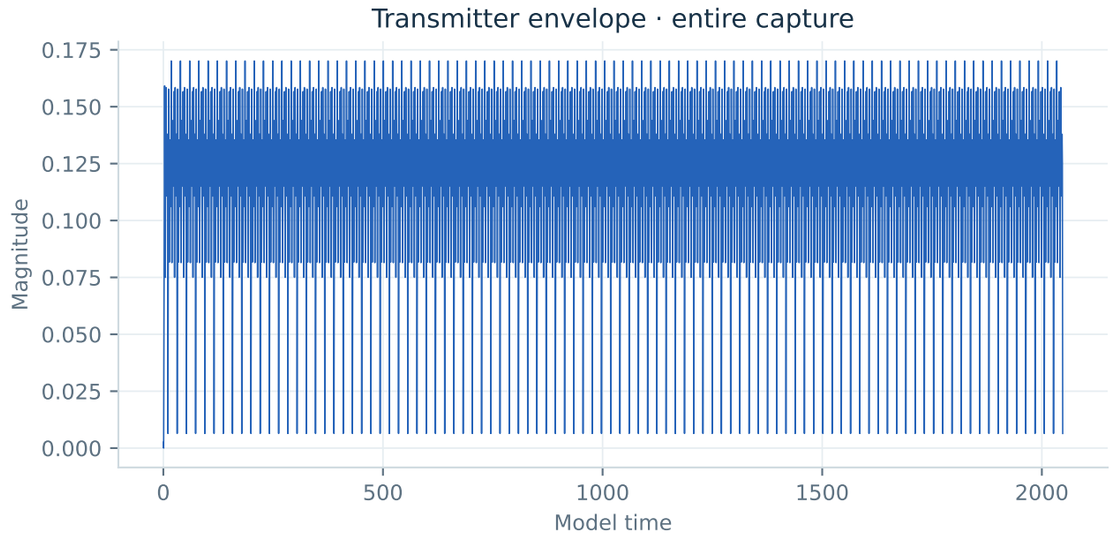


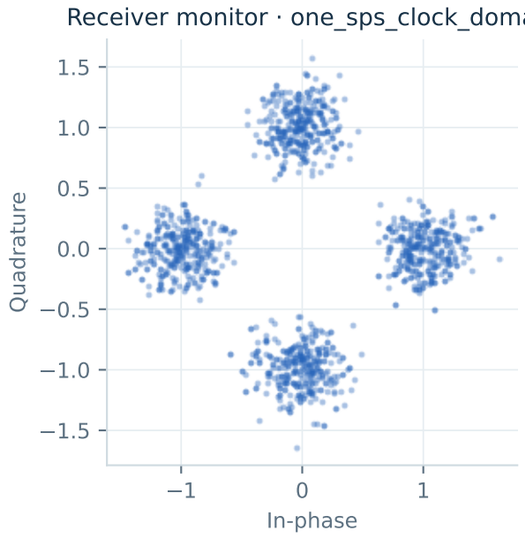
Source: saved portable regression vectors, replayed with the current Python core. The first 192 symbols are training and are excluded from BER/SER. Channel points are oversampled, not symbol decisions. Spectra share the TX peak reference. These figures do not establish original MATLAB receiver equivalence.

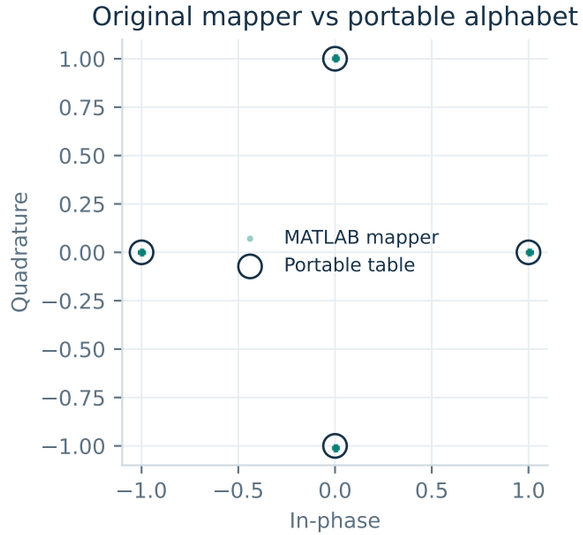
DQPSK · Farrow / fast AGC

ORIGINAL MATLAB CAPTURE / runtime PASS / exact repeat PASS / solver: FixedStepDiscrete





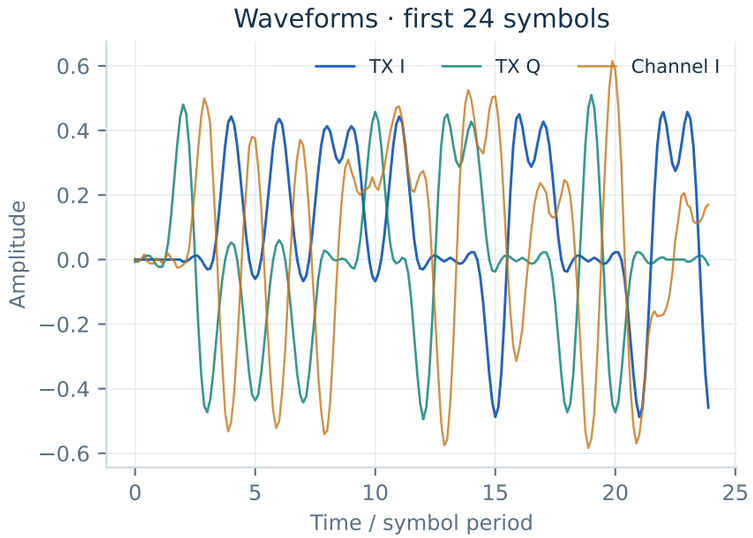


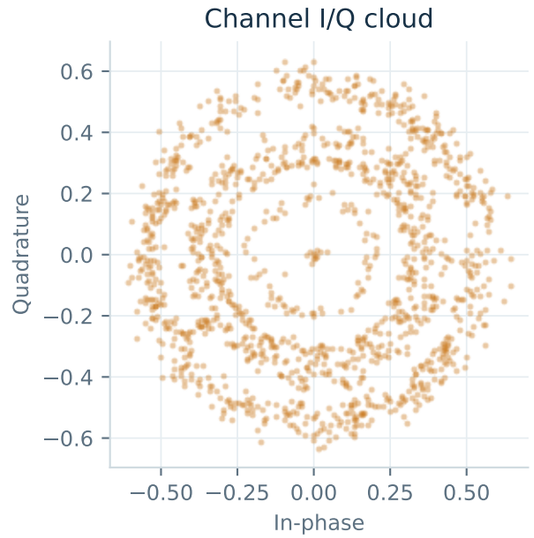


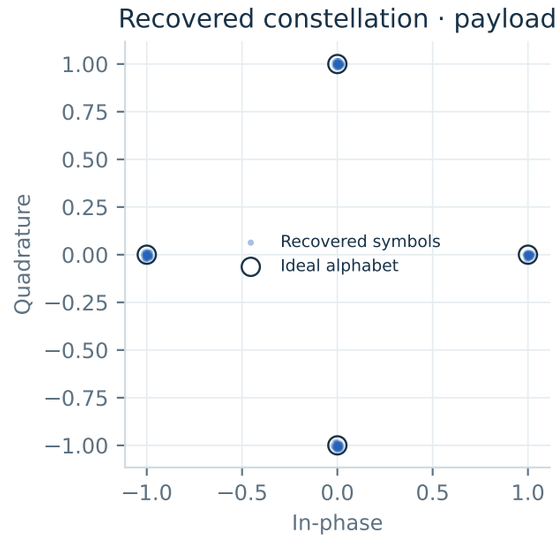
Actual R2020b signals with documented runtime settings. Native receiver monitors can contain held or oversampled values; their point clouds are not a BER measurement. Time is in the model's units. Simulation and repeatability do not establish receiver equivalence. Model: dqpsk_ff_symbol_farrow_sync_fast_agc_2008_05_12

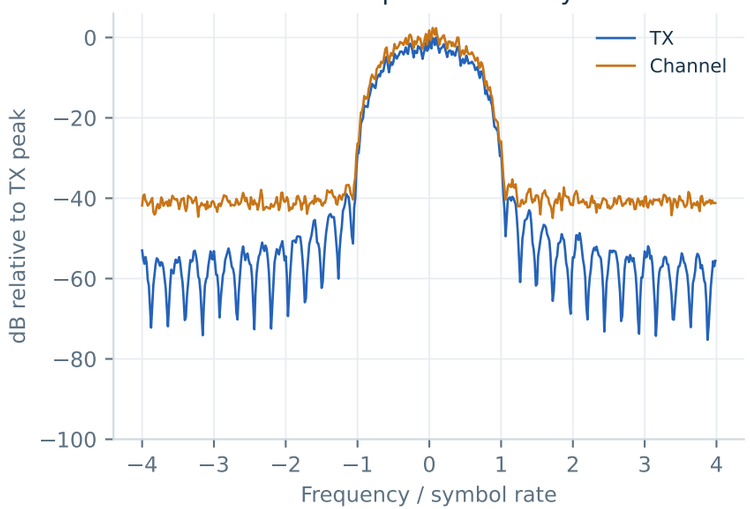
DQPSK · Farrow / fast AGC

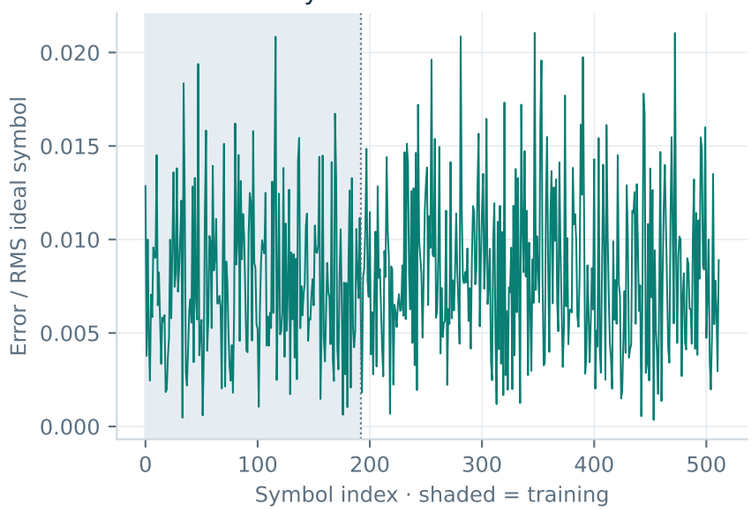
PORTABLE MODEL / Variant stress channel / 0 errors in 320 payload symbols

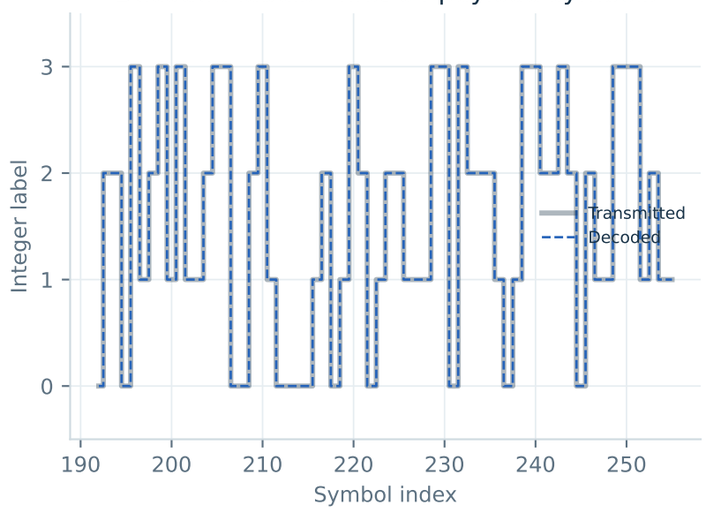








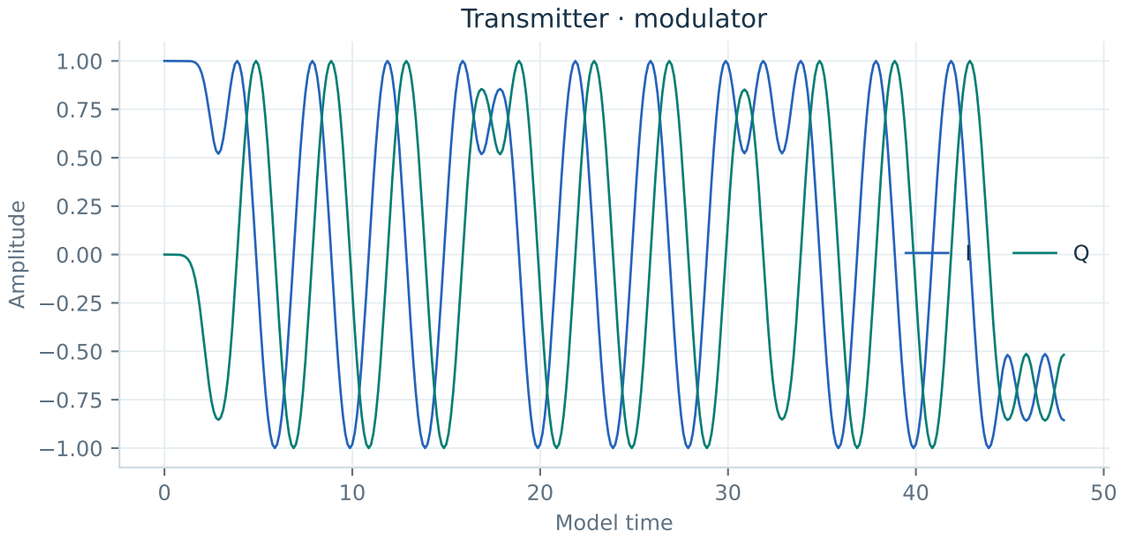


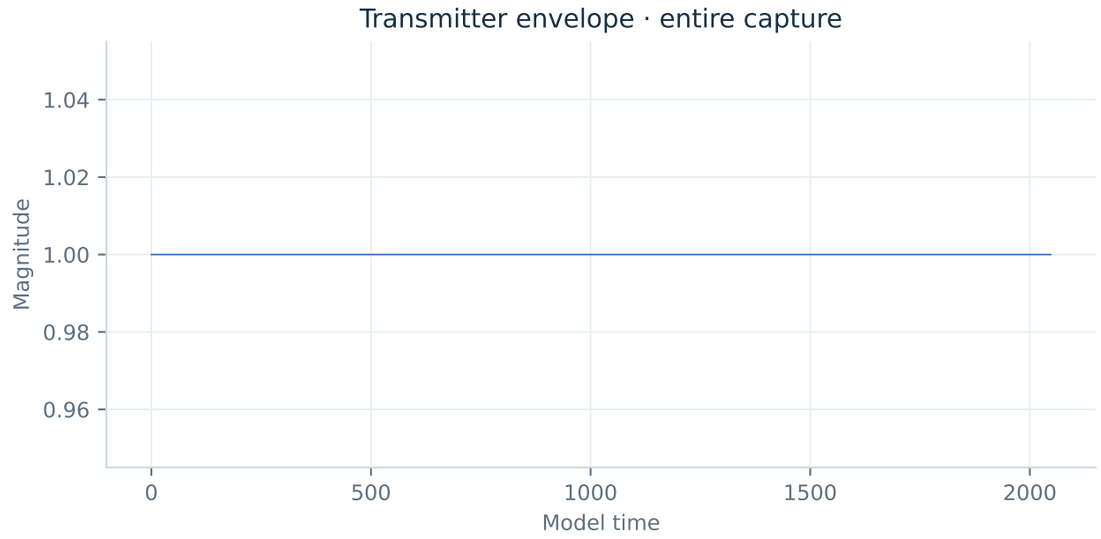


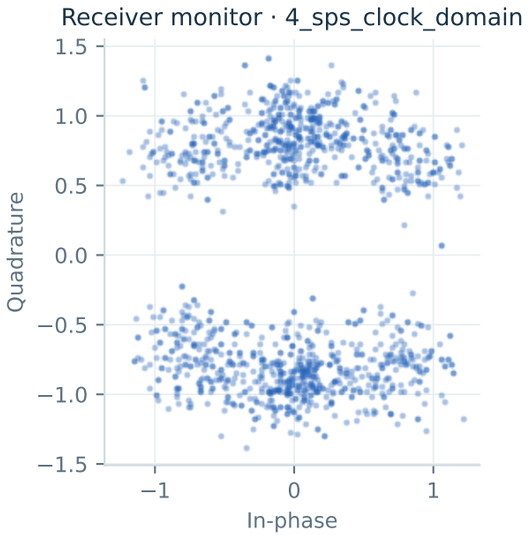
Source: saved portable regression vectors, replayed with the current Python core. The first 192 symbols are training and are excluded from BER/SER. Channel points are oversampled, not symbol decisions. Spectra share the TX peak reference. These figures do not establish original MATLAB receiver equivalence.

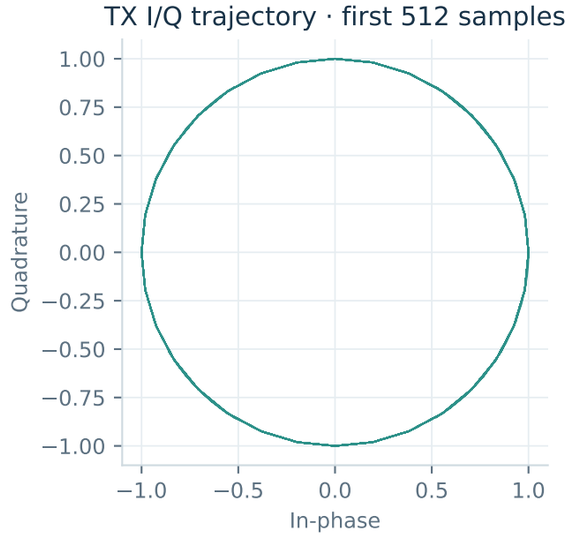
GMSK · coherent original

ORIGINAL MATLAB CAPTURE / runtime PASS / exact repeat PASS / solver: FixedStepDiscrete





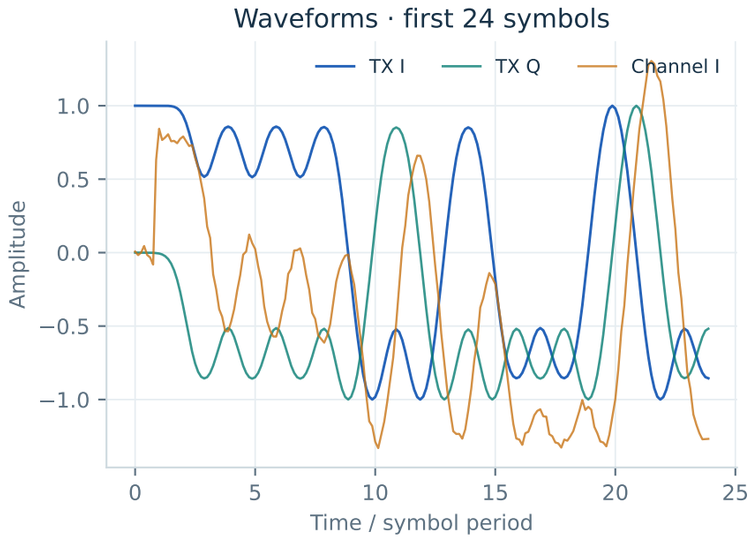


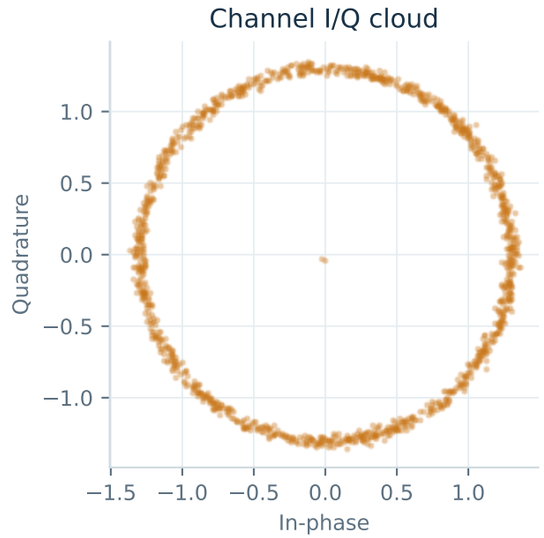


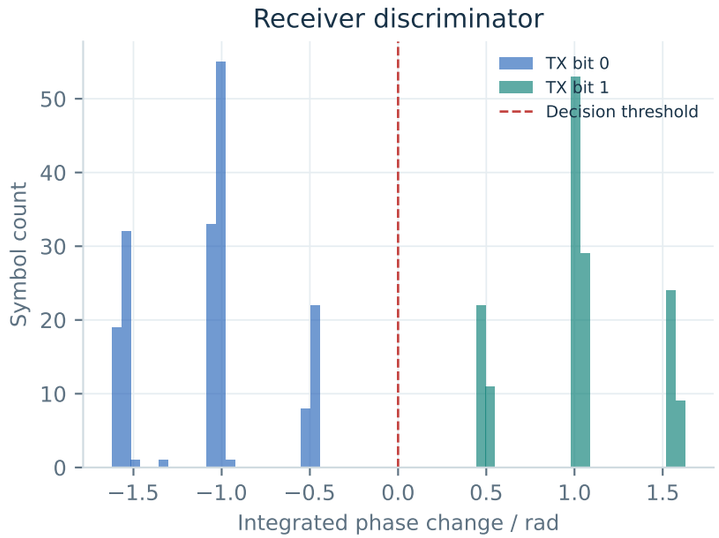
Actual R2020b signals with documented runtime settings. Native receiver monitors can contain held or oversampled values; their point clouds are not a BER measurement. Time is in the model's units. Simulation and repeatability do not establish receiver equivalence. Model: gmsk_coherent_demodulation_2010_10_03

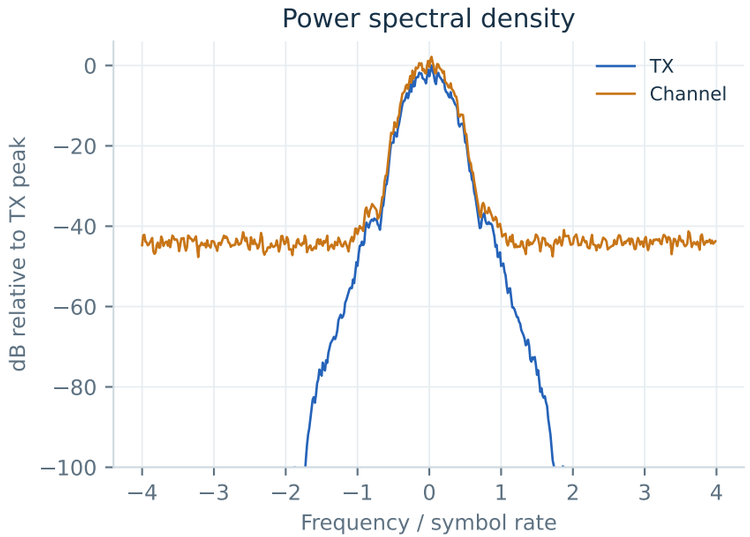
GMSK · coherent original

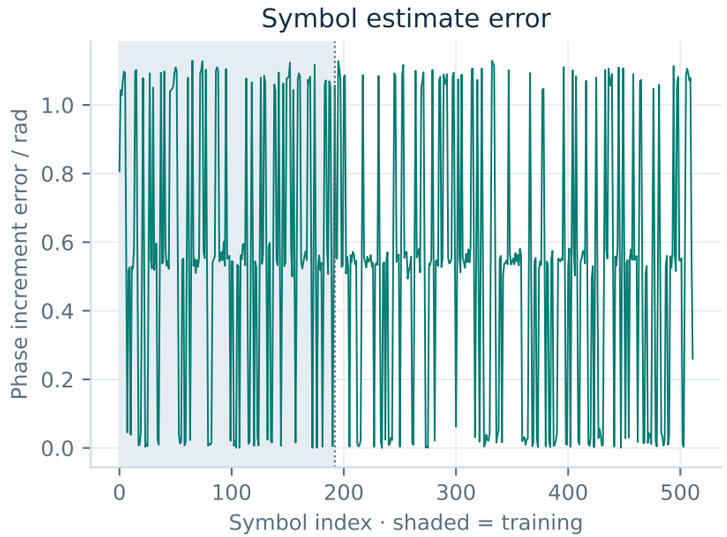
PORTABLE MODEL / Variant stress channel / 0 errors in 320 payload symbols

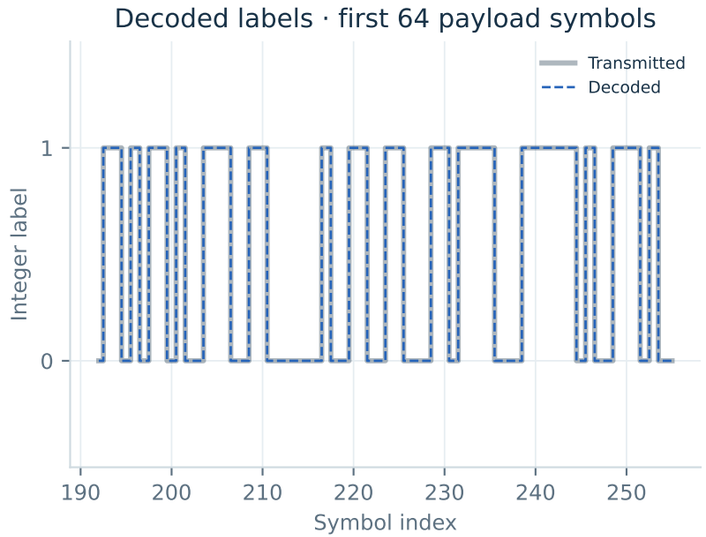








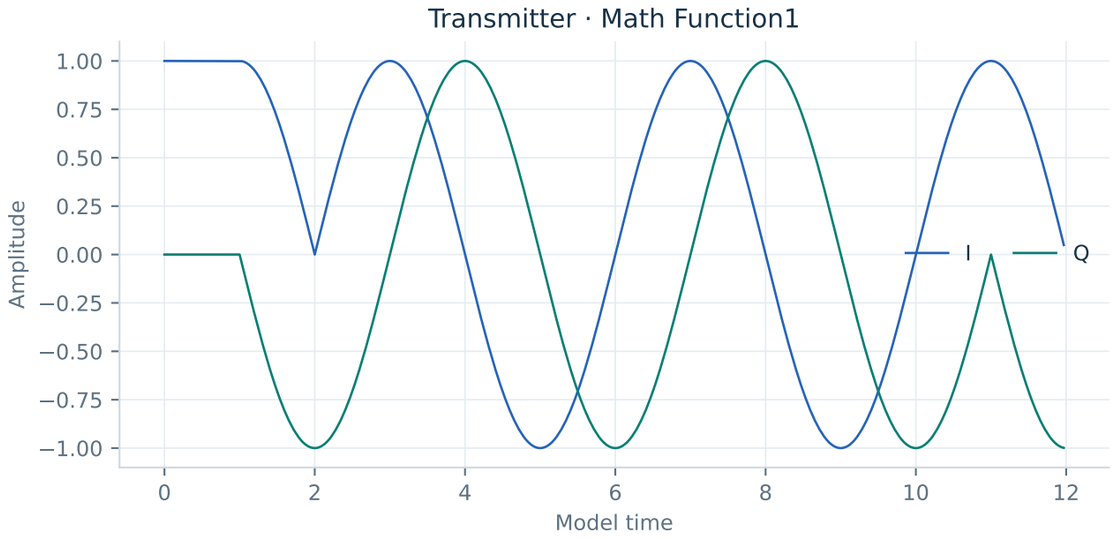


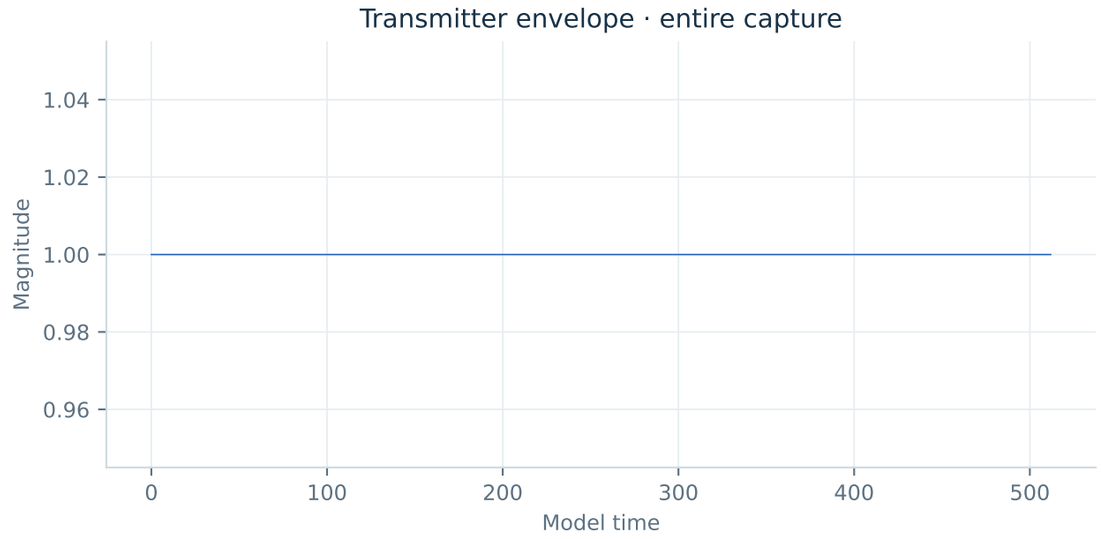


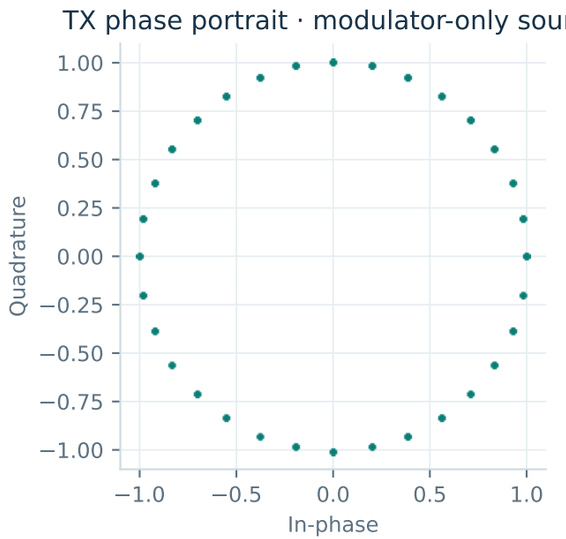
Source: saved portable regression vectors, replayed with the current Python core. The first 192 symbols are training and are excluded from BER/SER. Channel points are oversampled, not symbol decisions. Spectra share the TX peak reference. These figures do not establish original MATLAB receiver equivalence.

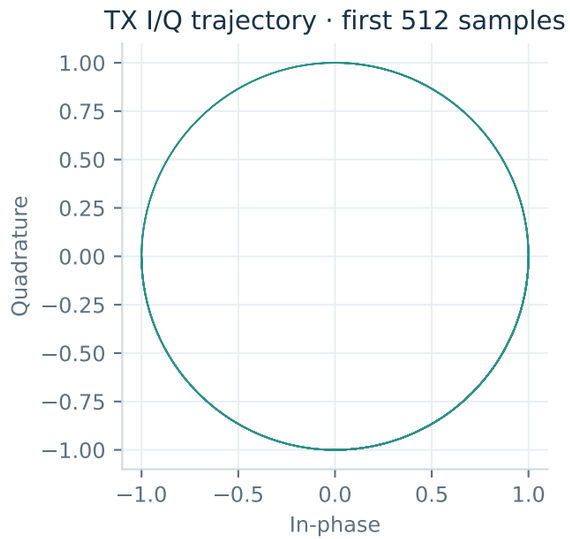
MSK · modulator

ORIGINAL MATLAB CAPTURE / runtime PASS / exact repeat PASS / solver: FixedStepDiscrete





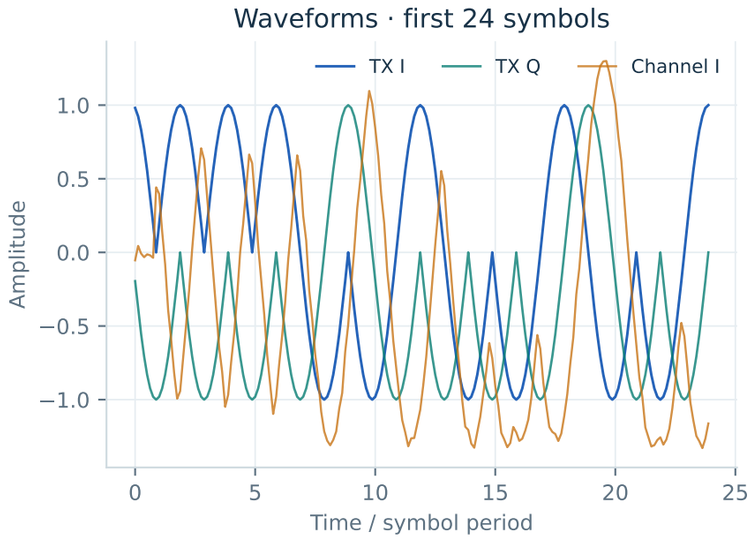


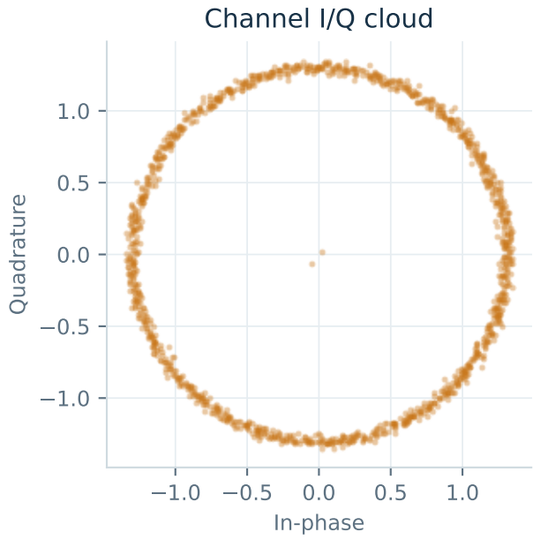


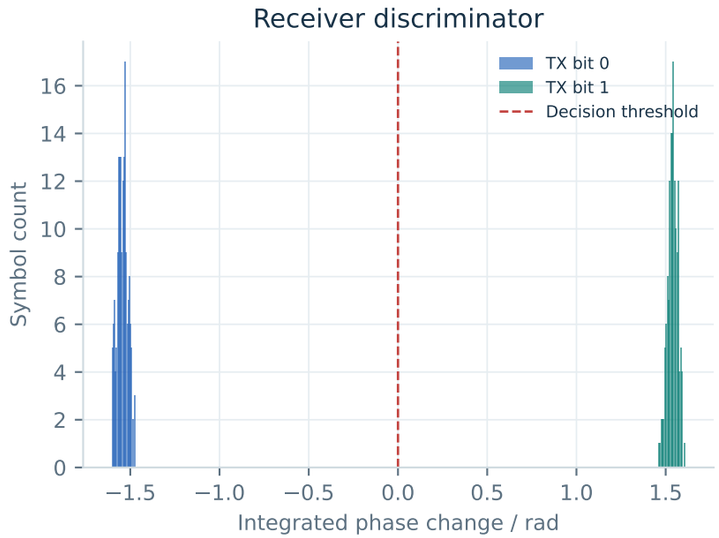
Actual R2020b signals with documented runtime settings. Native receiver monitors can contain held or oversampled values; their point clouds are not a BER measurement. Time is in the model's units. Simulation and repeatability do not establish receiver equivalence. Model: msk_modulator_2008_06_22

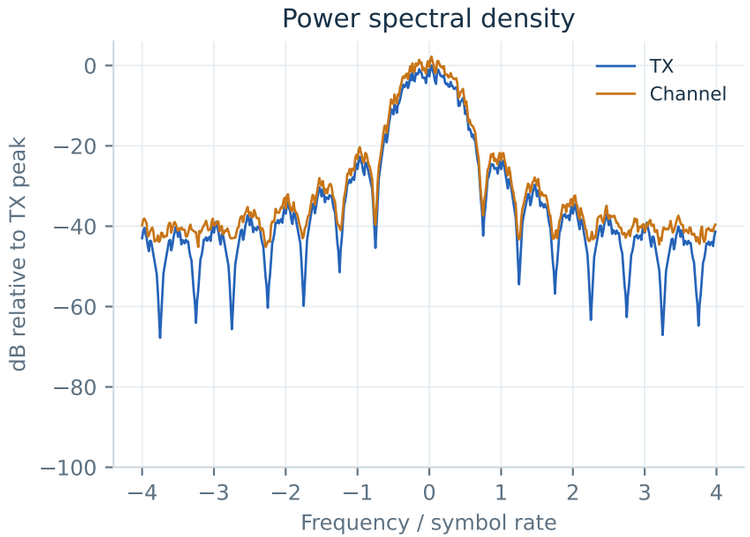
MSK · modulator

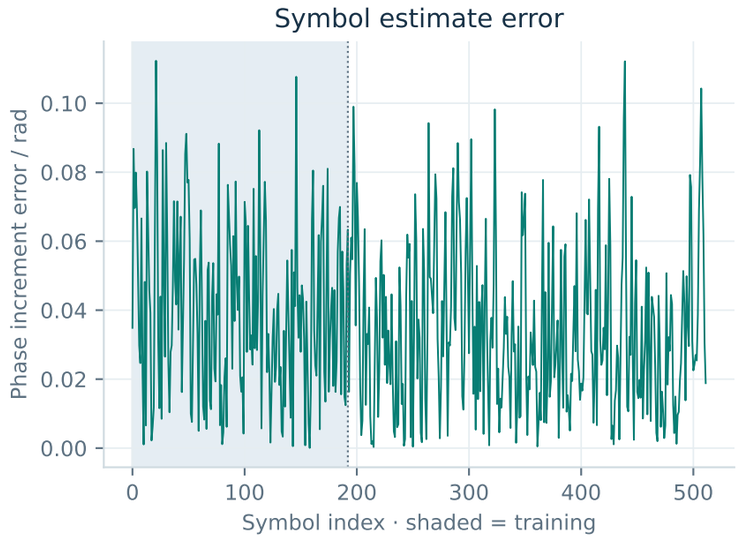
PORTABLE MODEL / Variant stress channel / 0 errors in 320 payload symbols

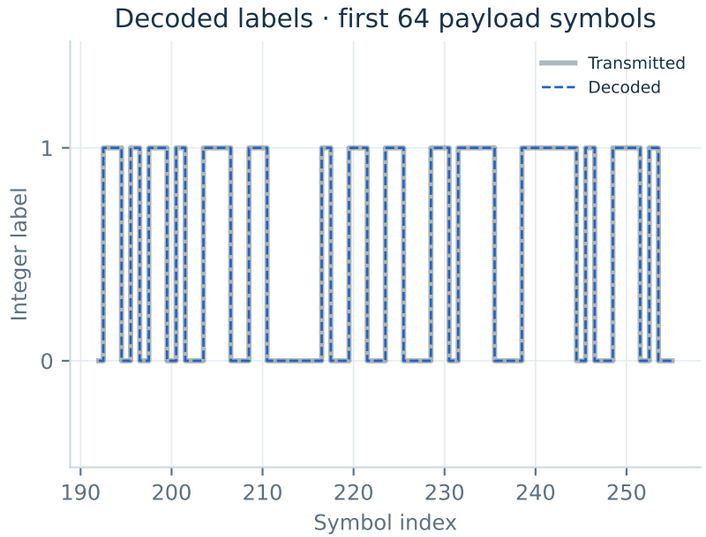








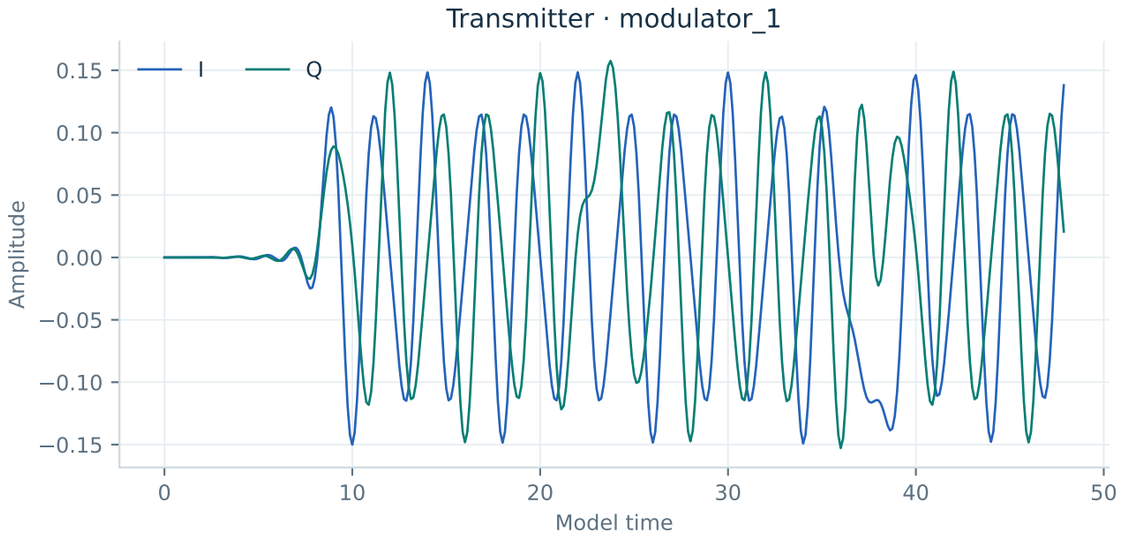


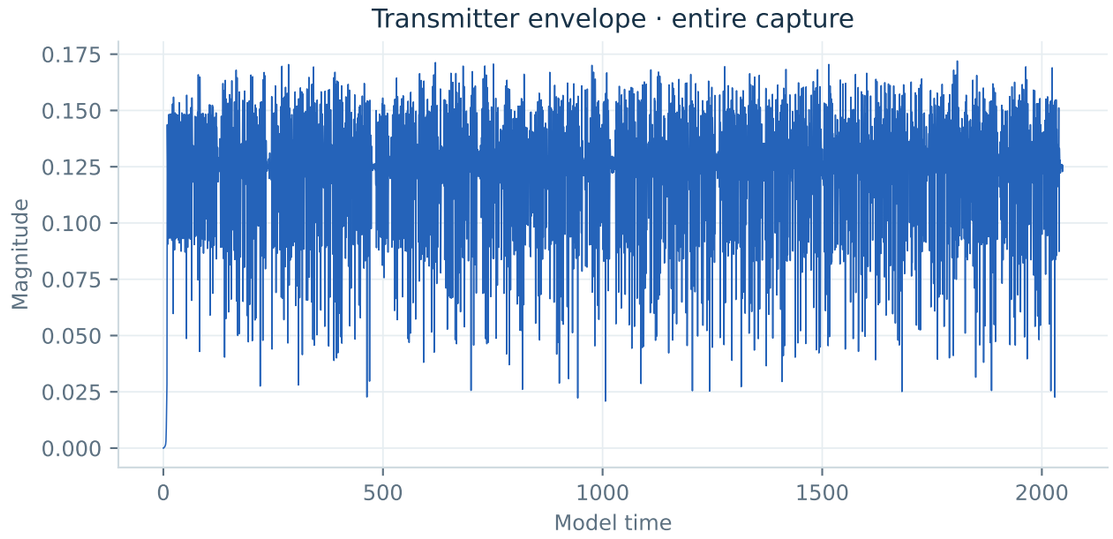


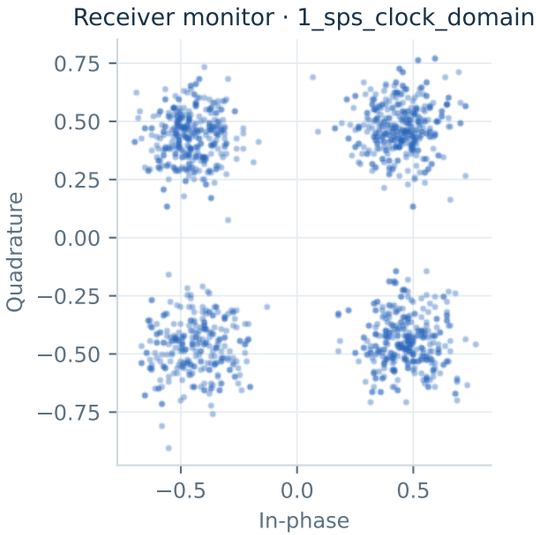
Source: saved portable regression vectors, replayed with the current Python core. The first 192 symbols are training and are excluded from BER/SER. Channel points are oversampled, not symbol decisions. Spectra share the TX peak reference. These figures do not establish original MATLAB receiver equivalence.

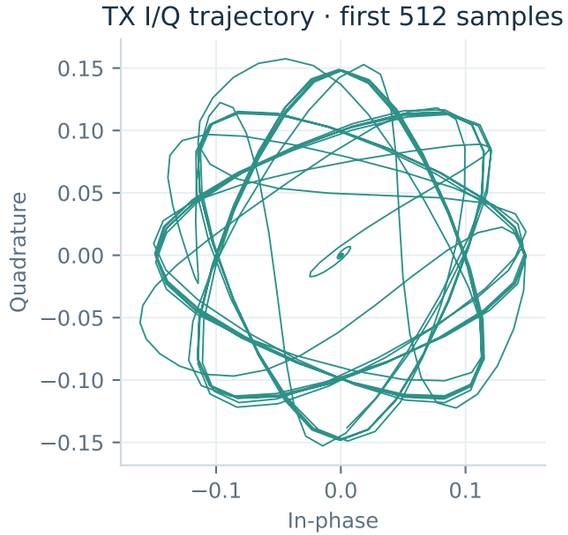
Pi/4 QPSK · 8 SPS / 2021

ORIGINAL MATLAB CAPTURE / runtime PASS / exact repeat PASS / solver: FixedStepDiscrete





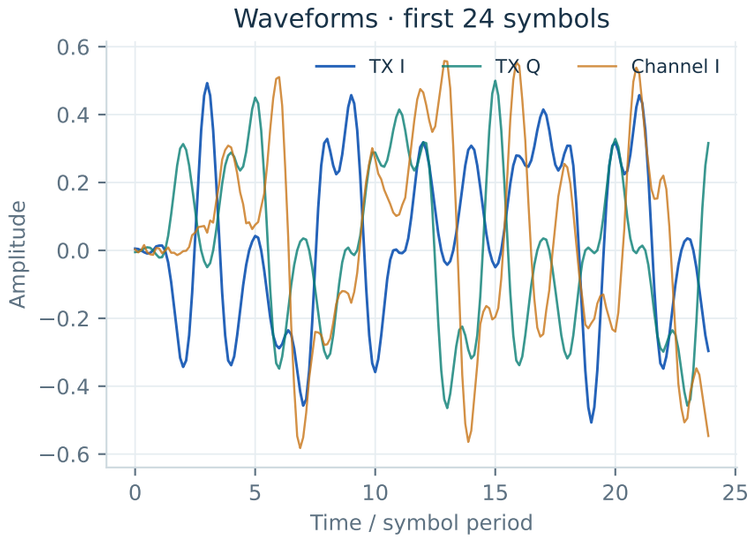


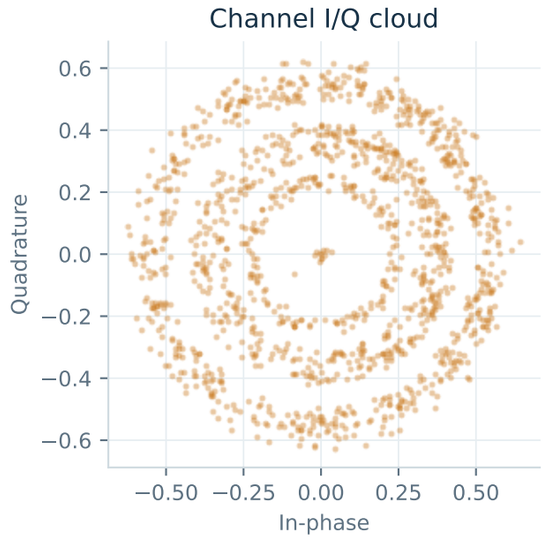


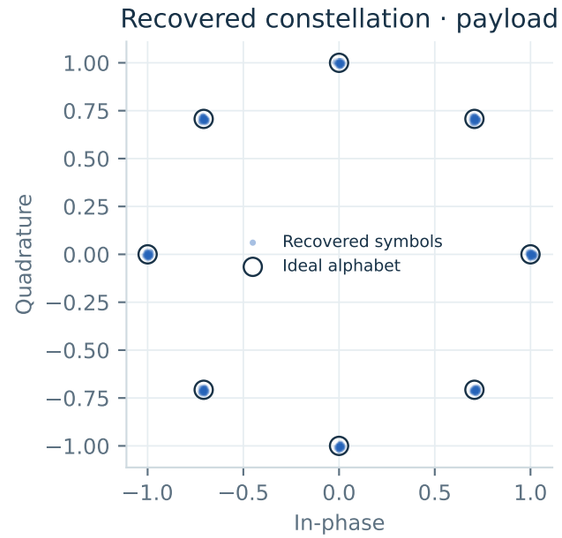
Actual R2020b signals with documented runtime settings. Native receiver monitors can contain held or oversampled values; their point clouds are not a BER measurement. Time is in the model's units. Simulation and repeatability do not establish receiver equivalence. Model: pi4_qpsk_8_sps_ff_sync_2021_05_27

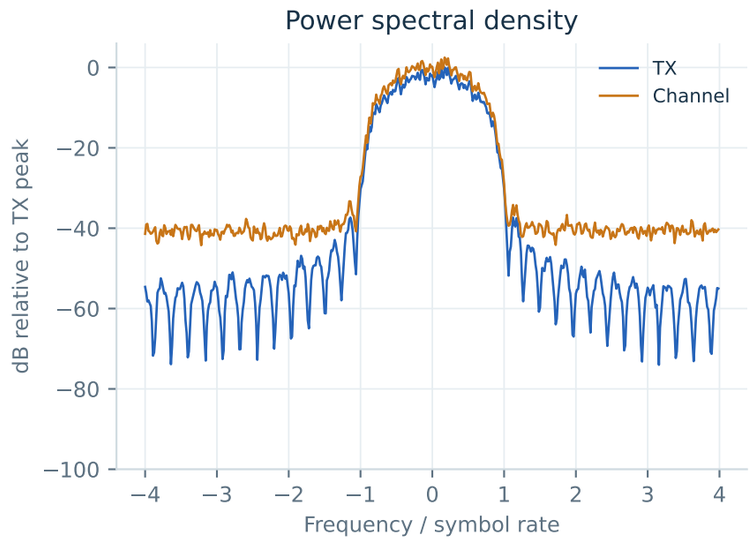
Pi/4 QPSK · 8 SPS / 2021

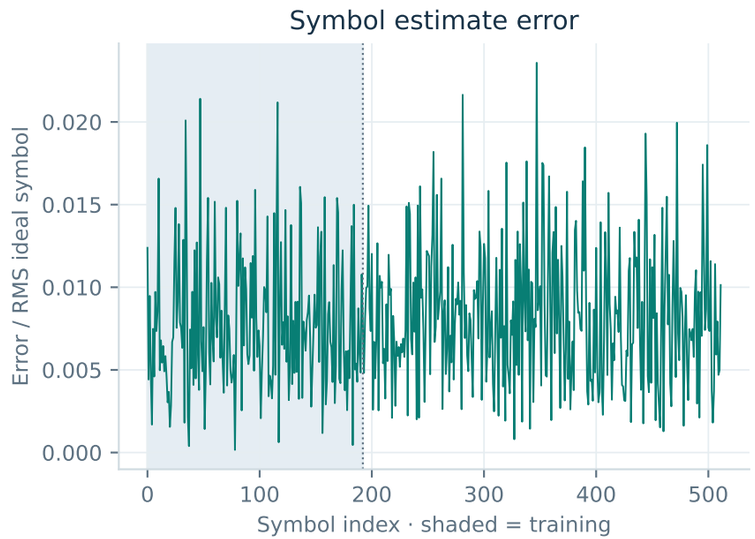
PORTABLE MODEL / Variant stress channel / 0 errors in 320 payload symbols

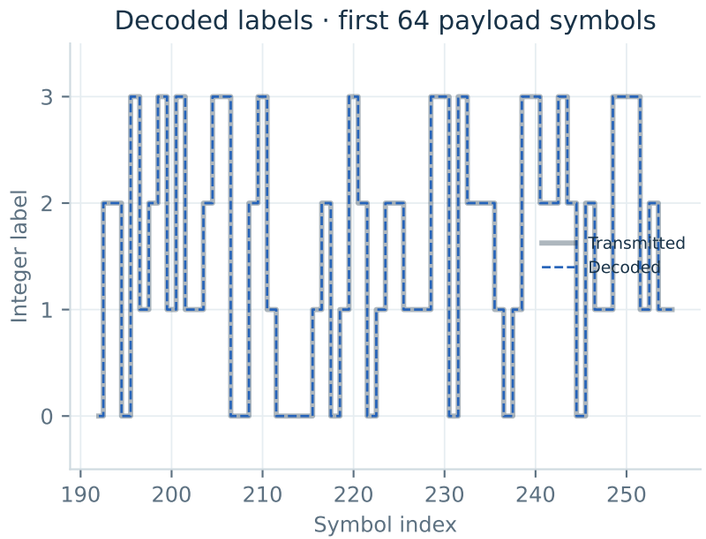








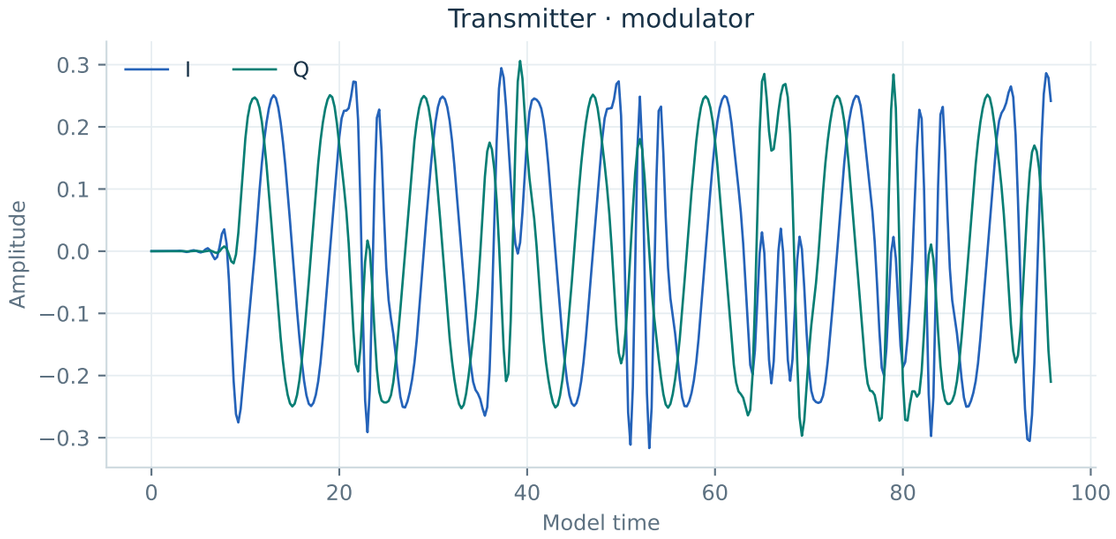


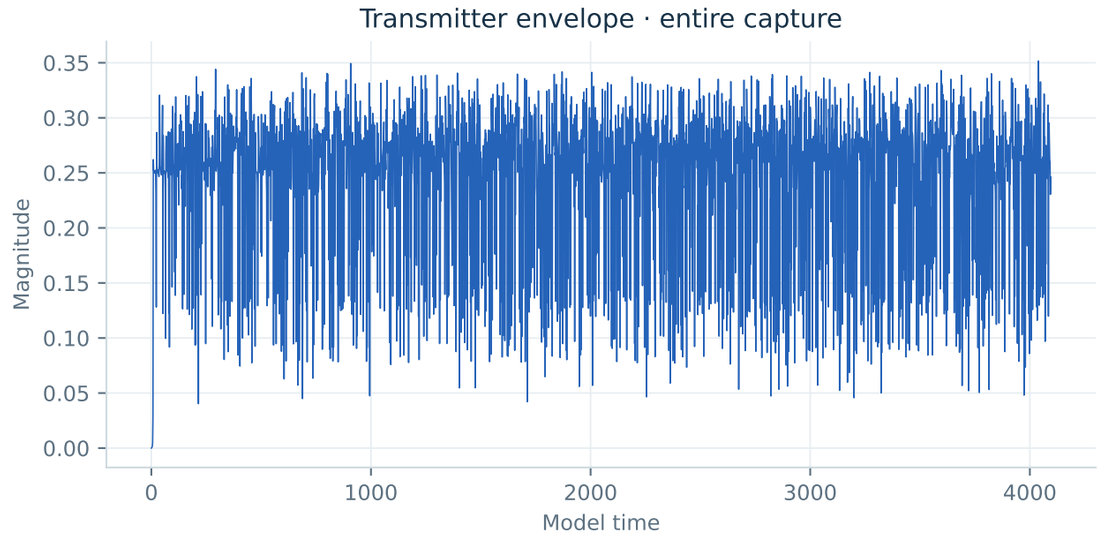


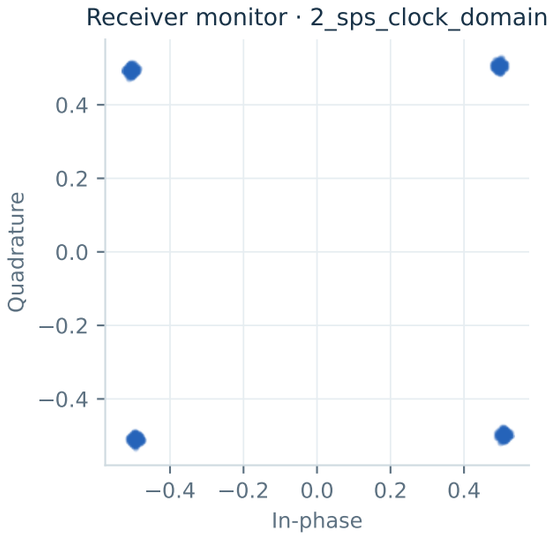
Source: saved portable regression vectors, replayed with the current Python core. The first 192 symbols are training and are excluded from BER/SER. Channel points are oversampled, not symbol decisions. Spectra share the TX peak reference. These figures do not establish original MATLAB receiver equivalence.

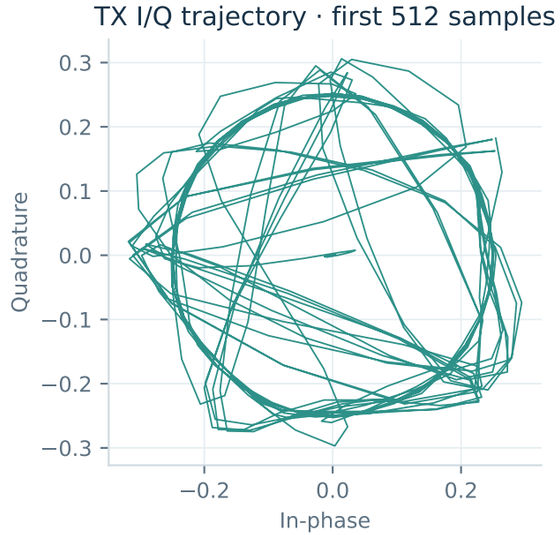
Pi/4 QPSK · fourth-power sync

ORIGINAL MATLAB CAPTURE / runtime PASS / exact repeat PASS / solver: FixedStepDiscrete





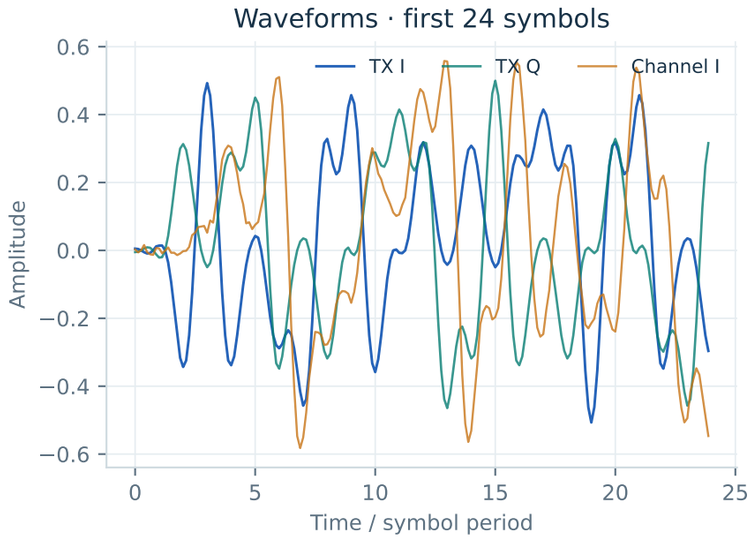


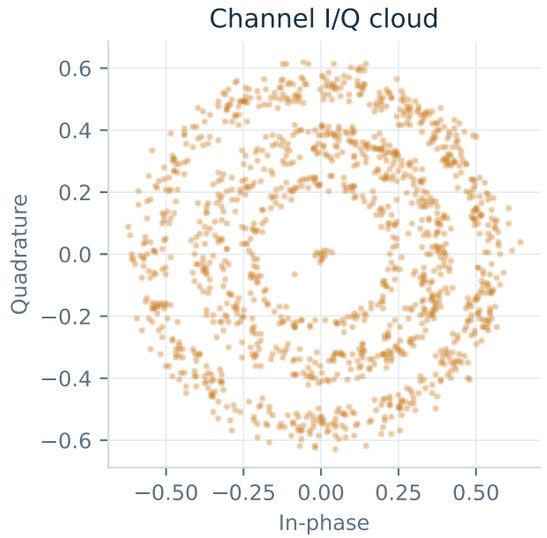


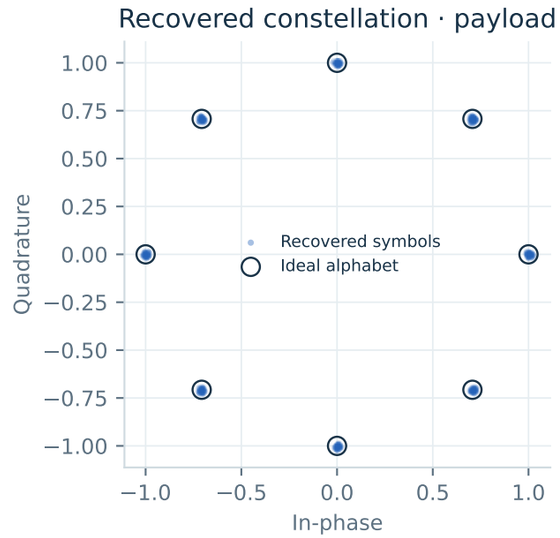
Actual R2020b signals with documented runtime settings. Native receiver monitors can contain held or oversampled values; their point clouds are not a BER measurement. Time is in the model's units. Simulation and repeatability do not establish receiver equivalence. Model: pi4_qpsk_feedforward_4_power_phase_sync_2010_07_15

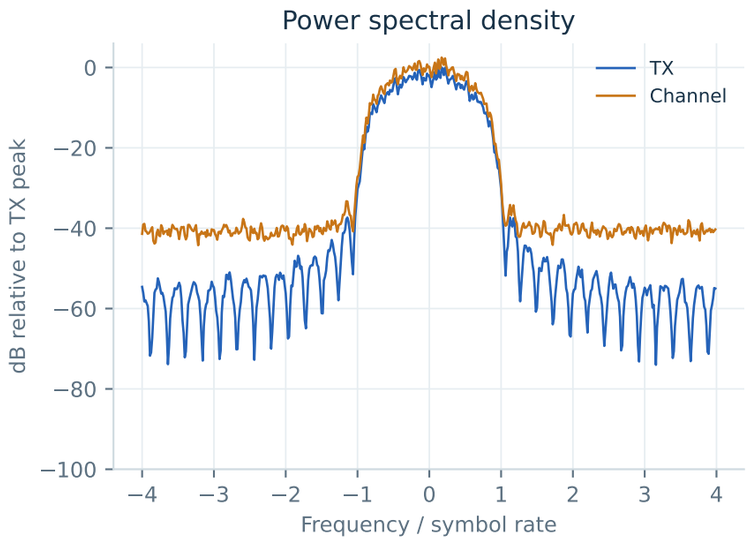
Pi/4 QPSK · fourth-power sync

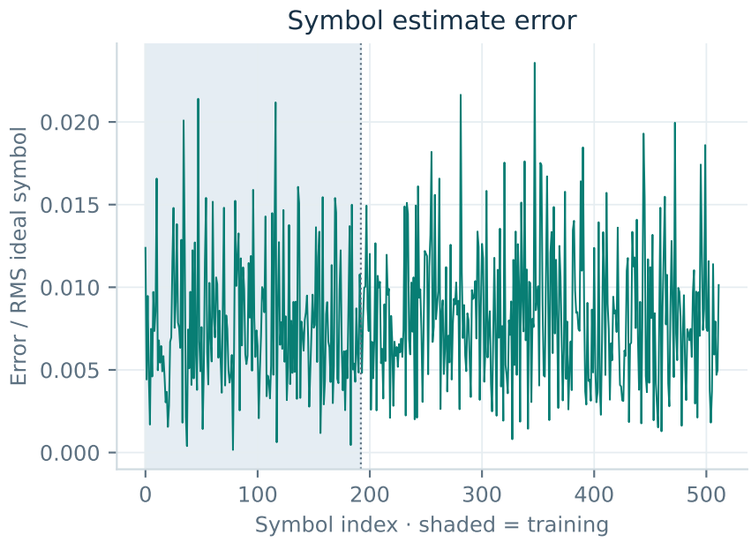
PORTABLE MODEL / Variant stress channel / 0 errors in 320 payload symbols

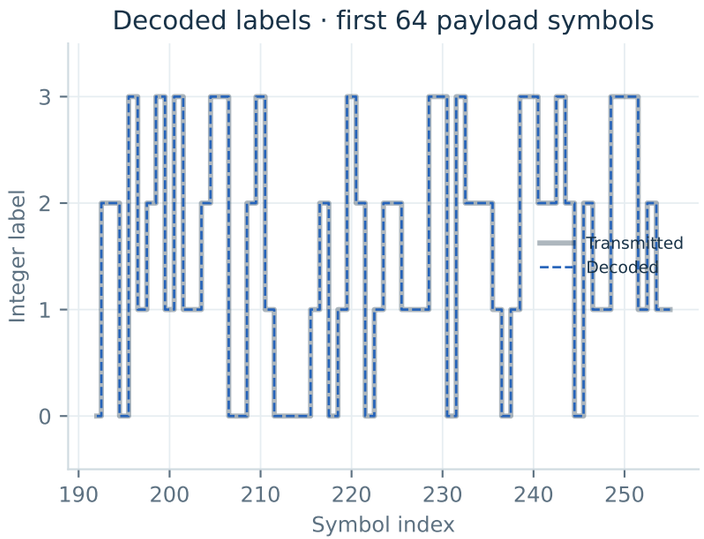








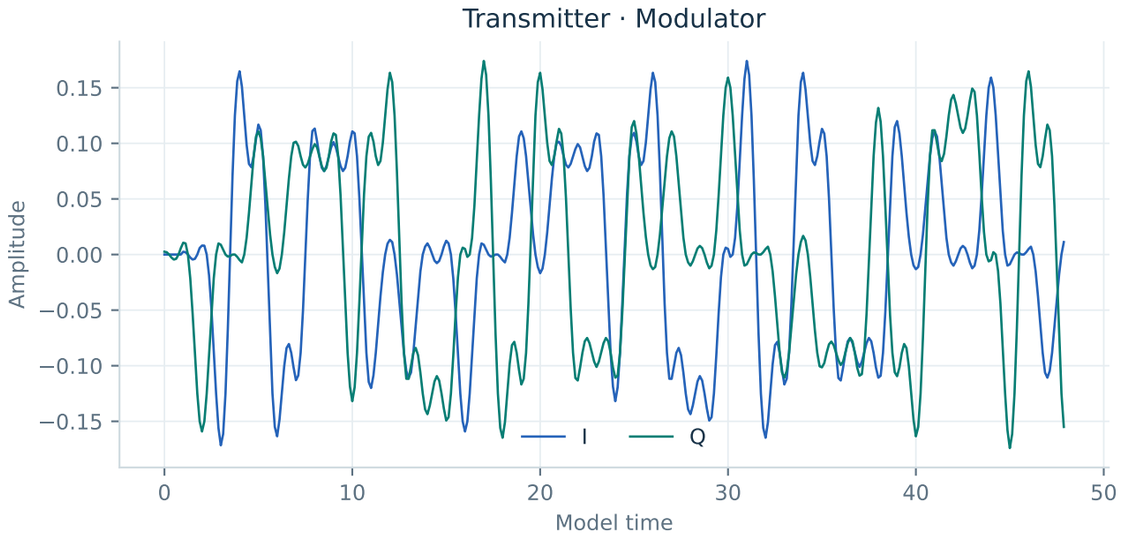


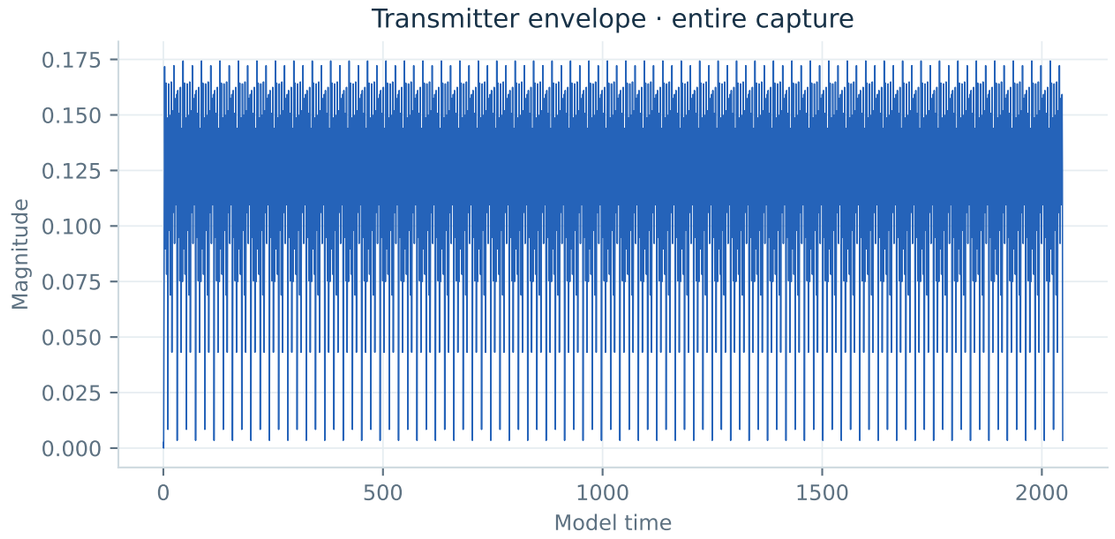


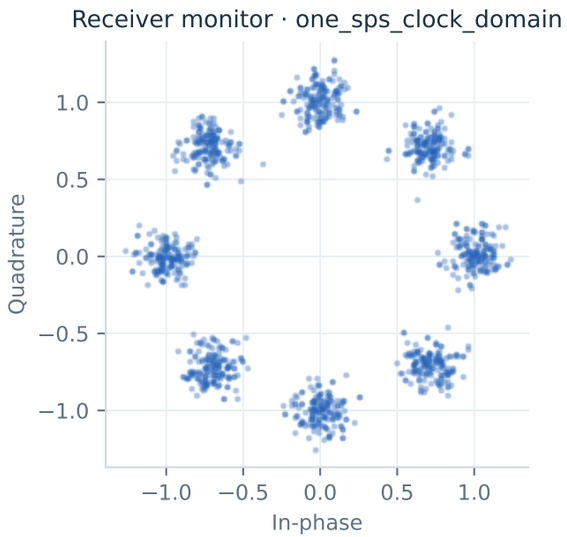
Source: saved portable regression vectors, replayed with the current Python core. The first 192 symbols are training and are excluded from BER/SER. Channel points are oversampled, not symbol decisions. Spectra share the TX peak reference. These figures do not establish original MATLAB receiver equivalence.

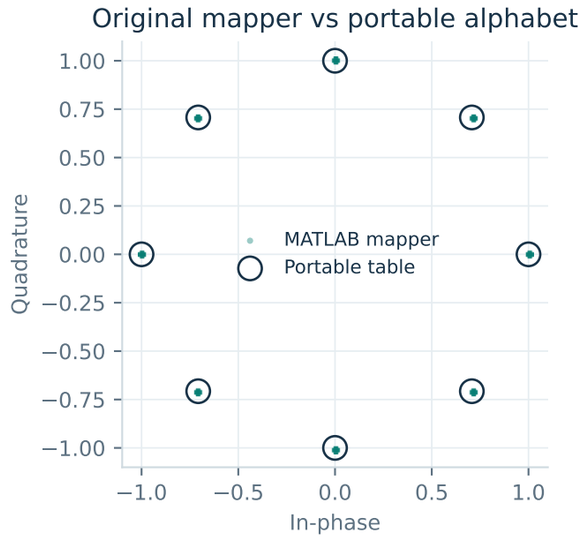
8-PSK · fast feedforward

ORIGINAL MATLAB CAPTURE / runtime PASS / exact repeat PASS / solver: FixedStepDiscrete





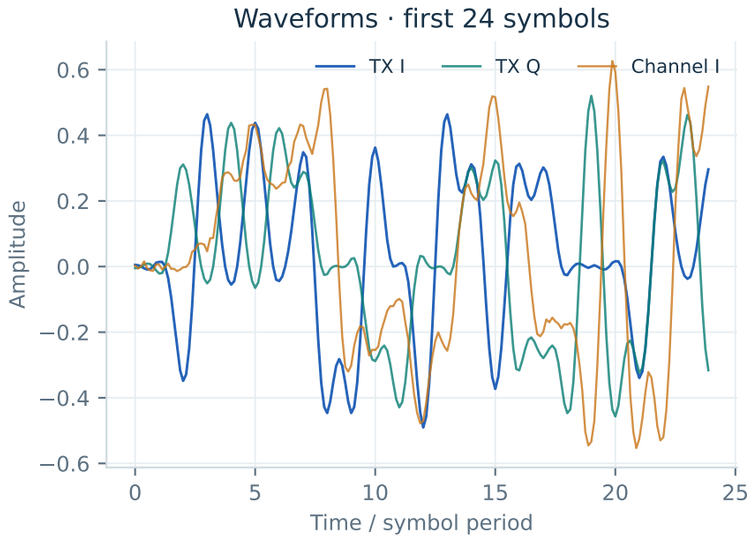


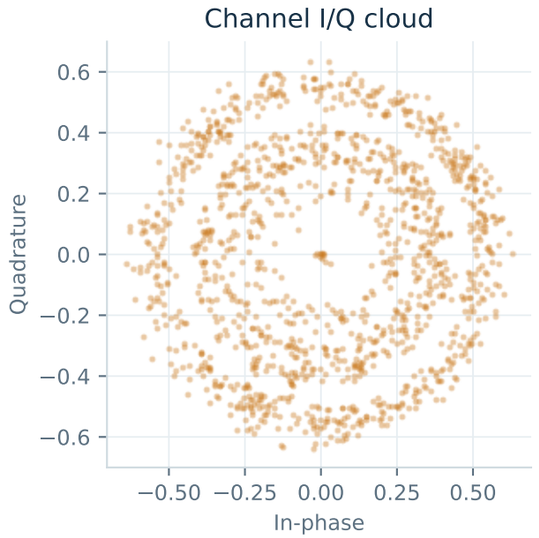


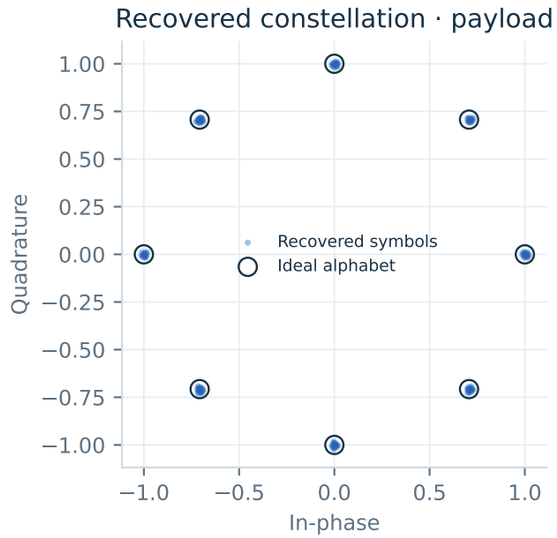
Actual R2020b signals with documented runtime settings. Native receiver monitors can contain held or oversampled values; their point clouds are not a BER measurement. Time is in the model's units. Simulation and repeatability do not establish receiver equivalence. Model: psk8_fast_ff_nda_symbol_sync_fast_ff_nda_phase_sync_2006_11_15

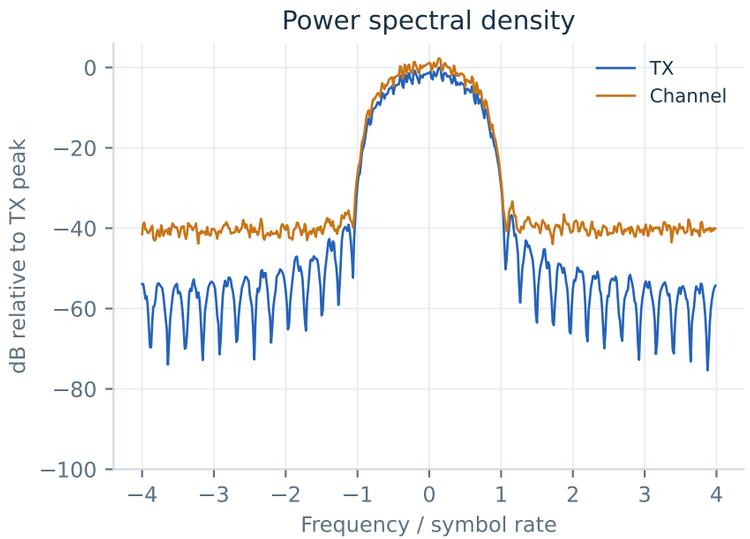
8-PSK · fast feedforward

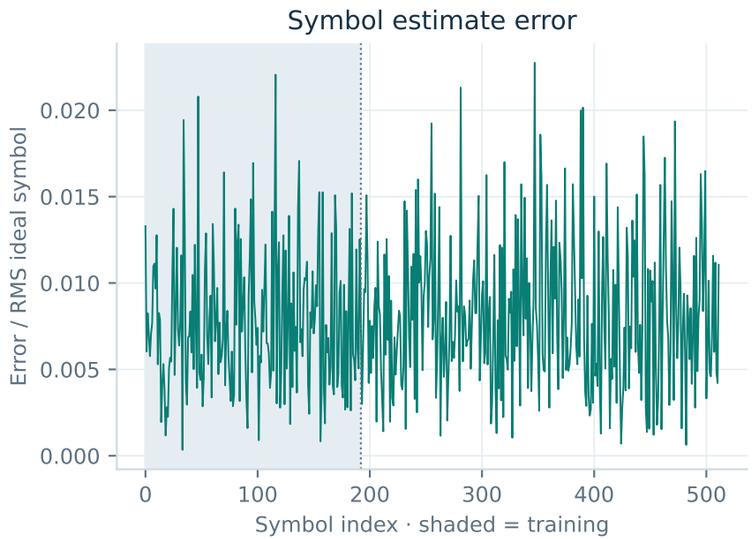
PORTABLE MODEL / Variant stress channel / 0 errors in 320 payload symbols

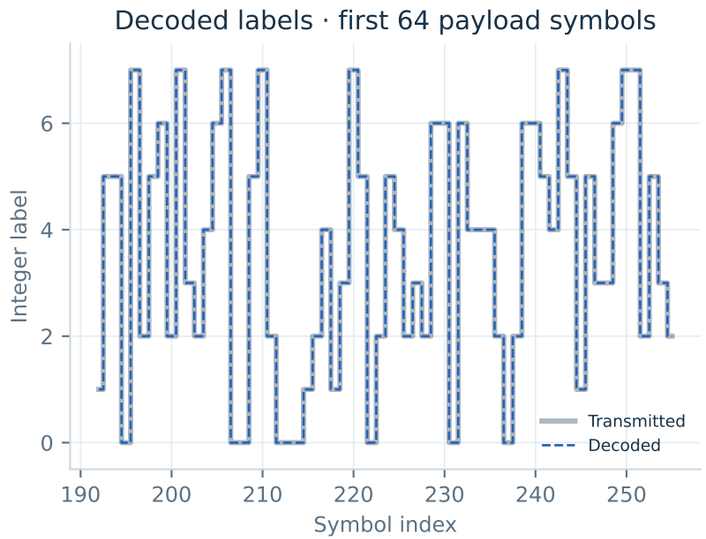








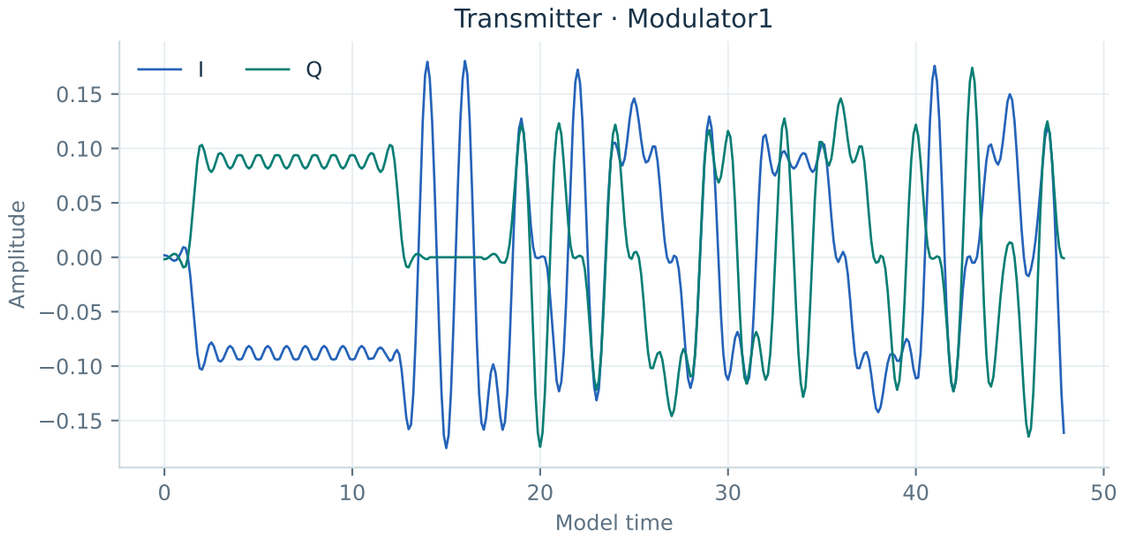


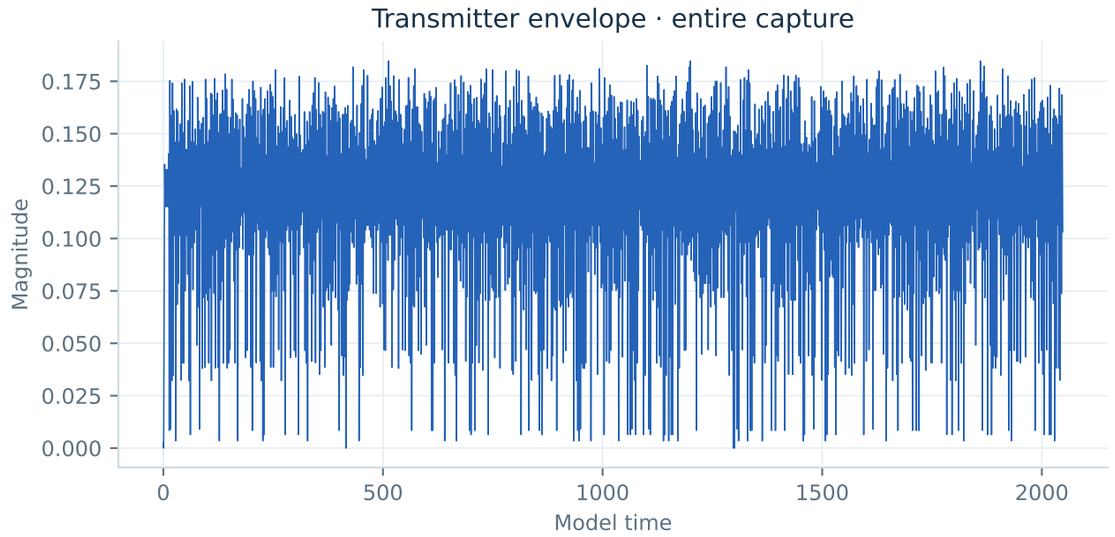


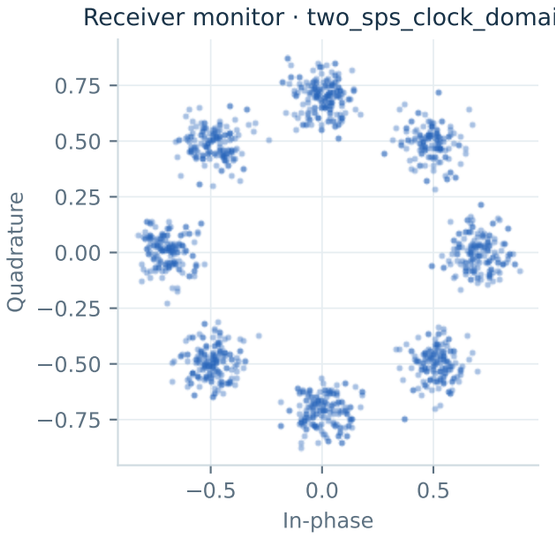
Source: saved portable regression vectors, replayed with the current Python core. The first 192 symbols are training and are excluded from BER/SER. Channel points are oversampled, not symbol decisions. Spectra share the TX peak reference. These figures do not establish original MATLAB receiver equivalence.

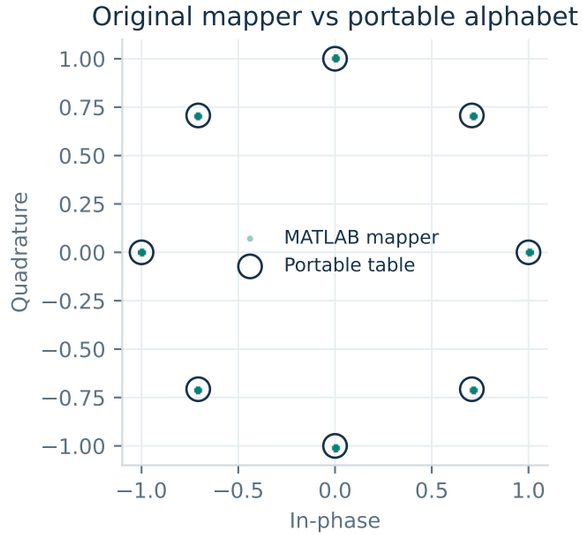
8-PSK · feedback / AGC

ORIGINAL MATLAB CAPTURE / runtime PASS / exact repeat PASS / solver: FixedStepDiscrete





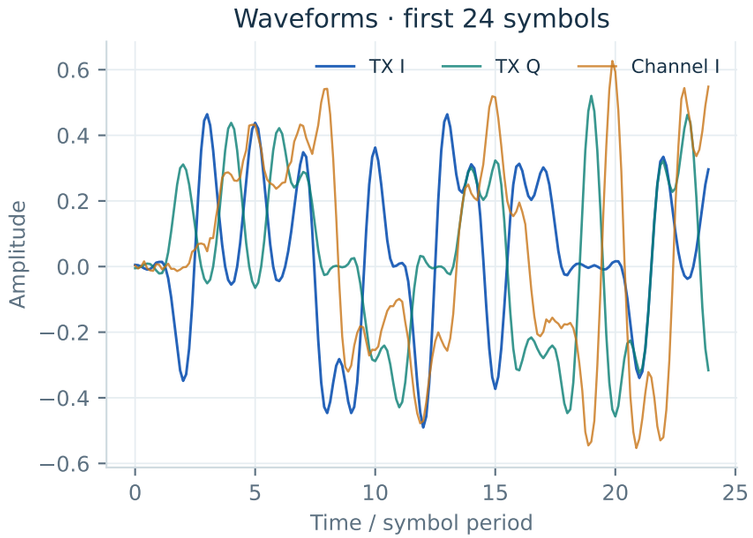


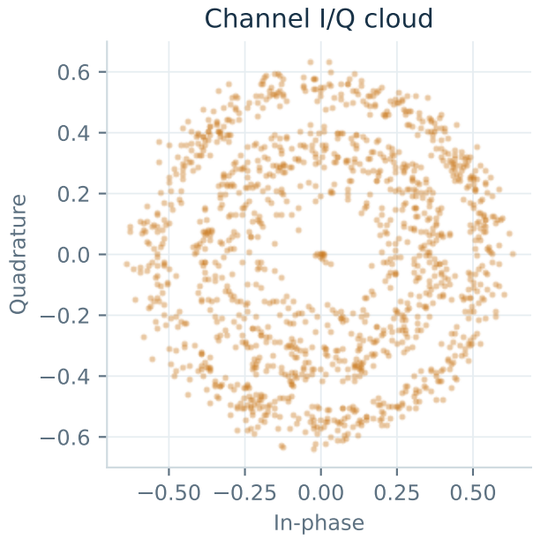


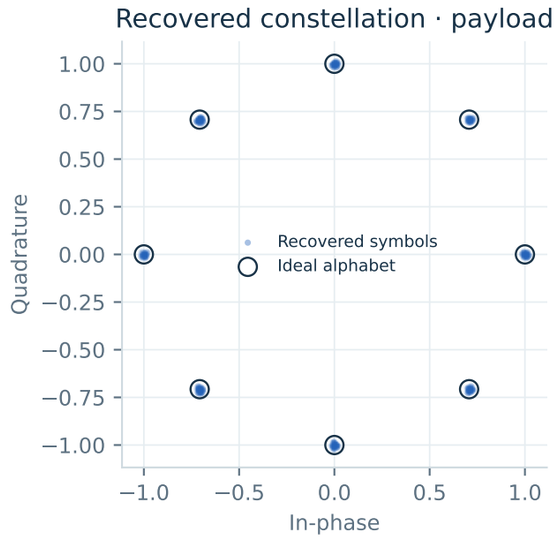
Actual R2020b signals with documented runtime settings. Native receiver monitors can contain held or oversampled values; their point clouds are not a BER measurement. Time is in the model's units. Simulation and repeatability do not establish receiver equivalence. Model: psk8_fb_symbol_sync_fb_phase_sync_agc_2009_08_19

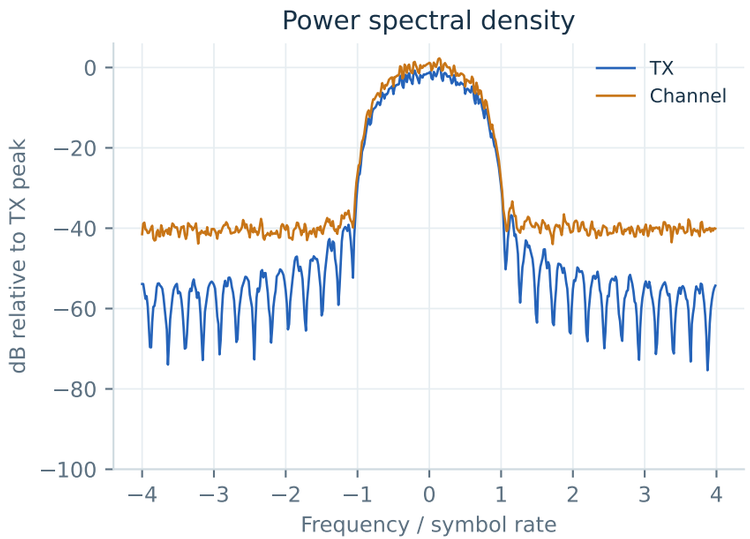
8-PSK · feedback / AGC

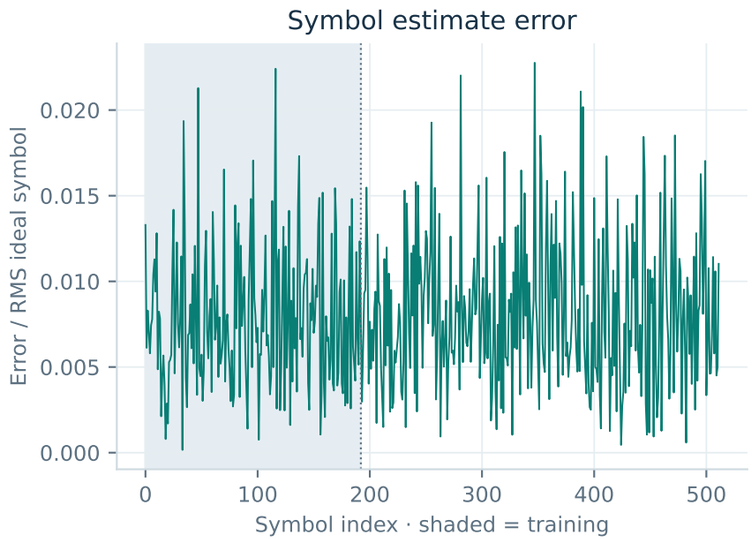
PORTABLE MODEL / Variant stress channel / 0 errors in 320 payload symbols

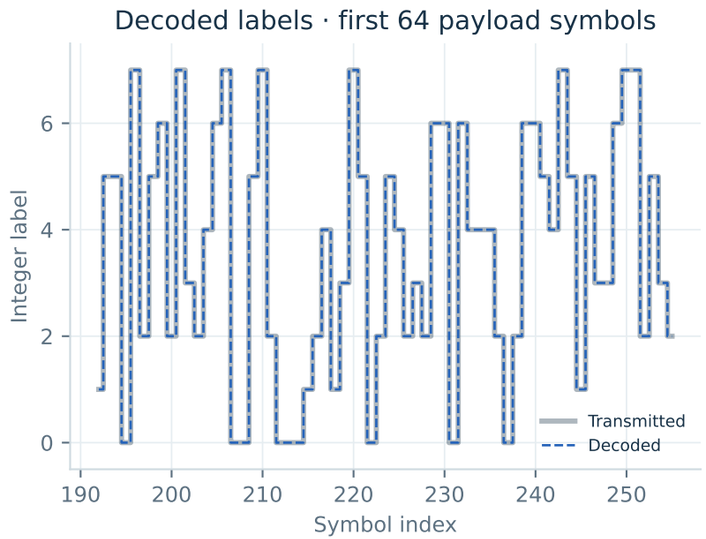








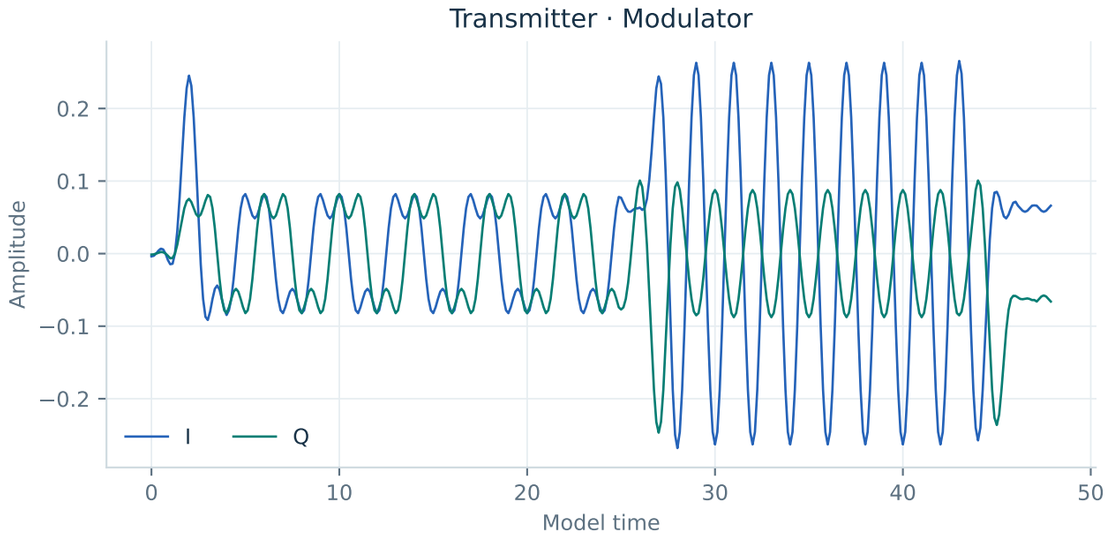


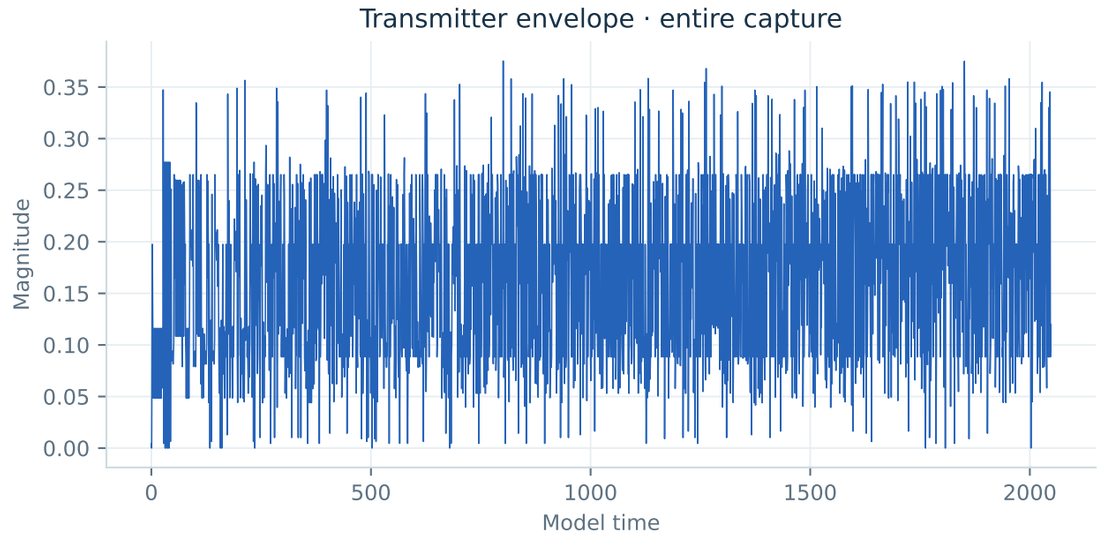


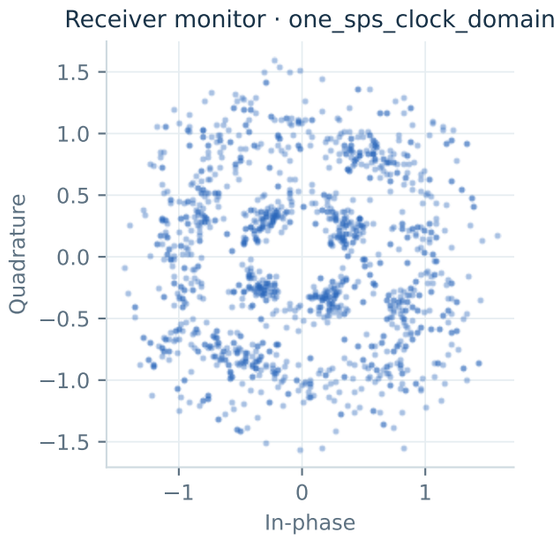
Source: saved portable regression vectors, replayed with the current Python core. The first 192 symbols are training and are excluded from BER/SER. Channel points are oversampled, not symbol decisions. Spectra share the TX peak reference. These figures do not establish original MATLAB receiver equivalence.

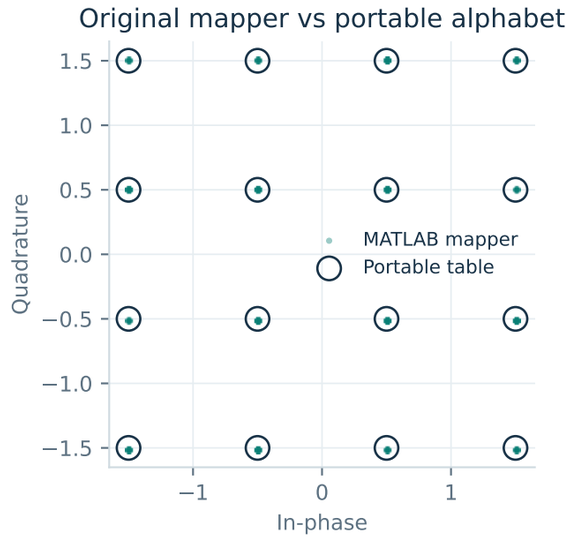
16-QAM · Gardner / Farrow

ORIGINAL MATLAB CAPTURE / runtime PASS / exact repeat PASS / solver: FixedStepDiscrete





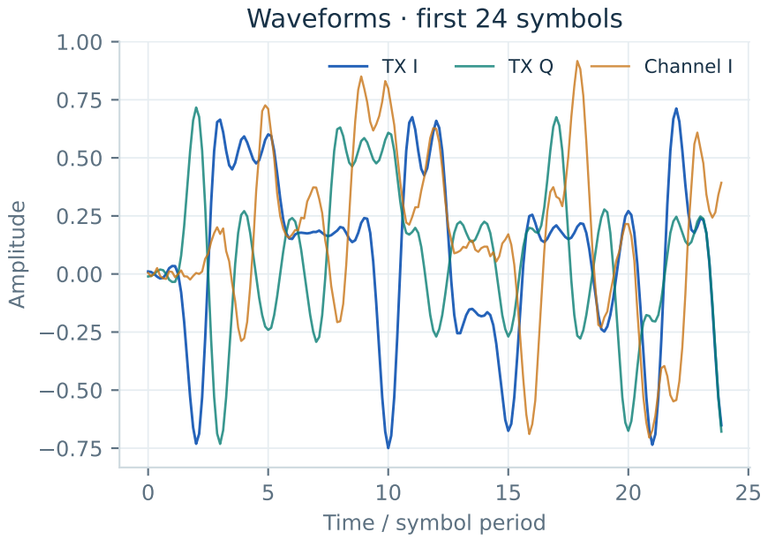


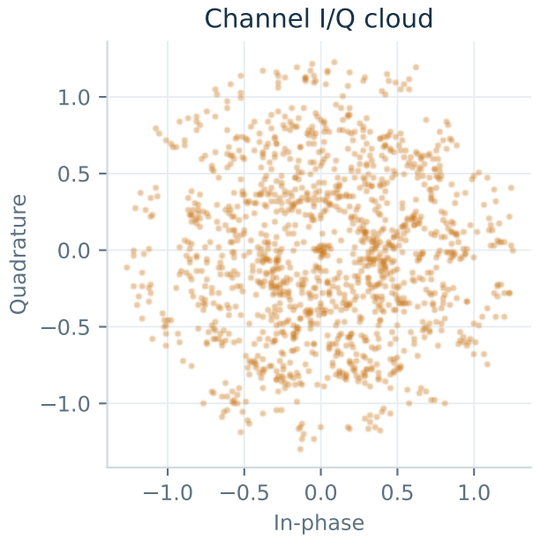


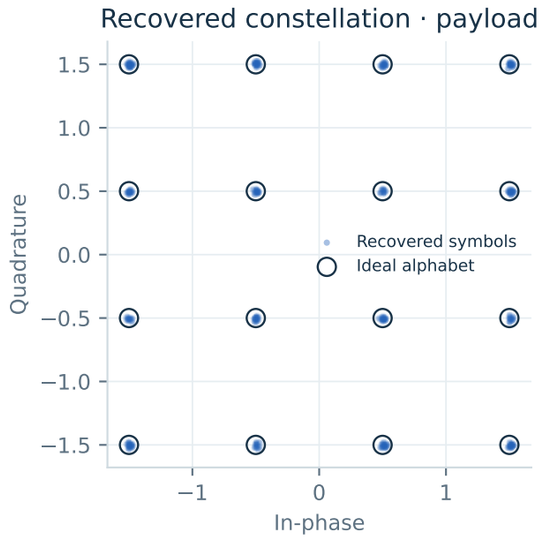
Actual R2020b signals with documented runtime settings. Native receiver monitors can contain held or oversampled values; their point clouds are not a BER measurement. Time is in the model's units. Simulation and repeatability do not establish receiver equivalence. Model: qam_fb_gardner_symbol_farrow_sync_fb_phase_sync_agc_2007_06_18

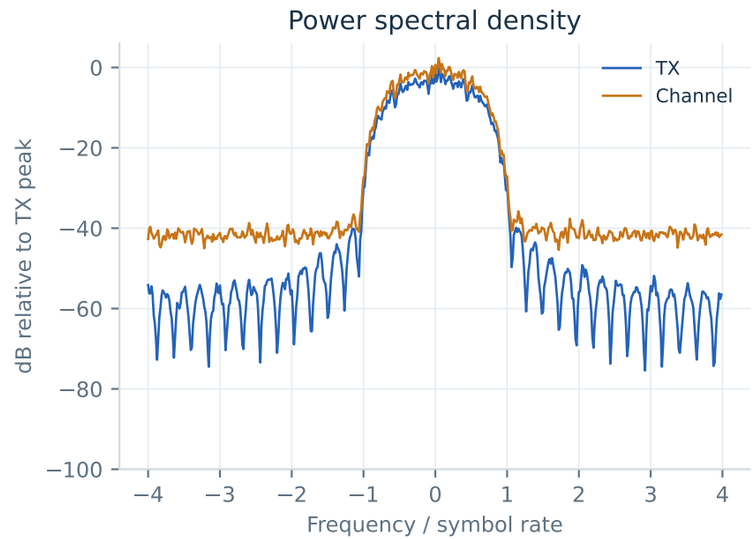
16-QAM · Gardner / Farrow

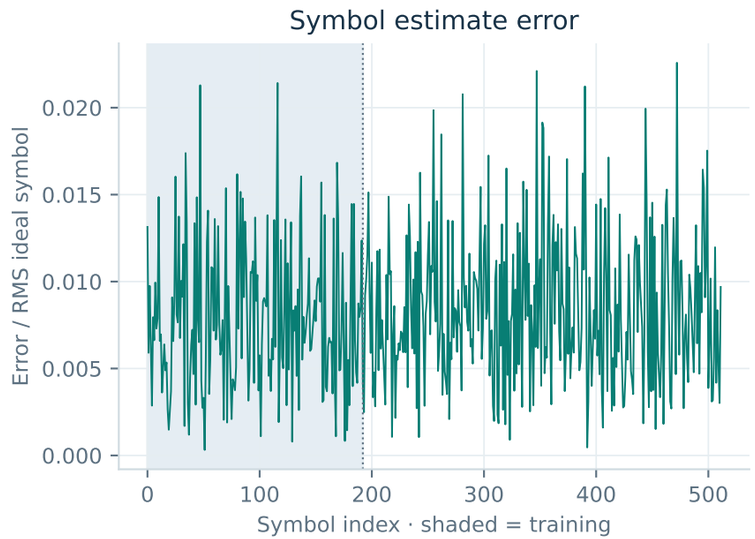
PORTABLE MODEL / Variant stress channel / 0 errors in 320 payload symbols

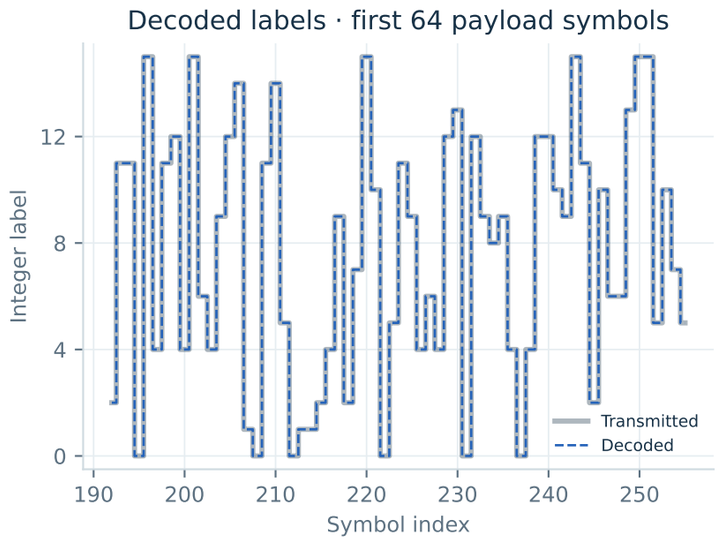








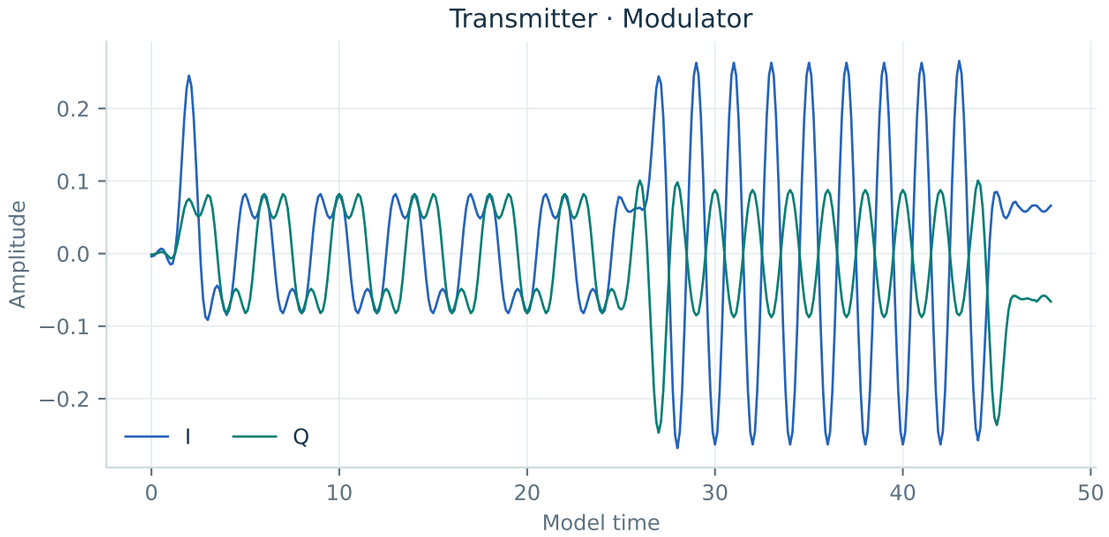


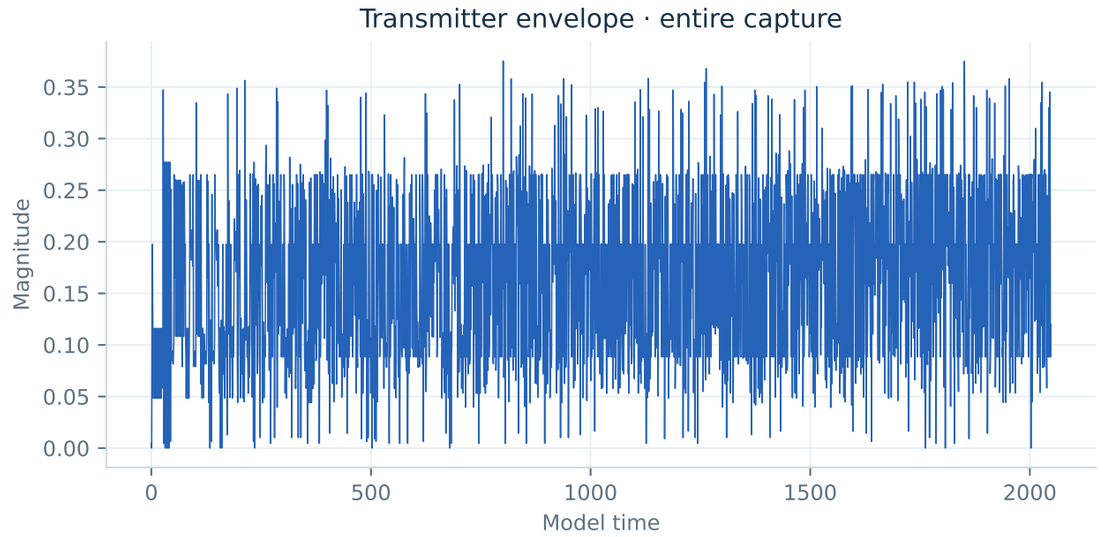


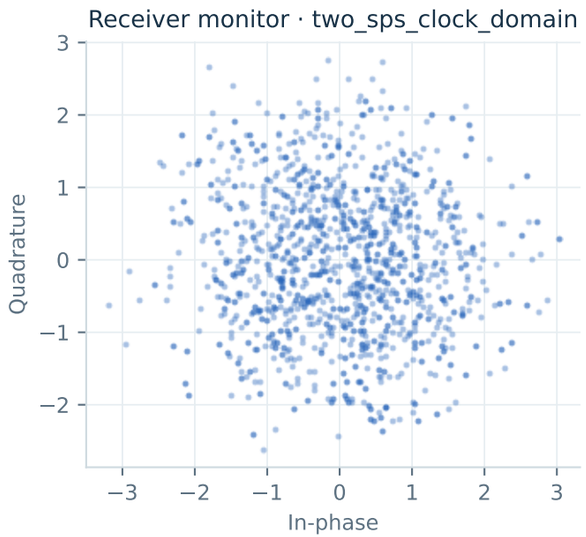
Source: saved portable regression vectors, replayed with the current Python core. The first 192 symbols are training and are excluded from BER/SER. Channel points are oversampled, not symbol decisions. Spectra share the TX peak reference. These figures do not establish original MATLAB receiver equivalence.

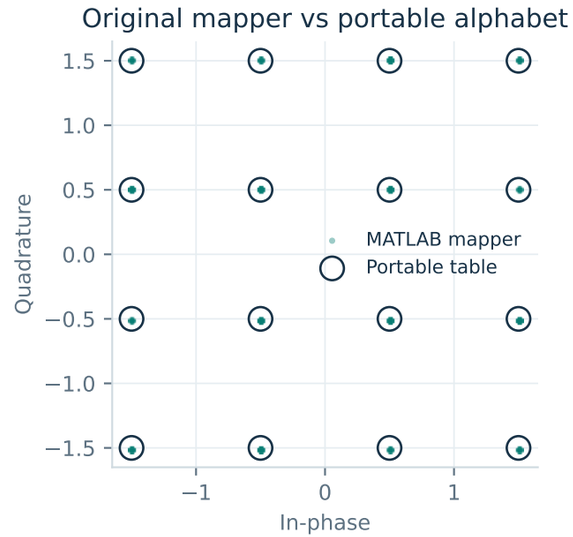
16-QAM · decision feedback EQ

ORIGINAL MATLAB CAPTURE / runtime PASS / exact repeat PASS / solver: FixedStepDiscrete





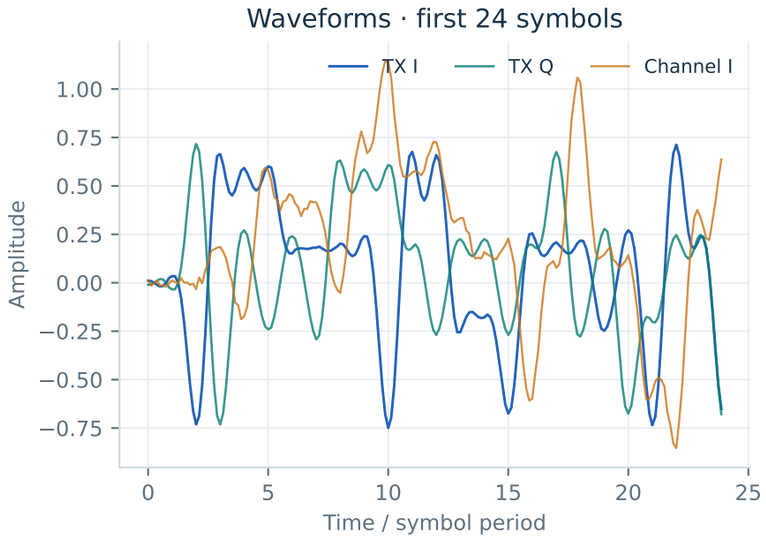


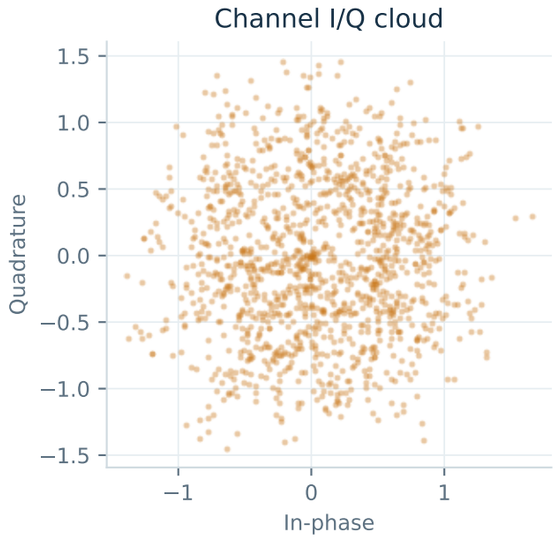


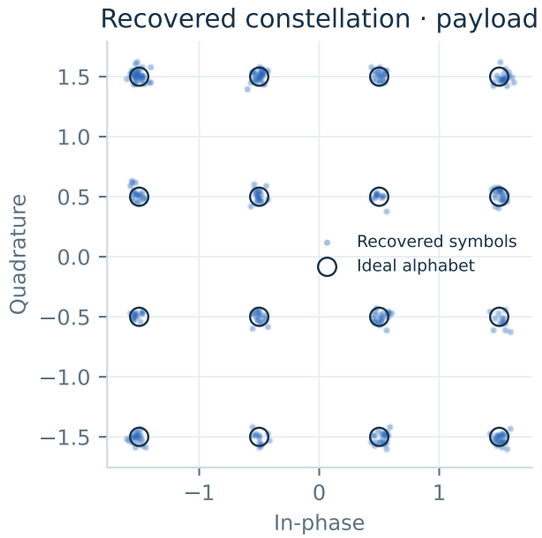
Actual R2020b signals with documented runtime settings. Native receiver monitors can contain held or oversampled values; their point clouds are not a BER measurement. Time is in the model's units. Simulation and repeatability do not establish receiver equivalence. Model: qam_fb_symbol_sync_fb_phase_sync_agc_dfe_2007_07_26

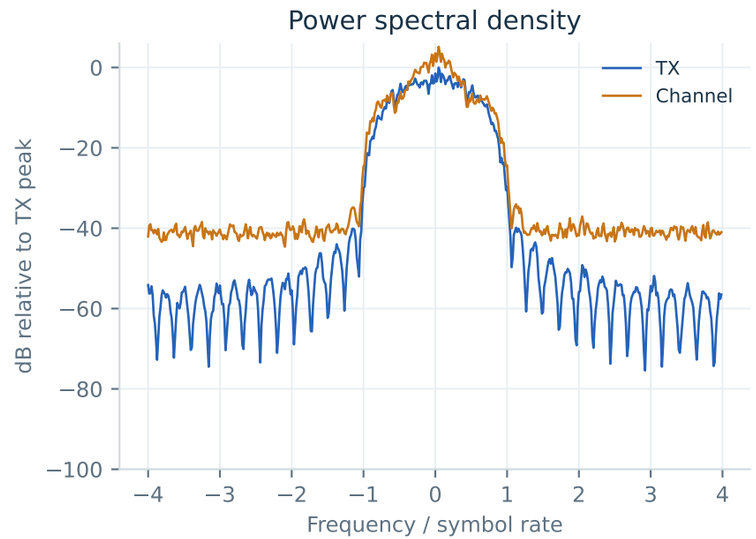
16-QAM · decision feedback EQ

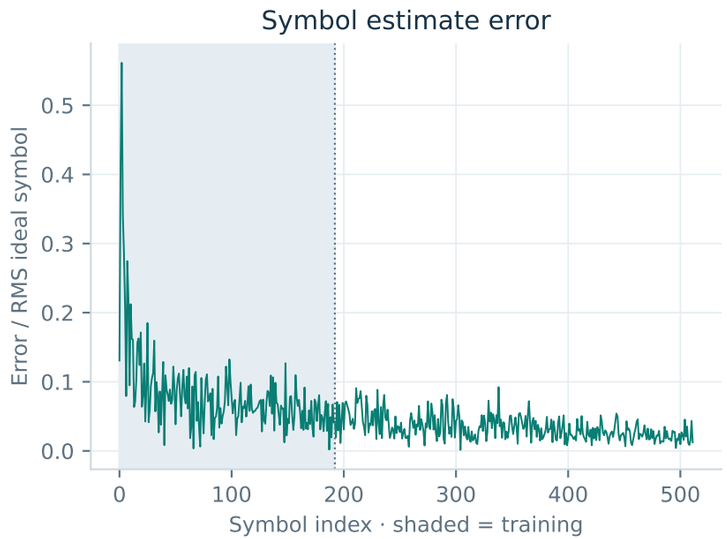
PORTABLE MODEL / Variant stress channel / 0 errors in 320 payload symbols

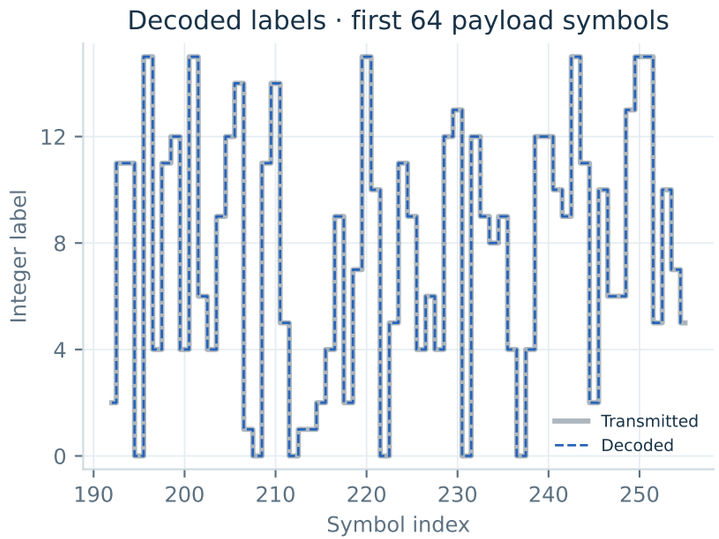








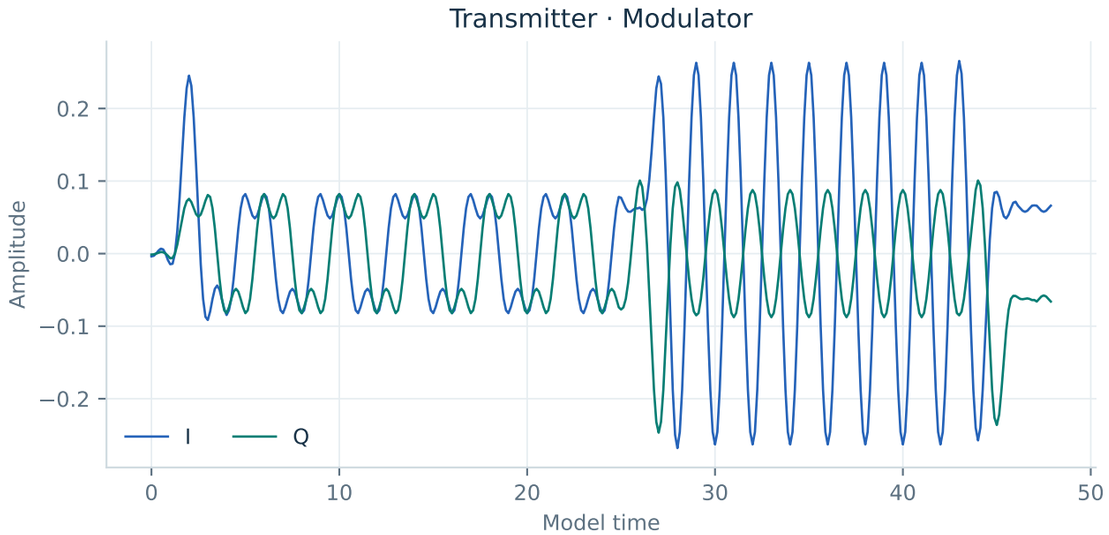


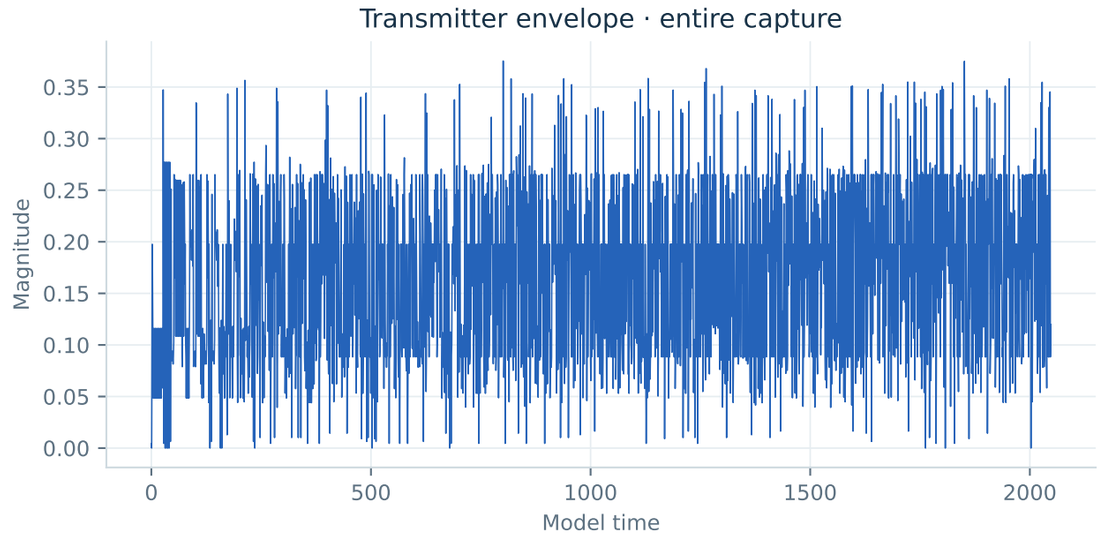


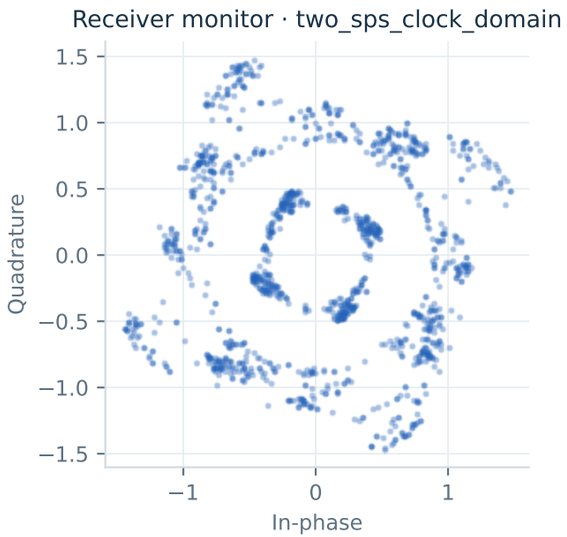
Source: saved portable regression vectors, replayed with the current Python core. The first 192 symbols are training and are excluded from BER/SER. Channel points are oversampled, not symbol decisions. Spectra share the TX peak reference. These figures do not establish original MATLAB receiver equivalence.

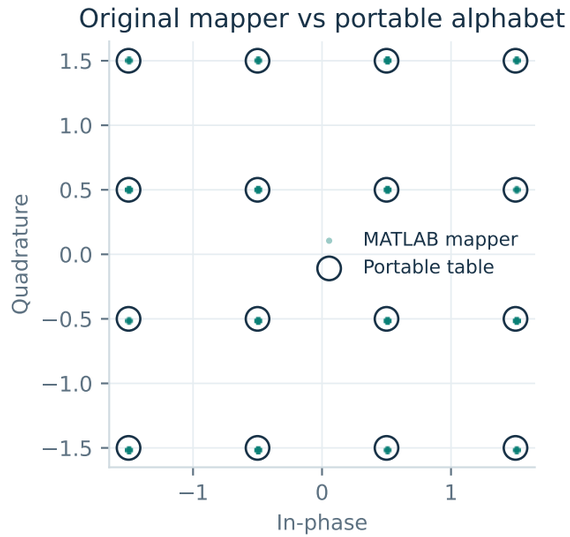
16-QAM · variable delay

ORIGINAL MATLAB CAPTURE / runtime PASS / exact repeat PASS / solver: FixedStepDiscrete





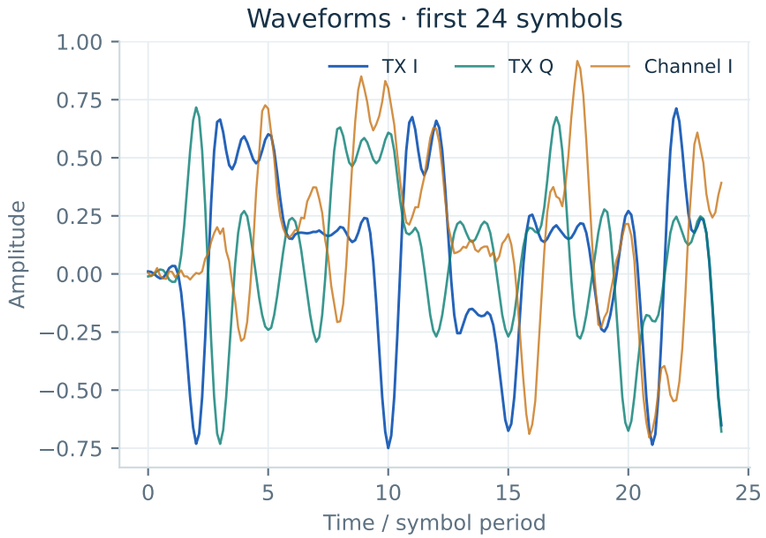


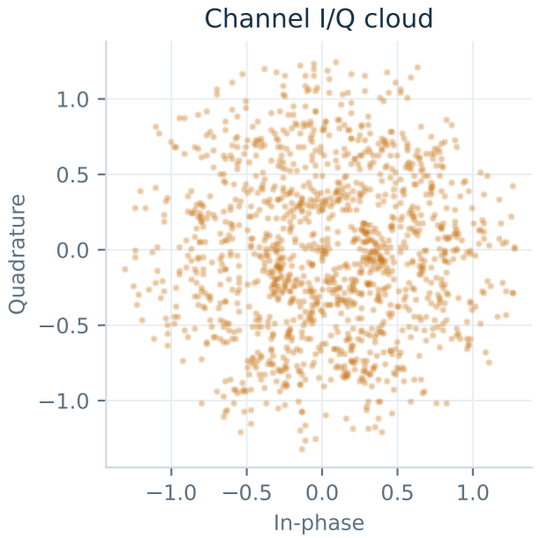


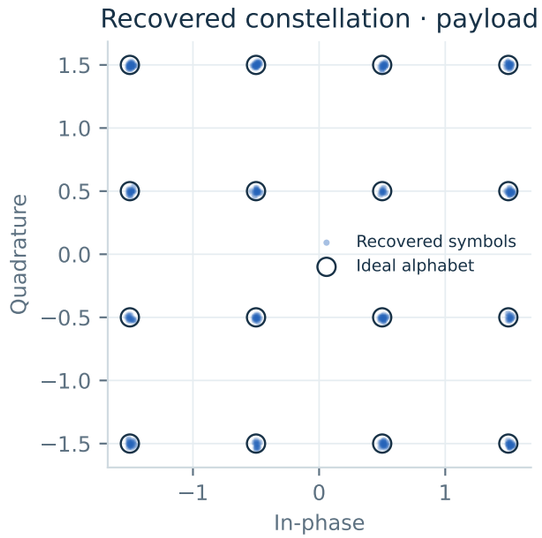
Actual R2020b signals with documented runtime settings. Native receiver monitors can contain held or oversampled values; their point clouds are not a BER measurement. Time is in the model's units. Simulation and repeatability do not establish receiver equivalence. Model: qam_fb_symbol_sync_fb_phase_sync_agc_var_ch_delay_2007_07_10

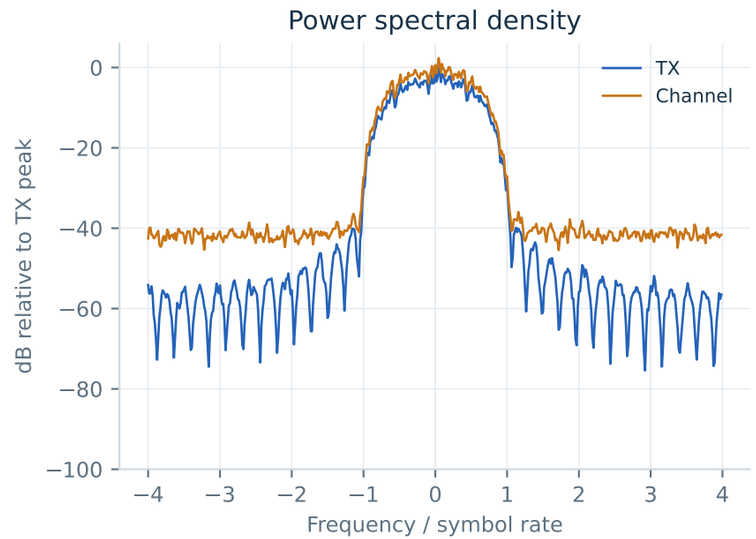
16-QAM · variable delay

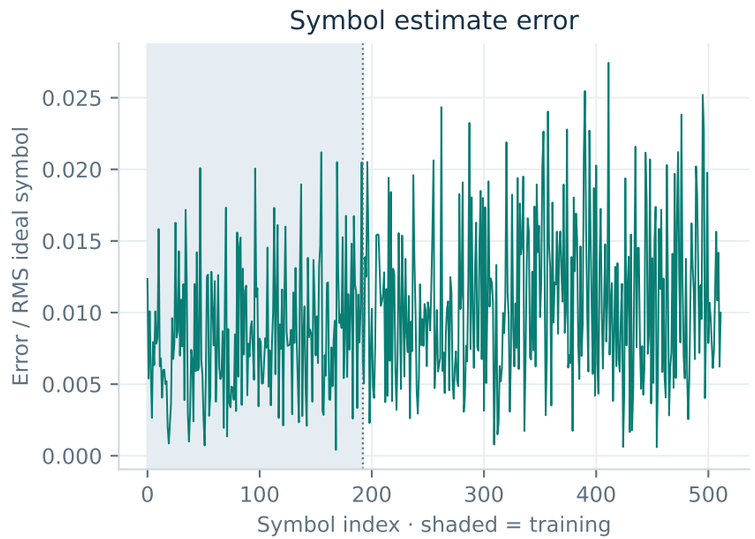
PORTABLE MODEL / Variant stress channel / 0 errors in 320 payload symbols

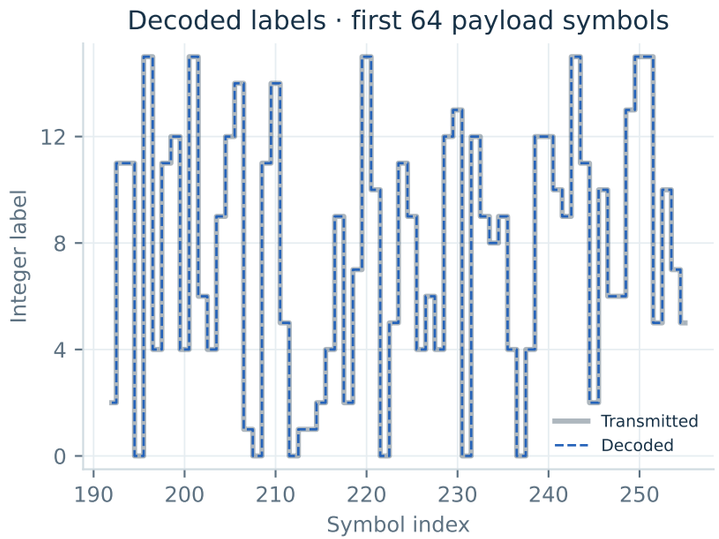








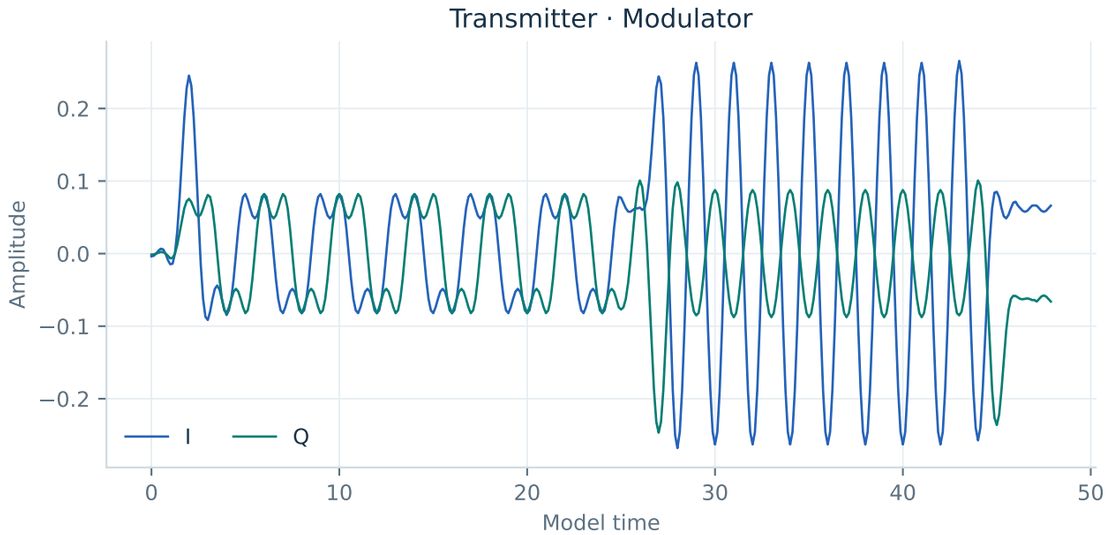


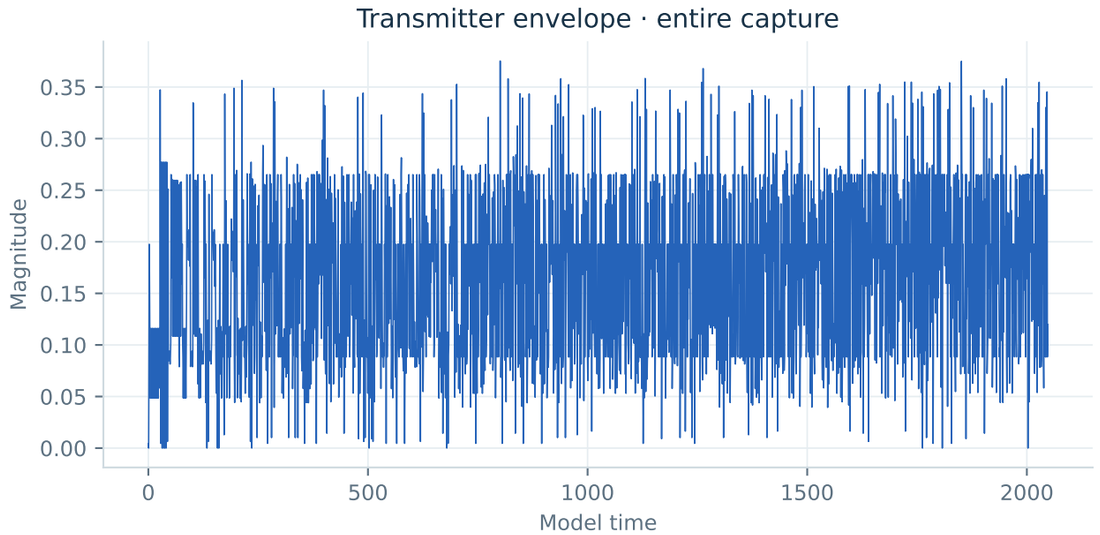


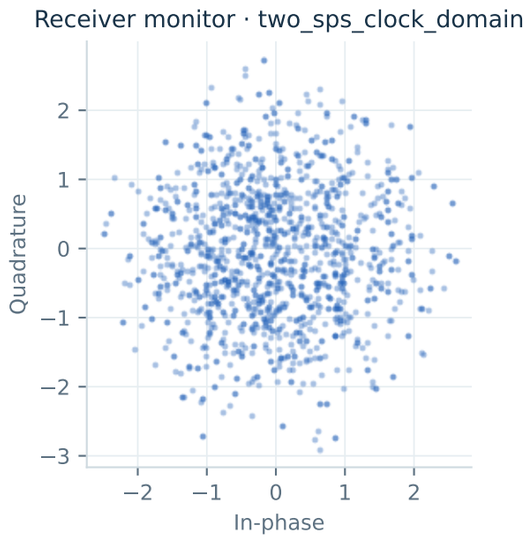
Source: saved portable regression vectors, replayed with the current Python core. The first 192 symbols are training and are excluded from BER/SER. Channel points are oversampled, not symbol decisions. Spectra share the TX peak reference. These figures do not establish original MATLAB receiver equivalence.

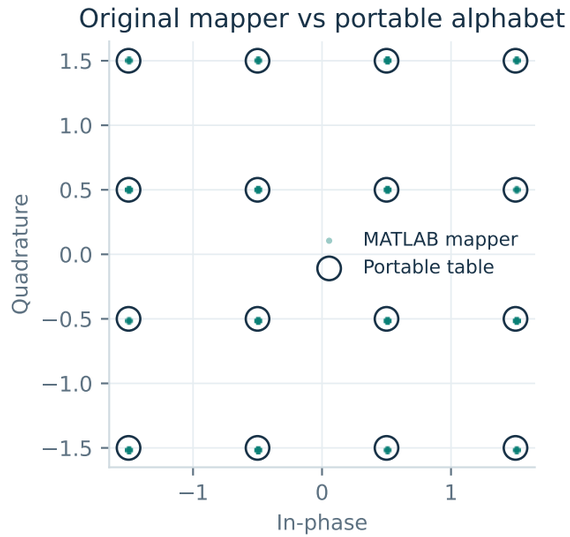
16-QAM · linear EQ

ORIGINAL MATLAB CAPTURE / runtime PASS / exact repeat PASS / solver: FixedStepDiscrete





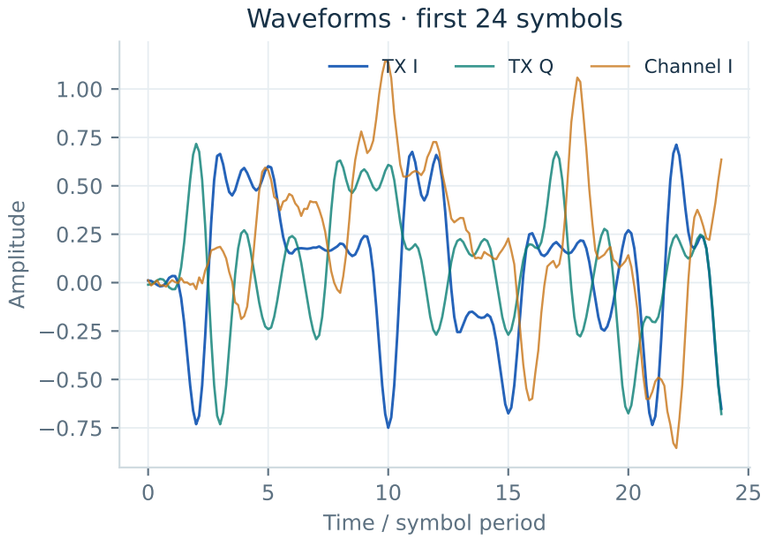


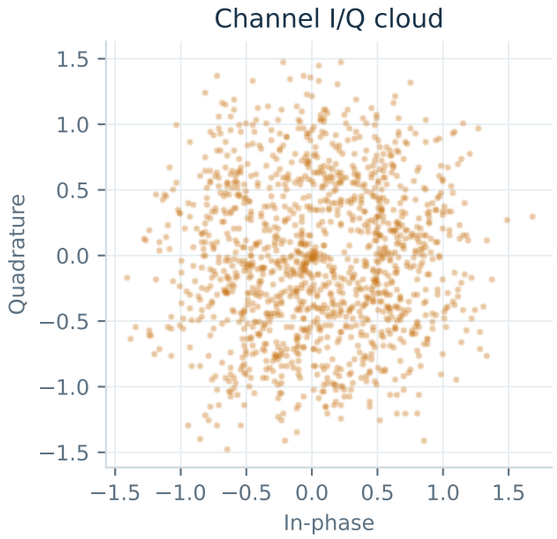


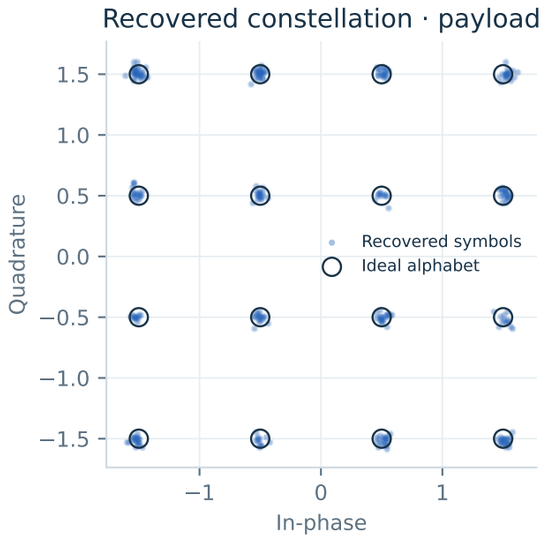
Actual R2020b signals with documented runtime settings. Native receiver monitors can contain held or oversampled values; their point clouds are not a BER measurement. Time is in the model's units. Simulation and repeatability do not establish receiver equivalence. Model: qam_fb_symbol_sync_fb_phase_sync_agc_var_ch_delay_eq_2007_07_21

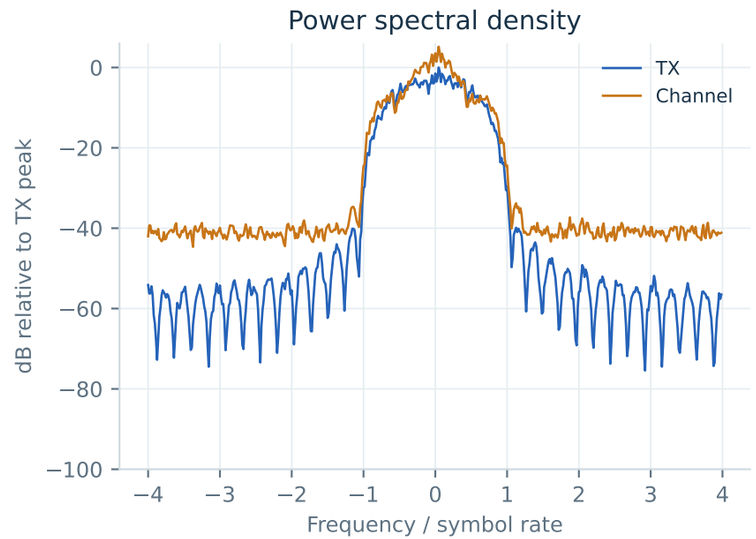
16-QAM · linear EQ

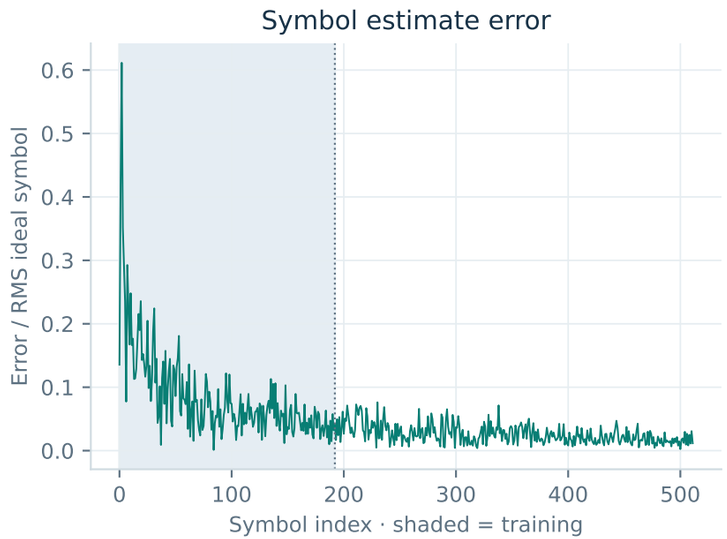
PORTABLE MODEL / Variant stress channel / 0 errors in 320 payload symbols

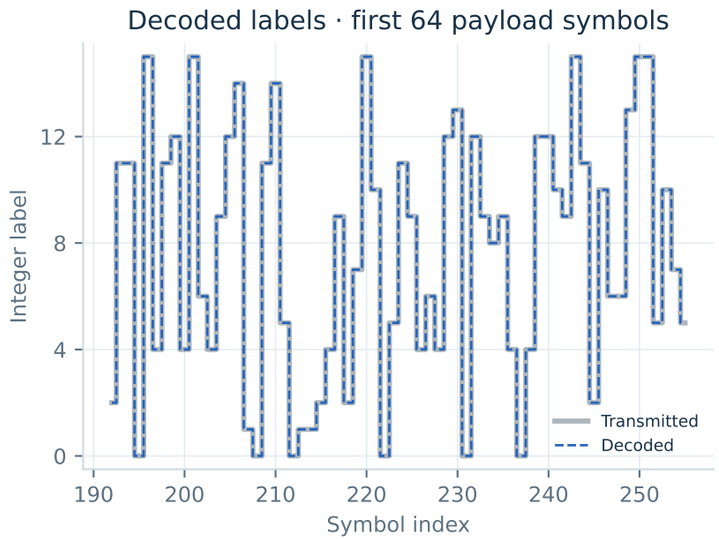








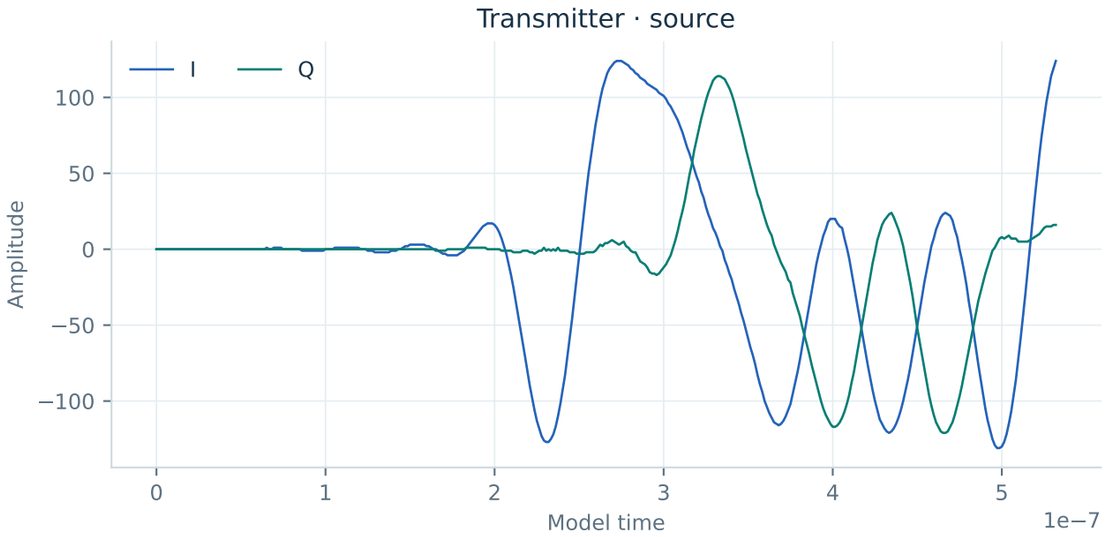


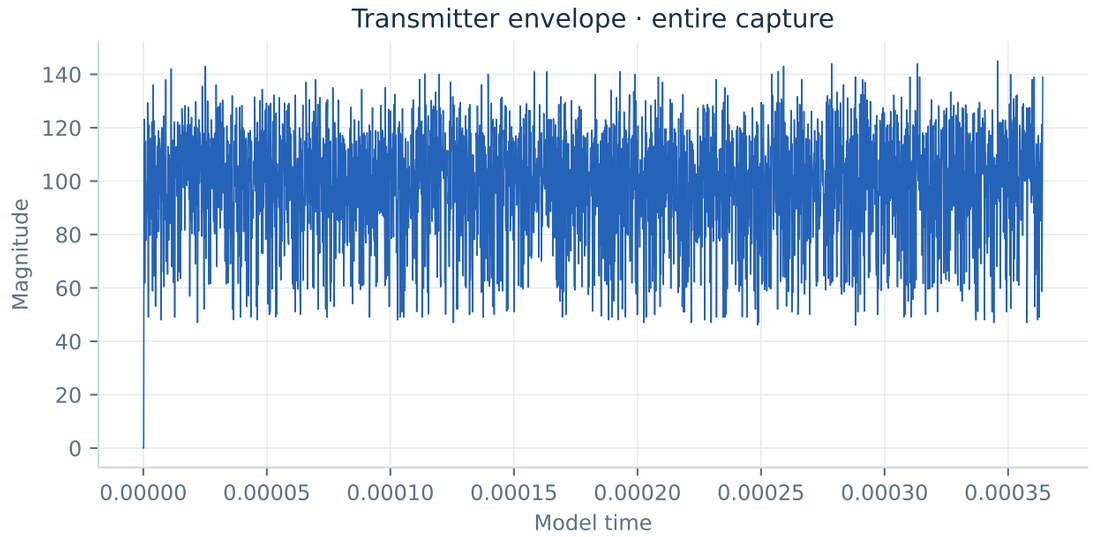


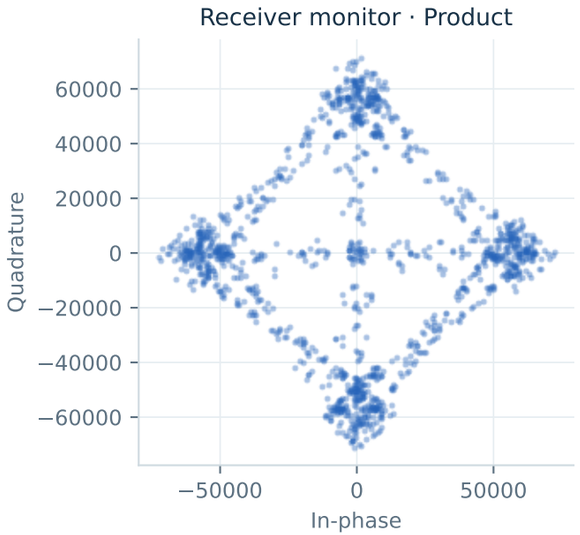
Source: saved portable regression vectors, replayed with the current Python core. The first 192 symbols are training and are excluded from BER/SER. Channel points are oversampled, not symbol decisions. Spectra share the TX peak reference. These figures do not establish original MATLAB receiver equivalence.

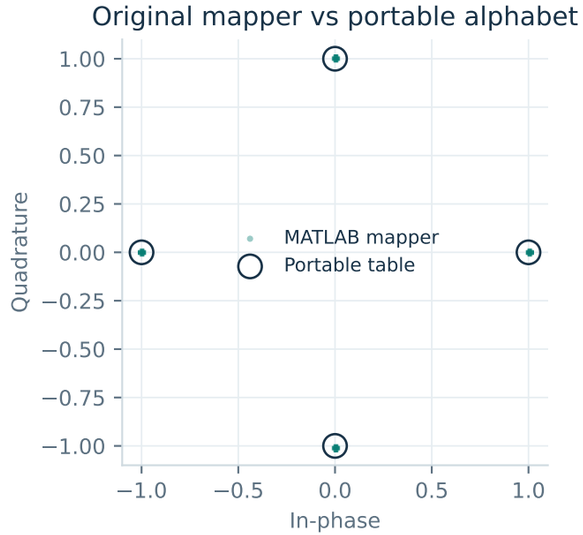
QPSK · phase noise / Costas

ORIGINAL MATLAB CAPTURE / runtime PASS / exact repeat PASS / solver: FixedStepDiscrete





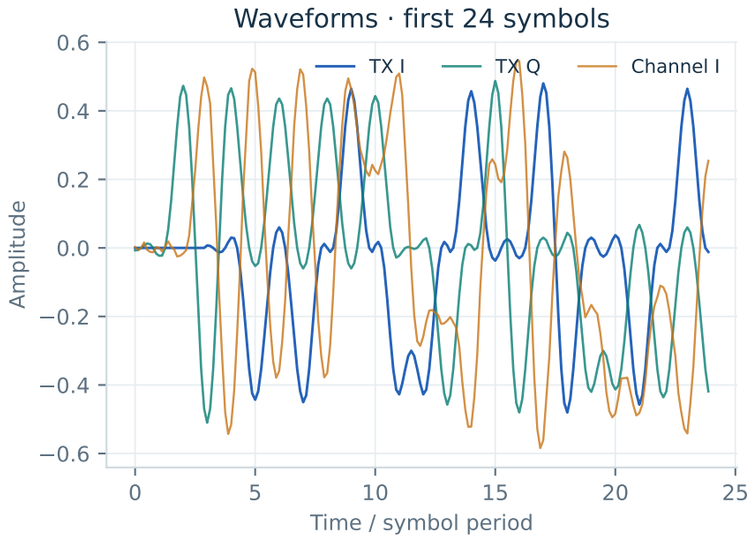


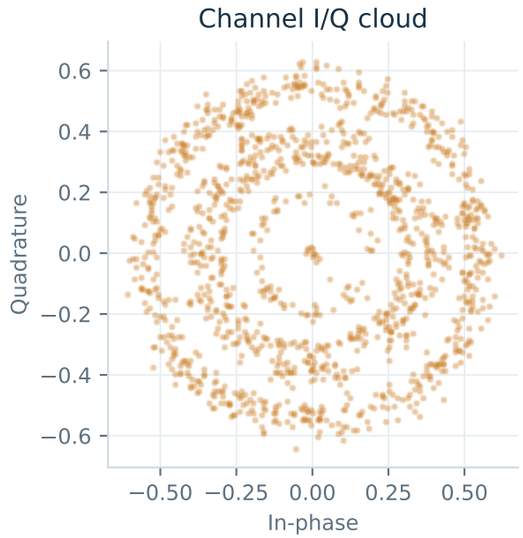


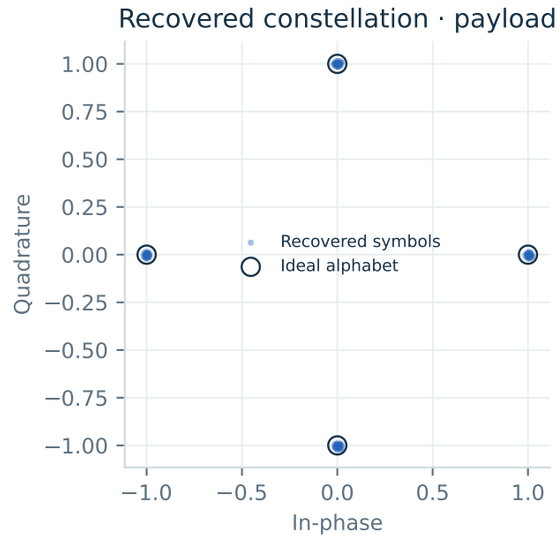
Actual R2020b signals with documented runtime settings. Native receiver monitors can contain held or oversampled values; their point clouds are not a BER measurement. Time is in the model's units. Simulation and repeatability do not establish receiver equivalence. Model: qpsk_demod_with_phase_noise

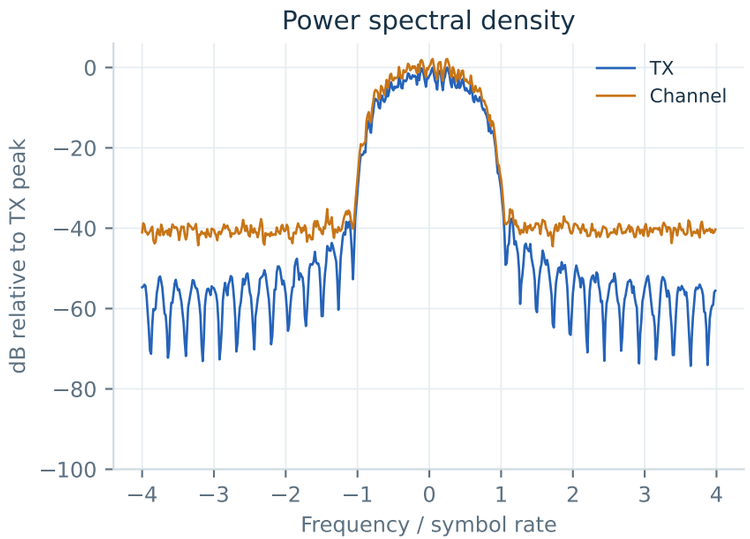
QPSK · phase noise / Costas

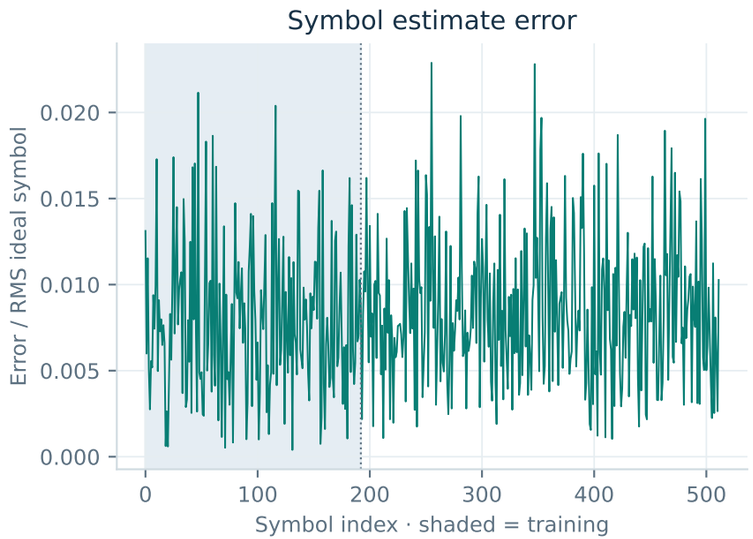
PORTABLE MODEL / Variant stress channel / 0 errors in 320 payload symbols

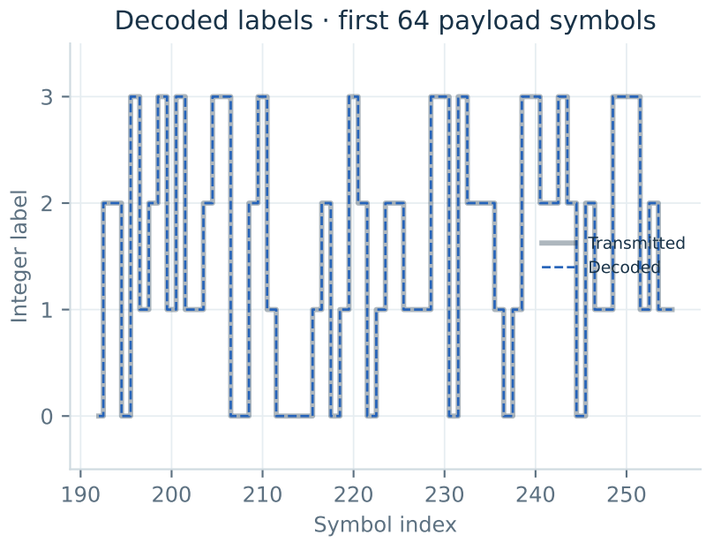








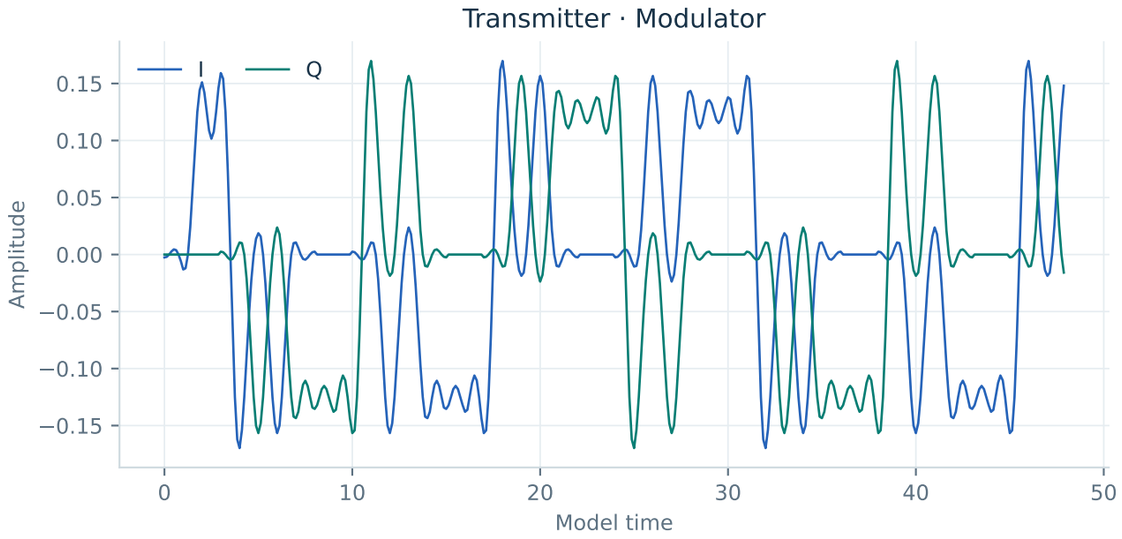


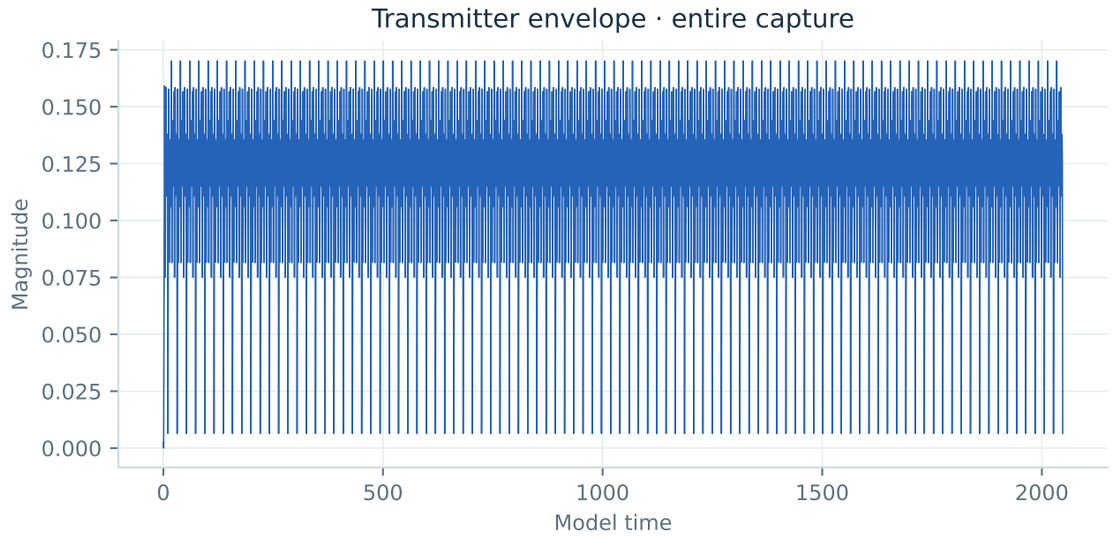


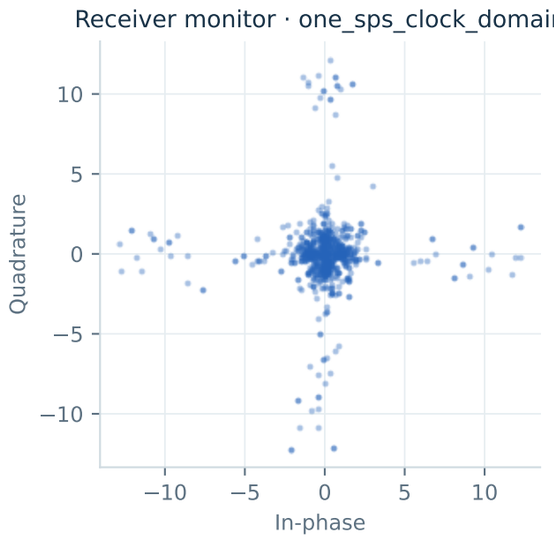
Source: saved portable regression vectors, replayed with the current Python core. The first 192 symbols are training and are excluded from BER/SER. Channel points are oversampled, not symbol decisions. Spectra share the TX peak reference. These figures do not establish original MATLAB receiver equivalence.

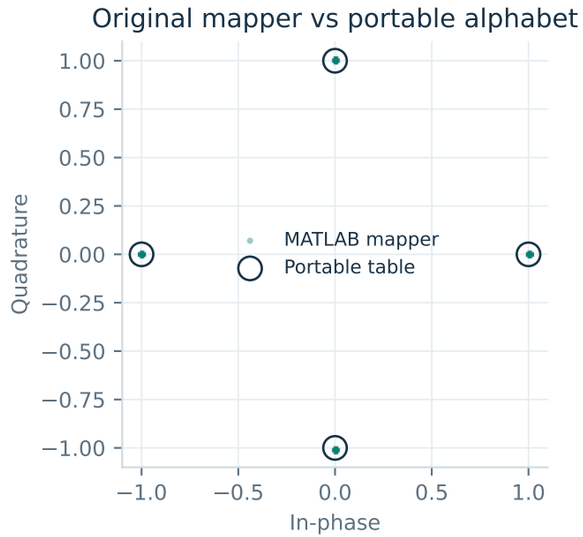
QPSK · fast acquisition

ORIGINAL MATLAB CAPTURE / runtime PASS / exact repeat PASS / solver: FixedStepDiscrete





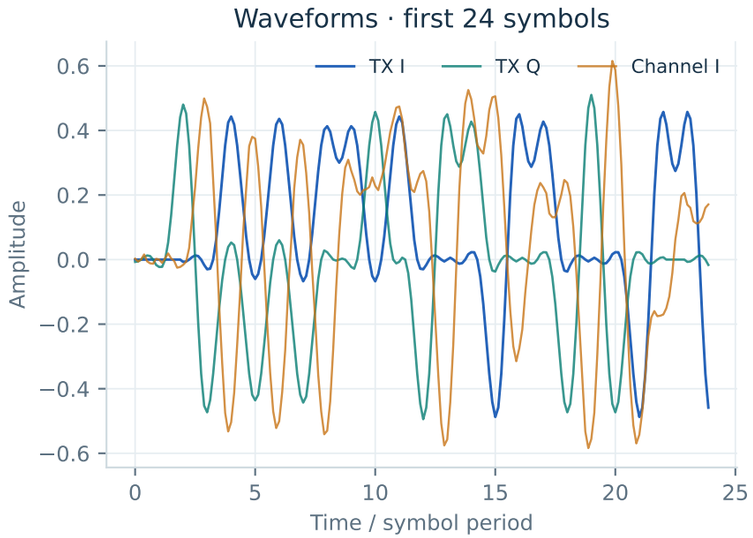


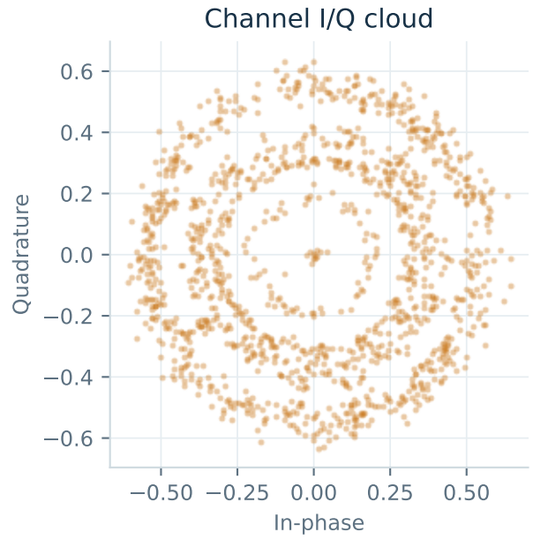


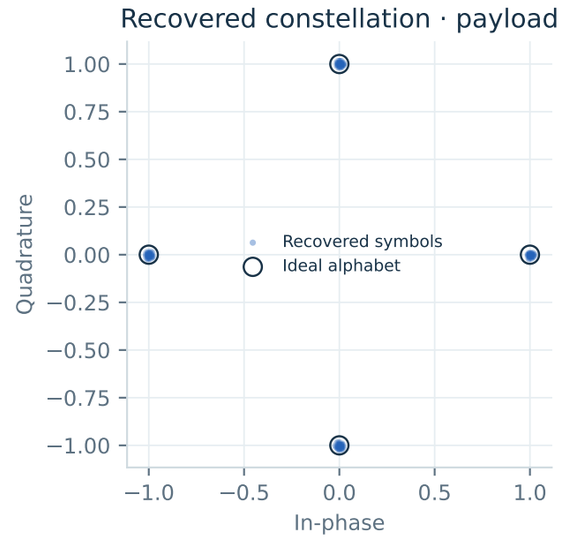
Actual R2020b signals with documented runtime settings. Native receiver monitors can contain held or oversampled values; their point clouds are not a BER measurement. Time is in the model's units. Simulation and repeatability do not establish receiver equivalence. Model: qpsk_fast_symbol_sync_fast_phase_sync_fast_agc_2007_03_06

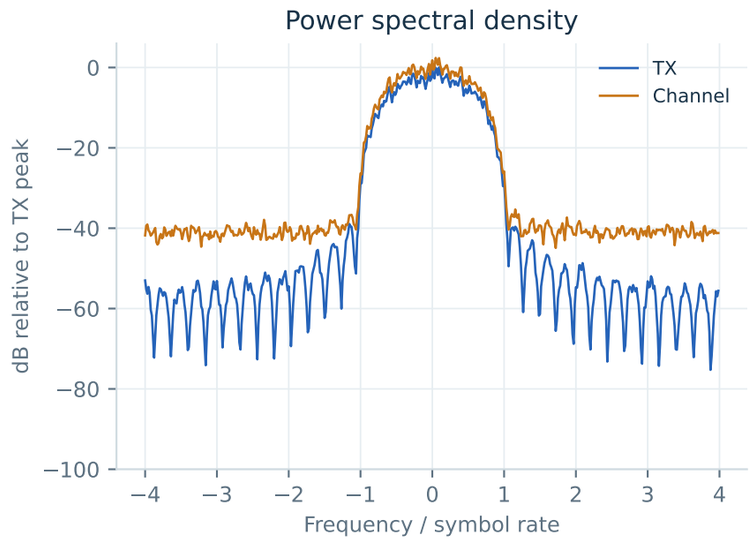
QPSK · fast acquisition

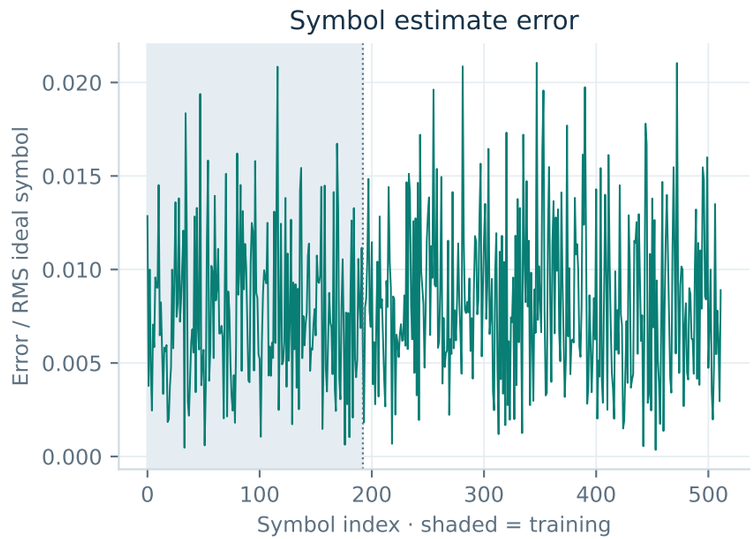
PORTABLE MODEL / Variant stress channel / 0 errors in 320 payload symbols

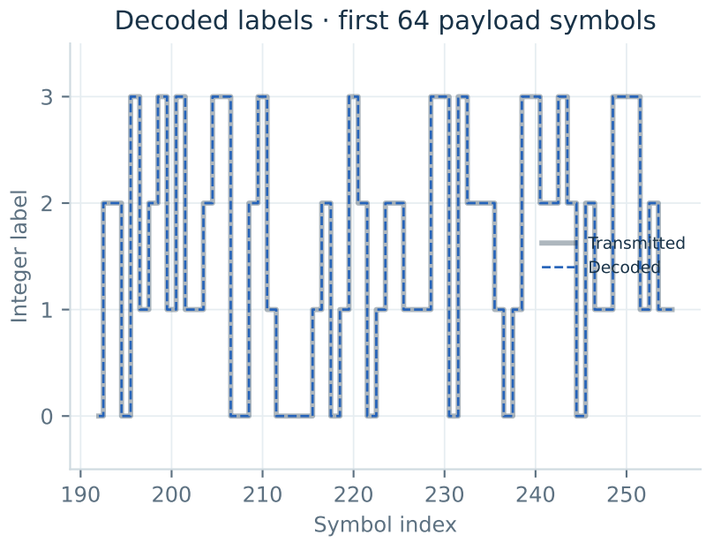








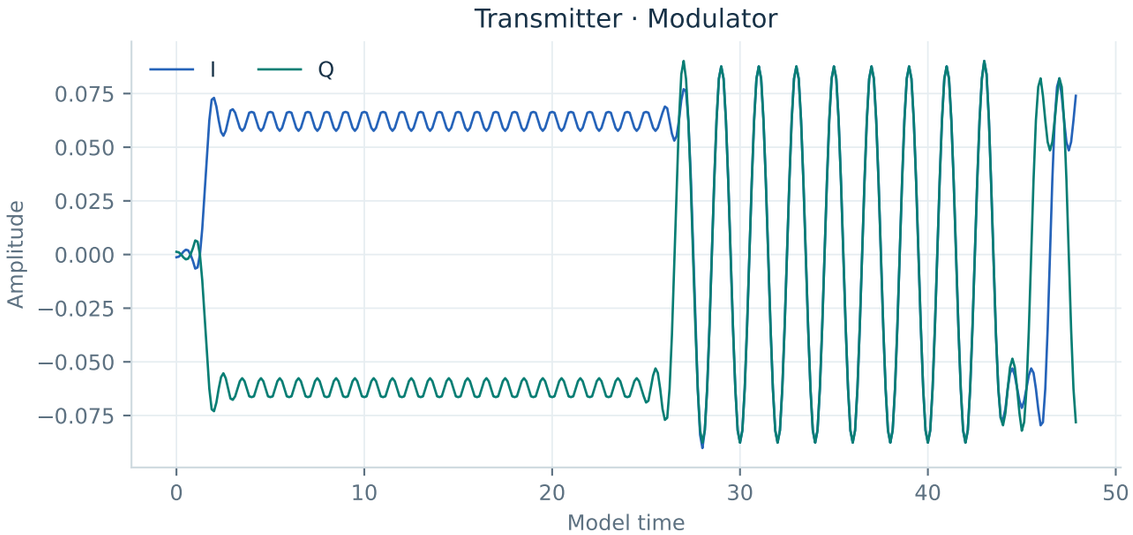


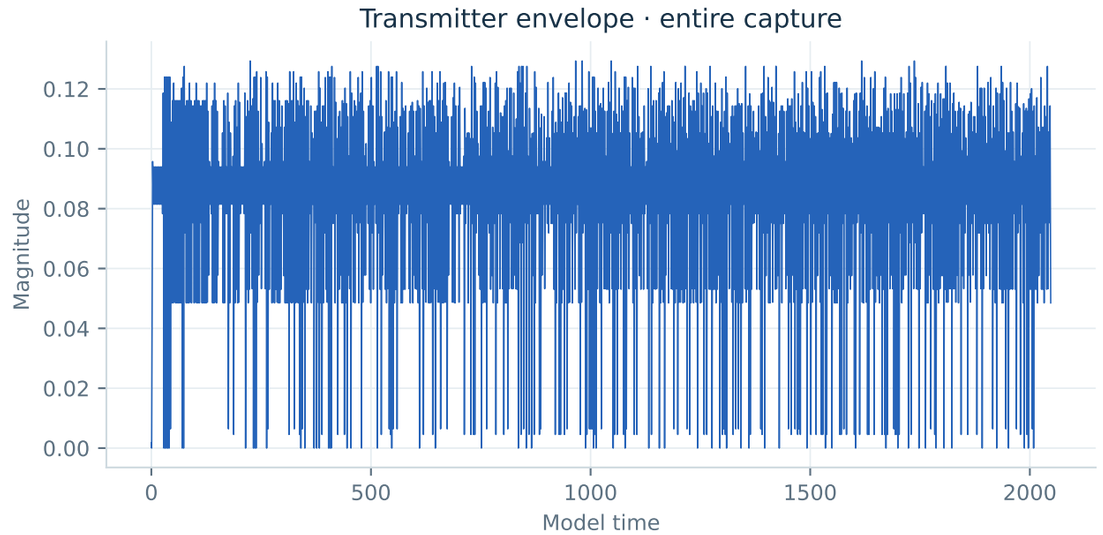


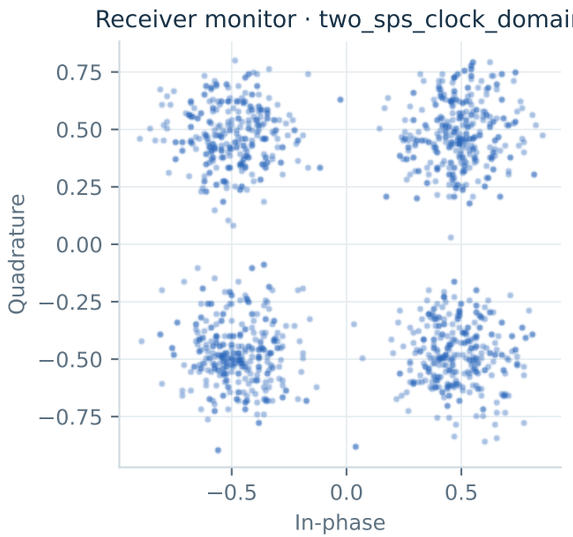
Source: saved portable regression vectors, replayed with the current Python core. The first 192 symbols are training and are excluded from BER/SER. Channel points are oversampled, not symbol decisions. Spectra share the TX peak reference. These figures do not establish original MATLAB receiver equivalence.

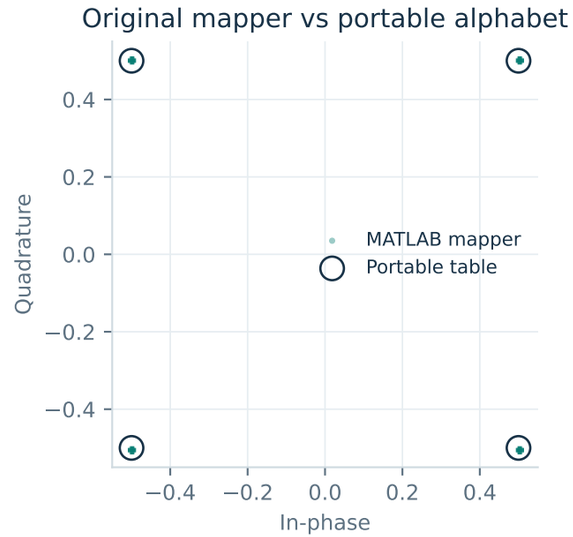
QPSK · feedback / AGC

ORIGINAL MATLAB CAPTURE / runtime PASS / exact repeat PASS / solver: FixedStepDiscrete





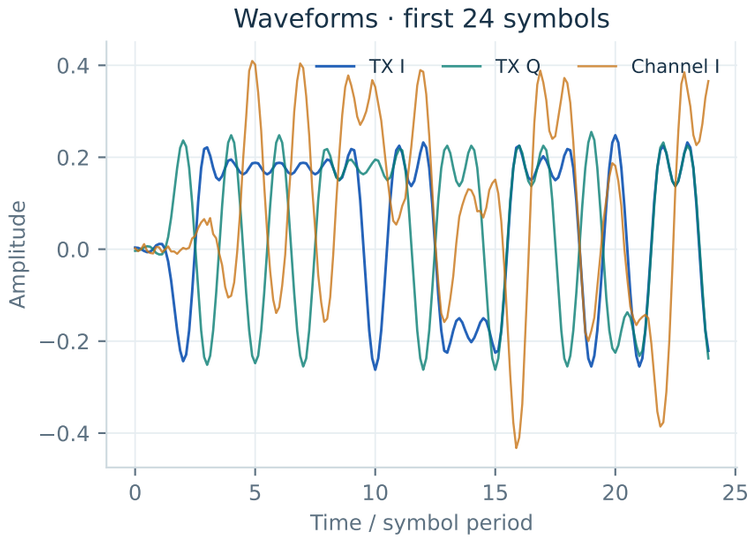


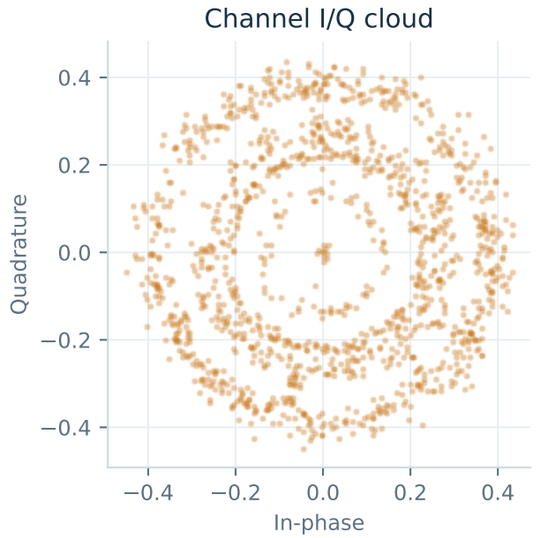


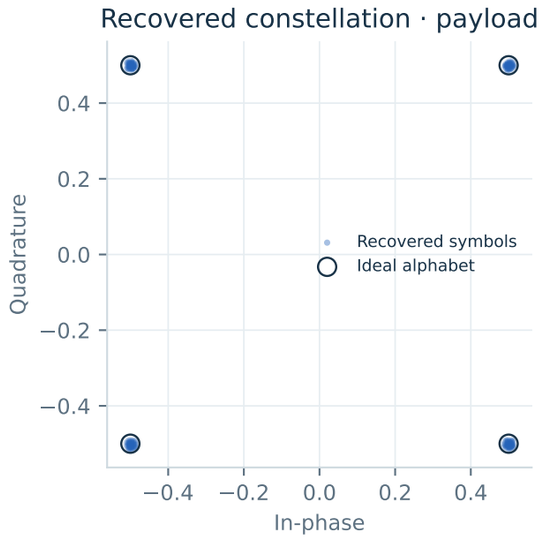
Actual R2020b signals with documented runtime settings. Native receiver monitors can contain held or oversampled values; their point clouds are not a BER measurement. Time is in the model's units. Simulation and repeatability do not establish receiver equivalence. Model: qpsk_fb_symbol_sync_fb_phase_sync_agc_2008_05_15

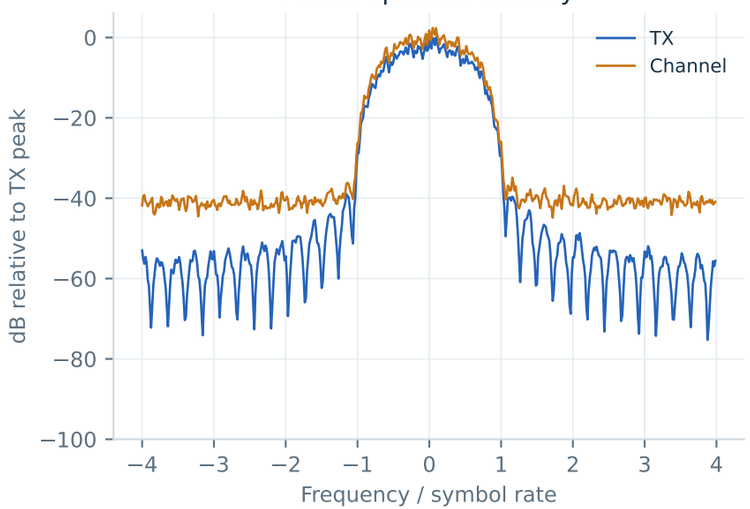
QPSK · feedback / AGC

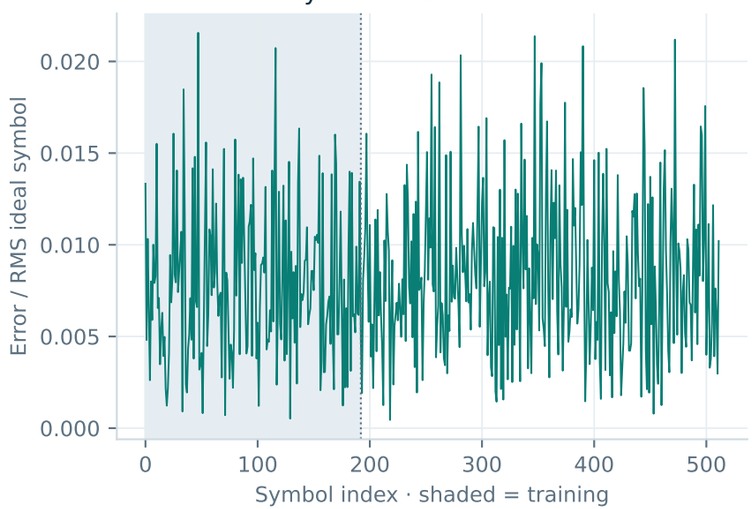
PORTABLE MODEL / Variant stress channel / 0 errors in 320 payload symbols

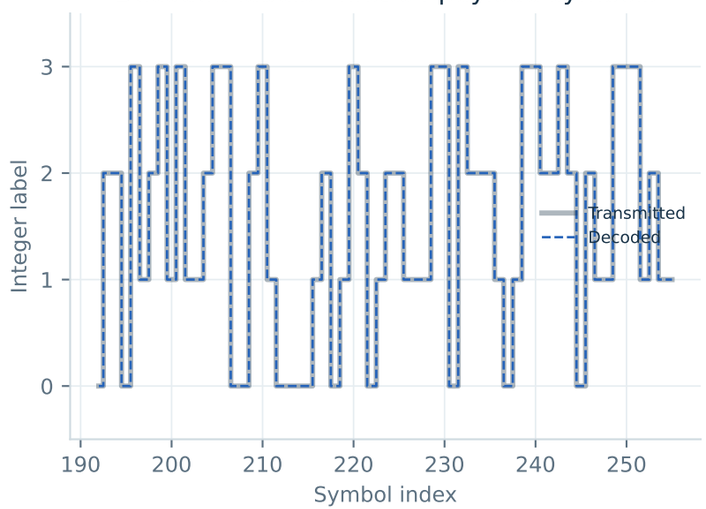








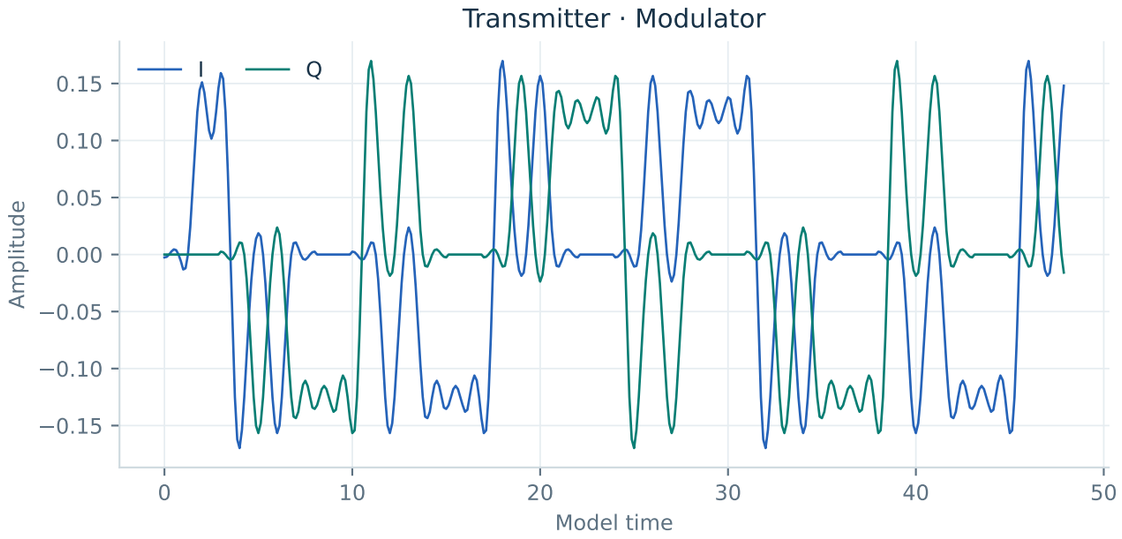


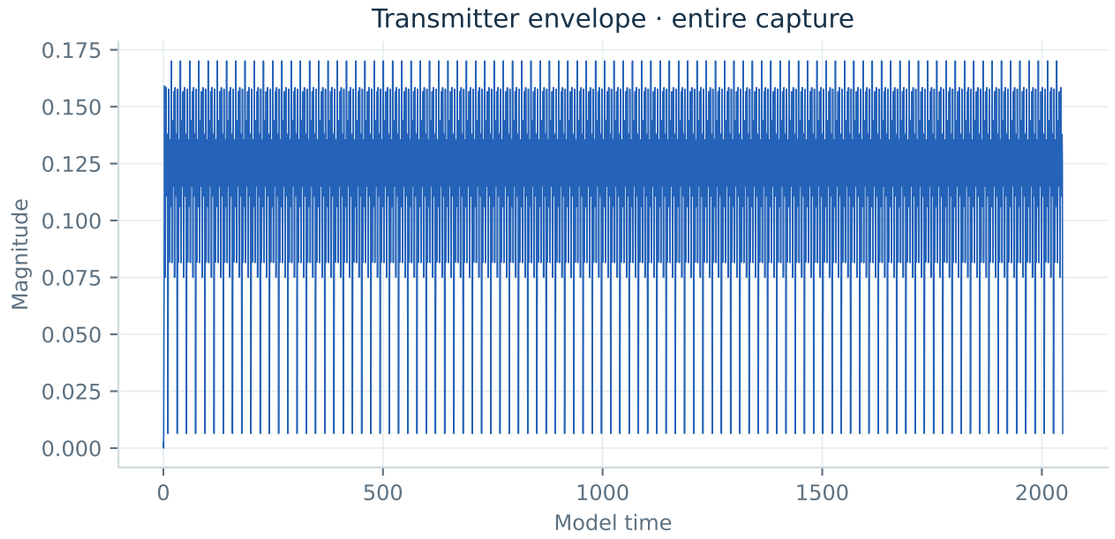


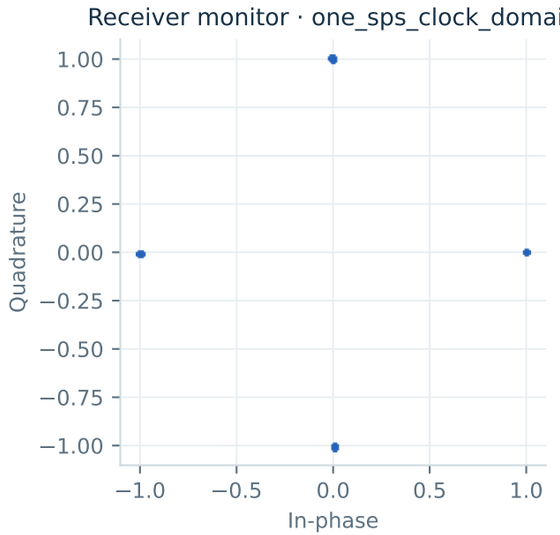
Source: saved portable regression vectors, replayed with the current Python core. The first 192 symbols are training and are excluded from BER/SER. Channel points are oversampled, not symbol decisions. Spectra share the TX peak reference. These figures do not establish original MATLAB receiver equivalence.

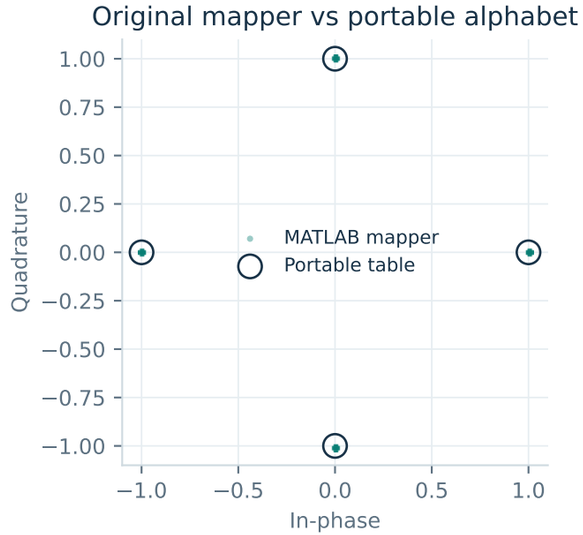
QPSK · feedforward / Farrow

ORIGINAL MATLAB CAPTURE / runtime PASS / exact repeat PASS / solver: FixedStepDiscrete





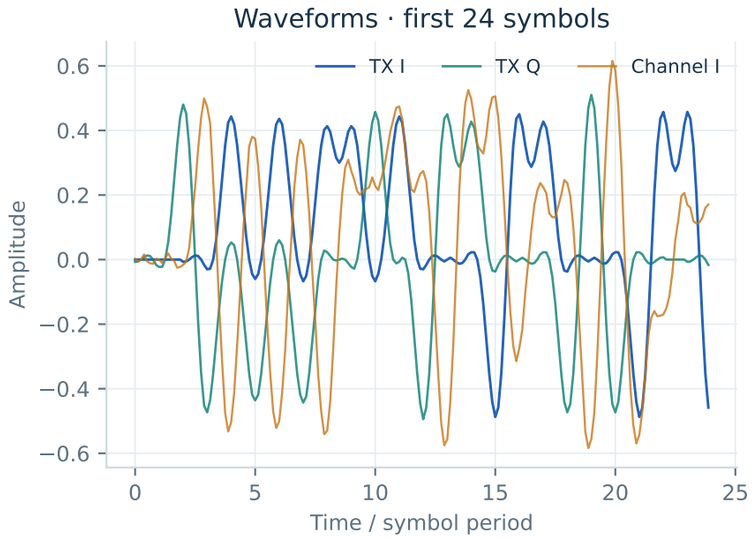


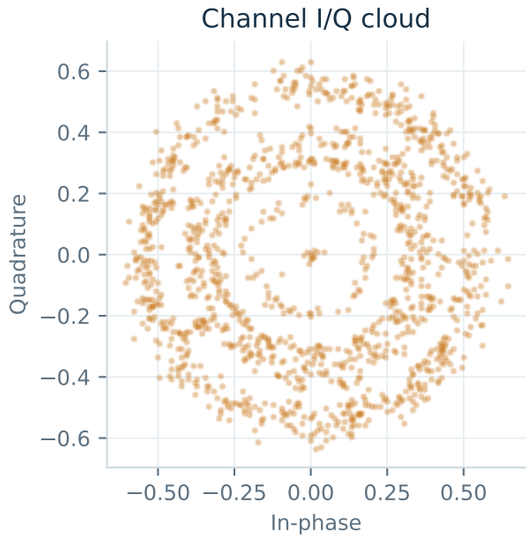


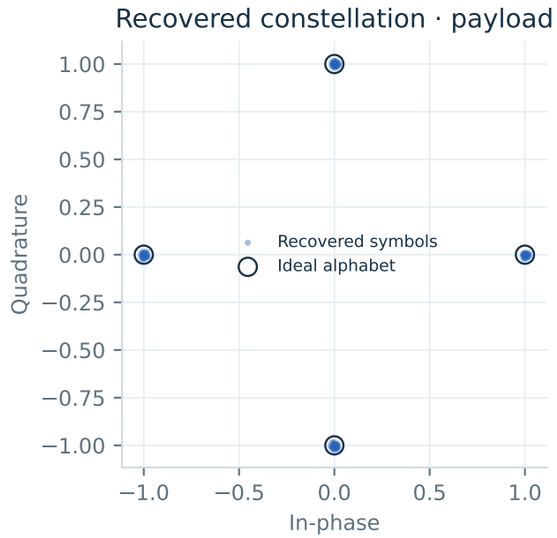
Actual R2020b signals with documented runtime settings. Native receiver monitors can contain held or oversampled values; their point clouds are not a BER measurement. Time is in the model's units. Simulation and repeatability do not establish receiver equivalence. Model: qpsk_ff_symbol_farrow_sync_ff_phase_sync_fast_agc_2007_05_20

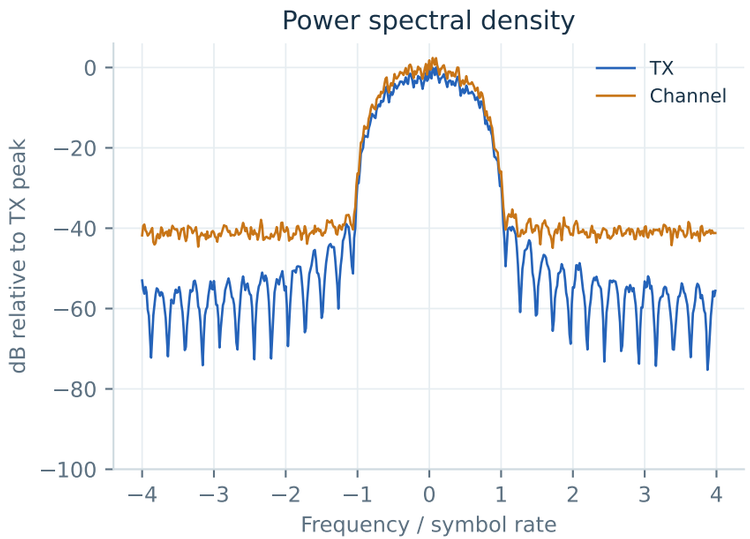
QPSK · feedforward / Farrow

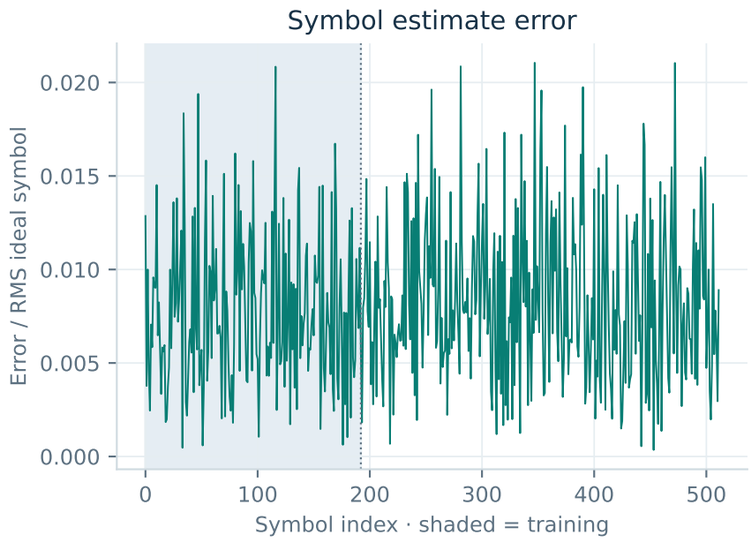
PORTABLE MODEL / Variant stress channel / 0 errors in 320 payload symbols

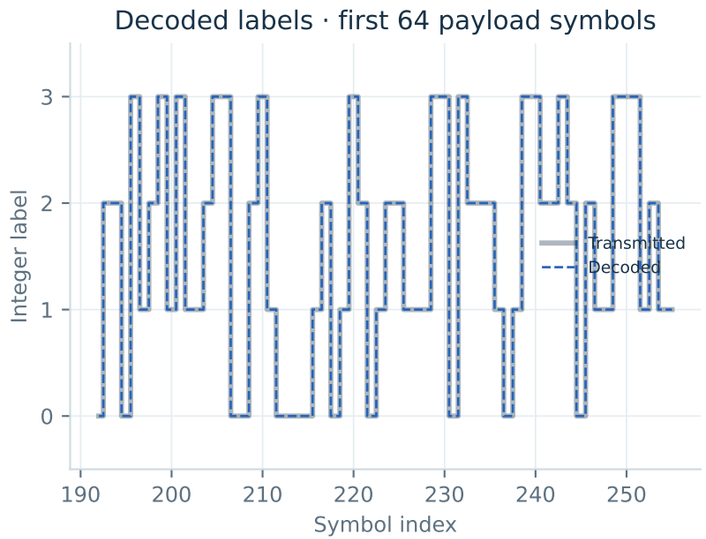








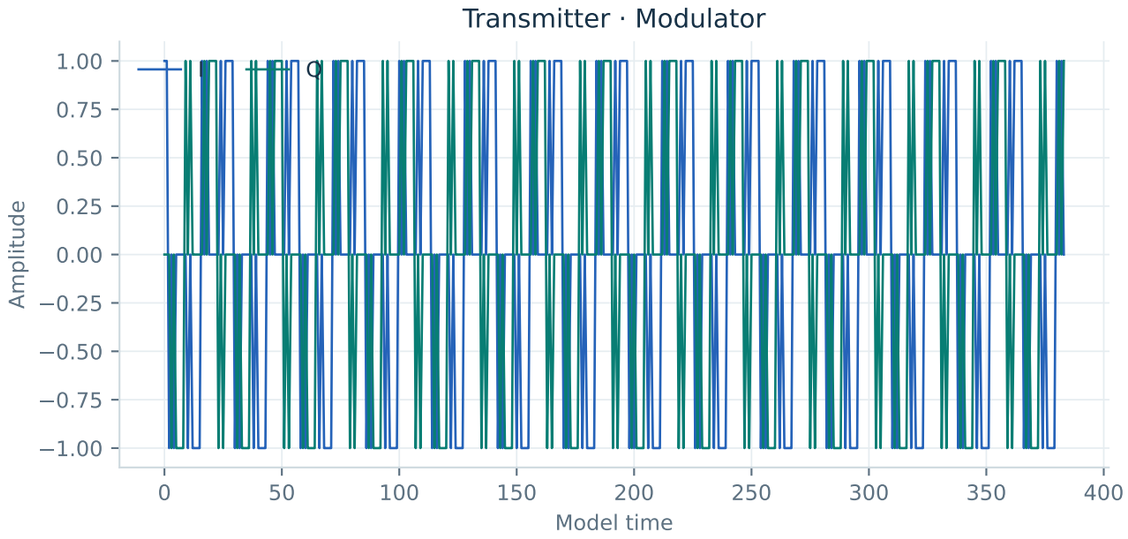


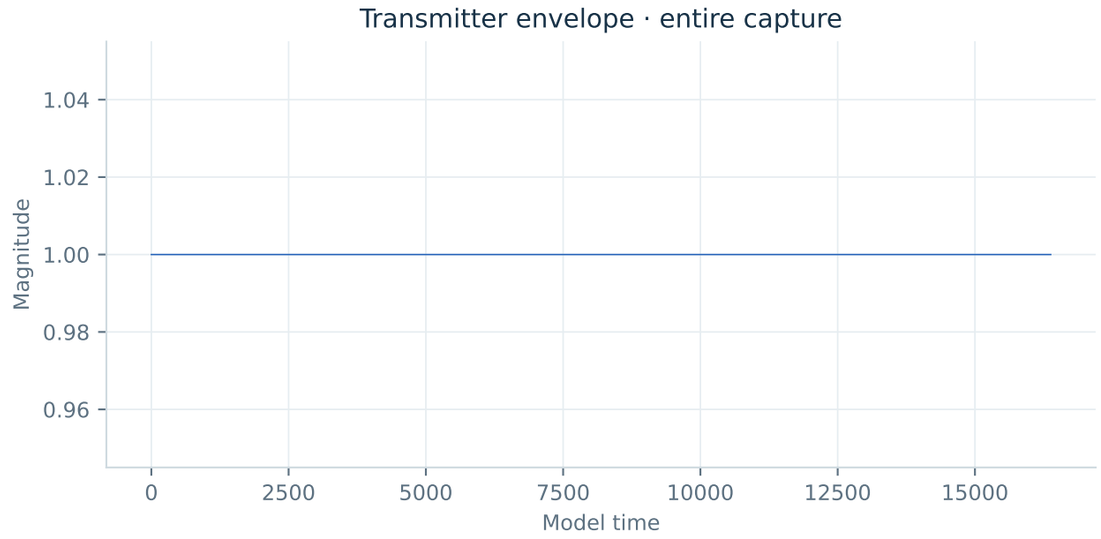


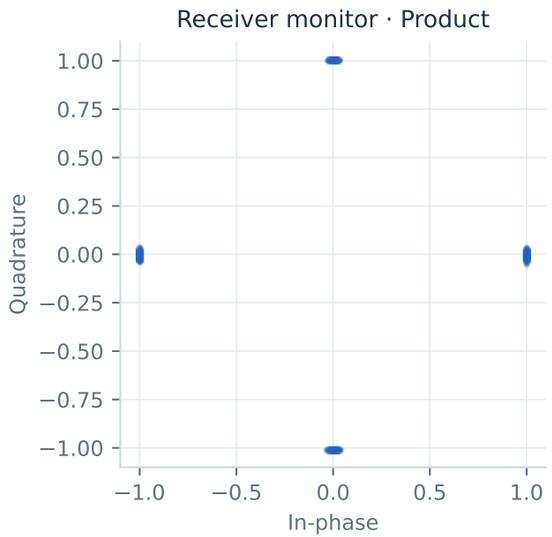
Source: saved portable regression vectors, replayed with the current Python core. The first 192 symbols are training and are excluded from BER/SER. Channel points are oversampled, not symbol decisions. Spectra share the TX peak reference. These figures do not establish original MATLAB receiver equivalence.

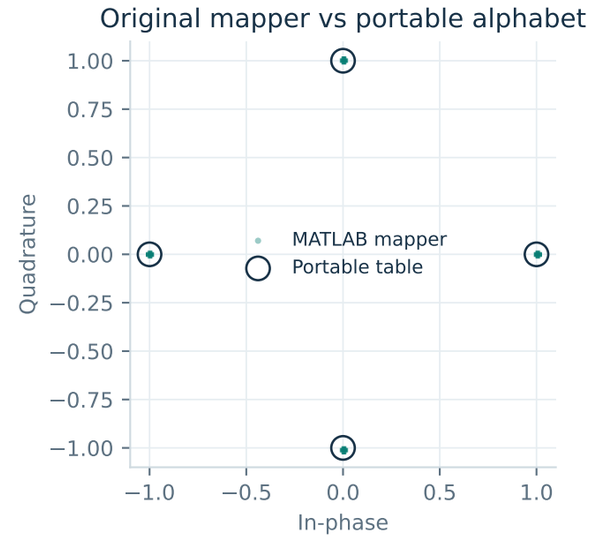
QPSK · phase tracking

ORIGINAL MATLAB CAPTURE / runtime PASS / exact repeat PASS / solver: FixedStepDiscrete





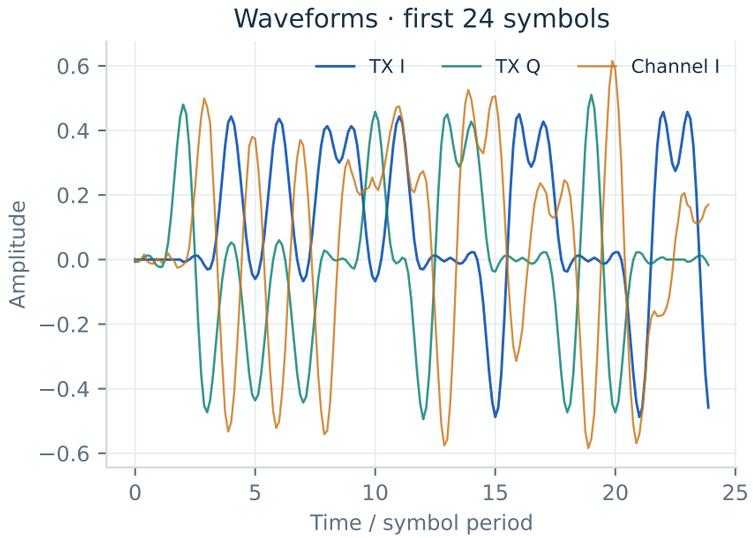


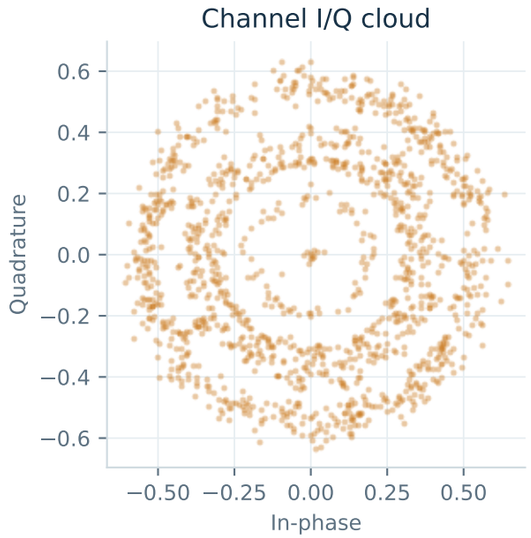


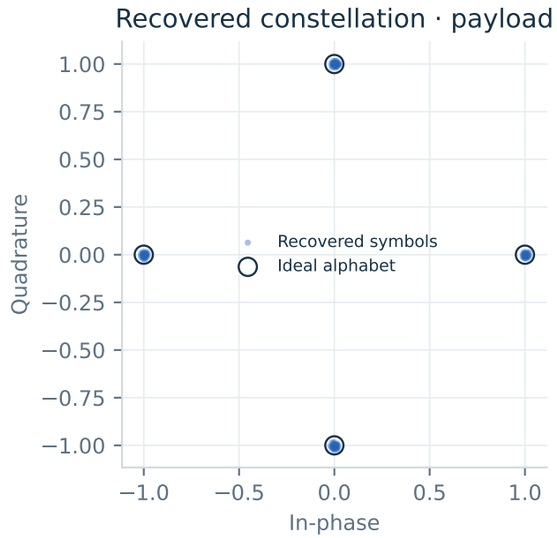
Actual R2020b signals with documented runtime settings. Native receiver monitors can contain held or oversampled values; their point clouds are not a BER measurement. Time is in the model's units. Simulation and repeatability do not establish receiver equivalence. Model: qpsk_phase_noise_phase_sync_2009_04_12

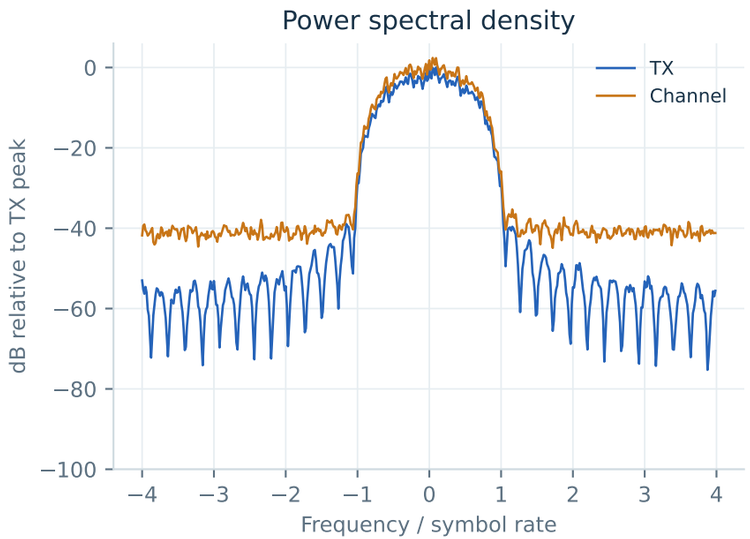
QPSK · phase tracking

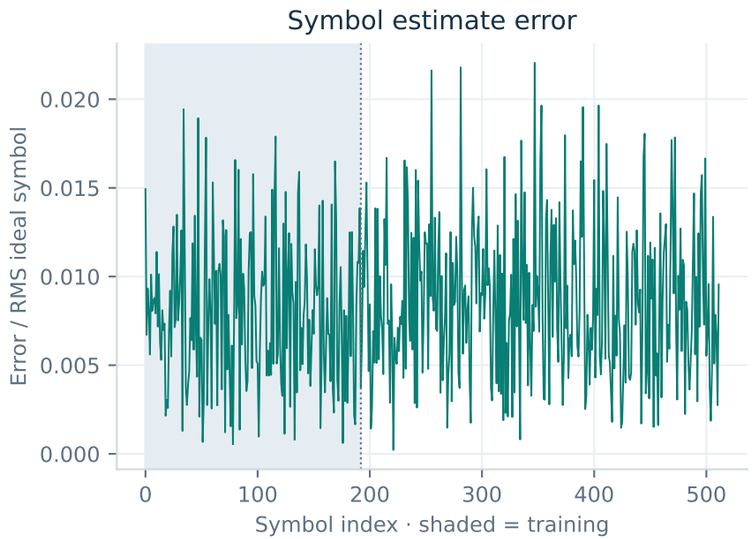
PORTABLE MODEL / Variant stress channel / 0 errors in 320 payload symbols

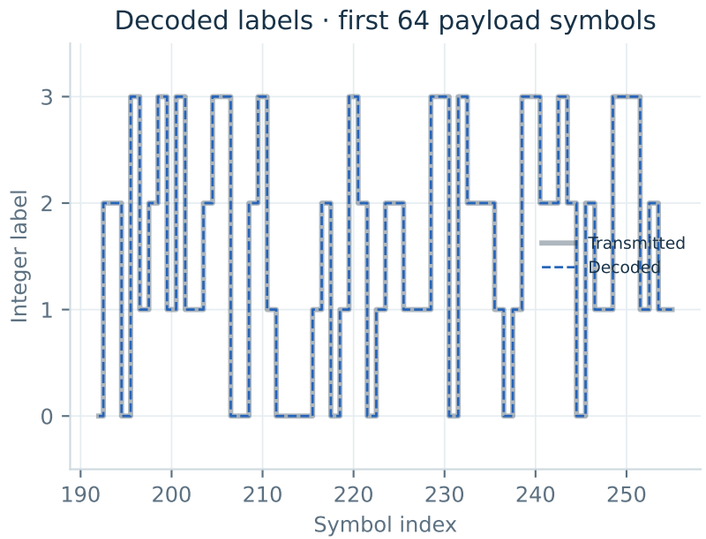








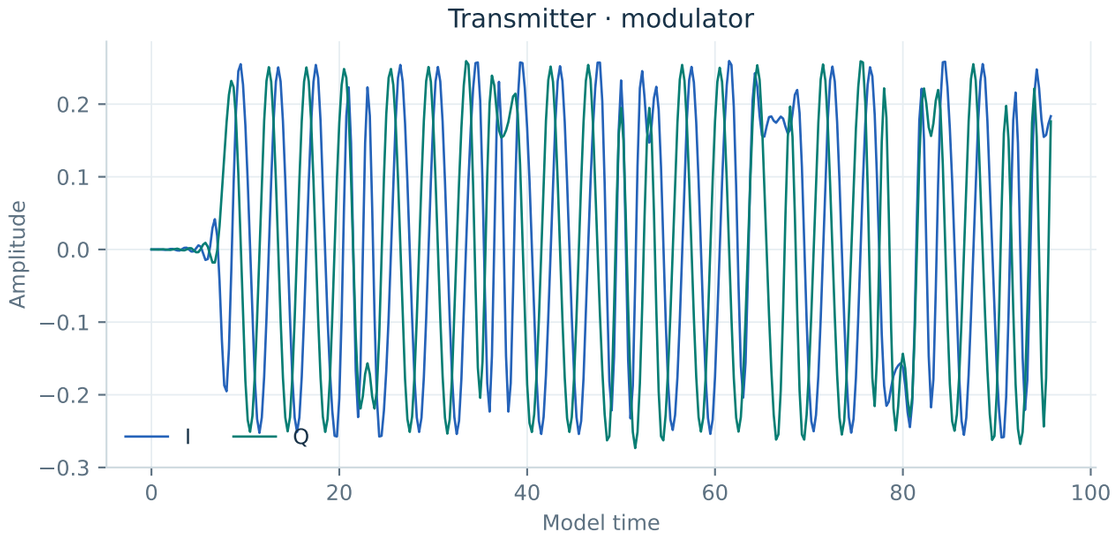


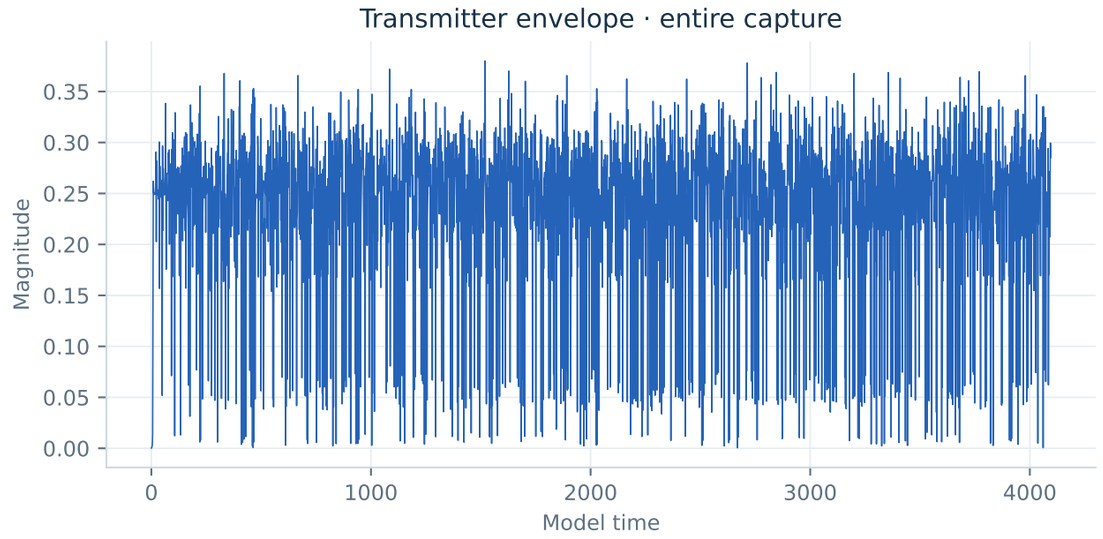


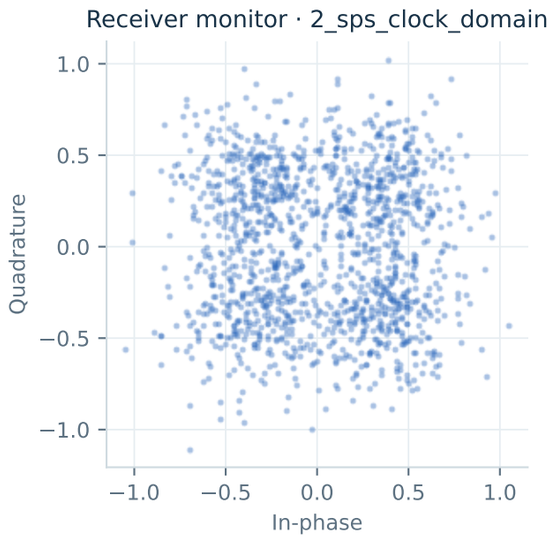
Source: saved portable regression vectors, replayed with the current Python core. The first 192 symbols are training and are excluded from BER/SER. Channel points are oversampled, not symbol decisions. Spectra share the TX peak reference. These figures do not establish original MATLAB receiver equivalence.

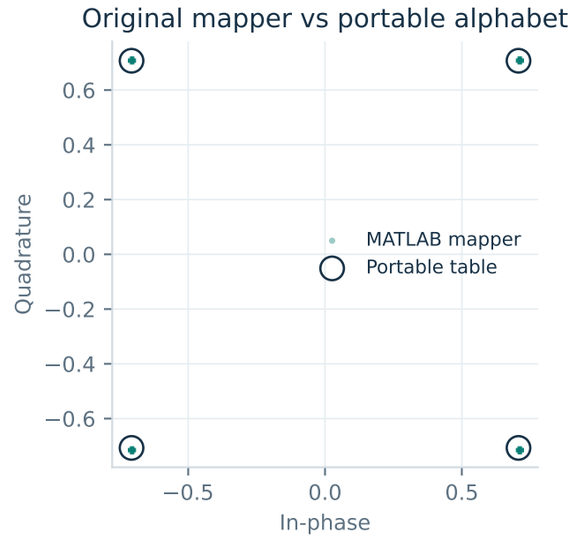
SIMO · maximal ratio combining

ORIGINAL MATLAB CAPTURE / runtime PASS / exact repeat PASS / solver: FixedStepDiscrete





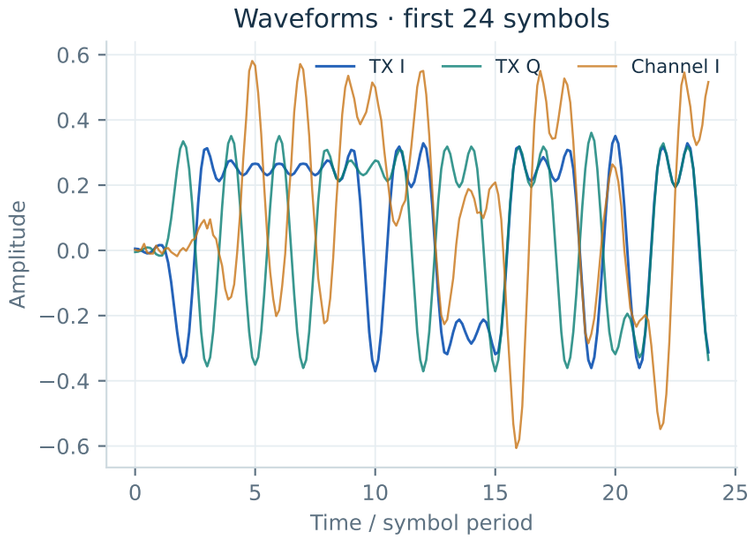


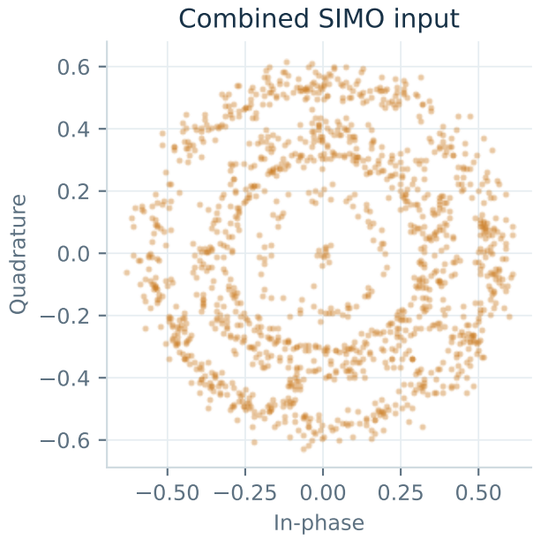


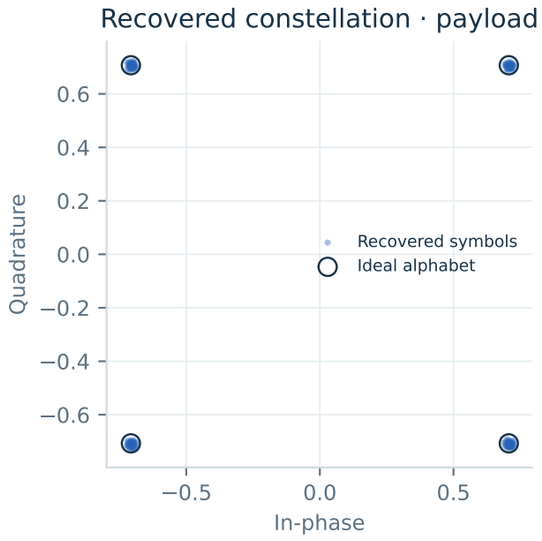
Actual R2020b signals with documented runtime settings. Native receiver monitors can contain held or oversampled values; their point clouds are not a BER measurement. Time is in the model's units. Simulation and repeatability do not establish receiver equivalence. Model: simo_maximal_ratio_combining_2010_08_27

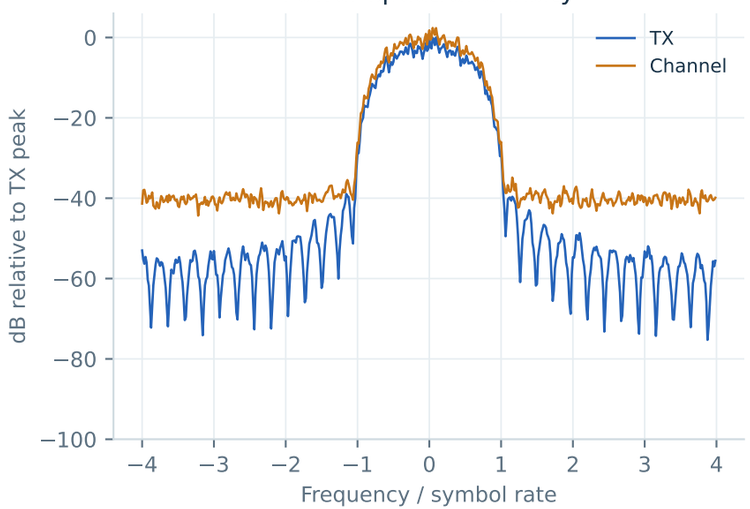
SIMO · maximal ratio combining

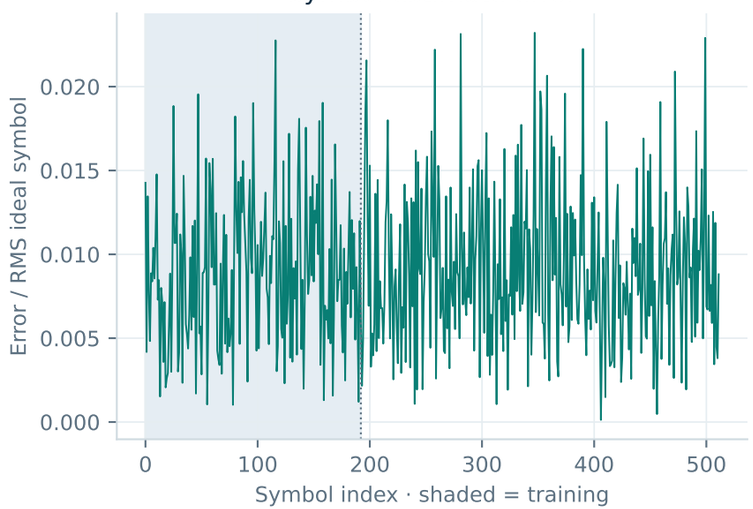
PORTABLE MODEL / Variant stress channel / 0 errors in 320 payload symbols

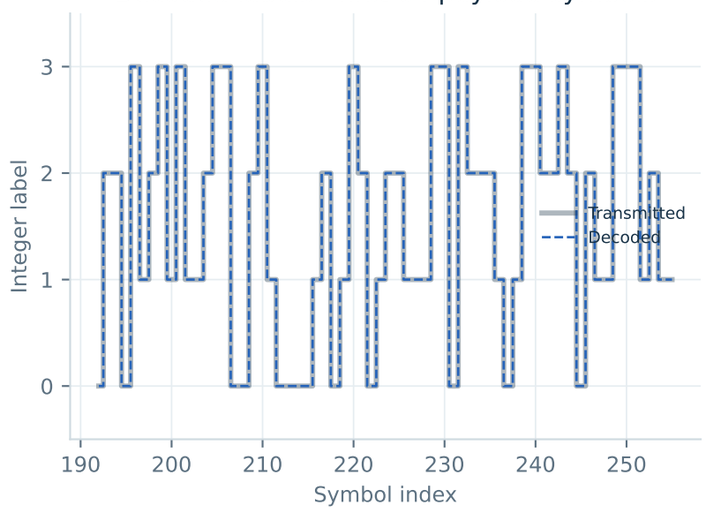












Source: saved portable regression vectors, replayed with the current Python core. The first 192 symbols are training and are excluded from BER/SER. Channel points are oversampled, not symbol decisions. Spectra share the TX peak reference. These figures do not establish original MATLAB receiver equivalence.